NIHVA is the global leader in door opening solutions, dedicated to satisfying end-user needs for security, safety and convenience

NIHVA

NIHVA TECHNOLOGIES PVT. LTD.

NIHVA High Speed doors HS-Series
Product range 2018

NIHVA is the global leader in door opening solutions, dedicated to satisfying end-user needs for security, safety and convenience

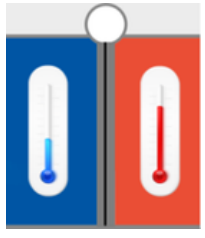
Supplied more than 100 high speed doors to SKODA in Czech Republic two factories in Mlada Boleslav and Krasimi

NIHVA is the global leader in door opening solutions, dedicated to satisfying end-user needs for security, safety and convenience

The global leader in door opening solutions

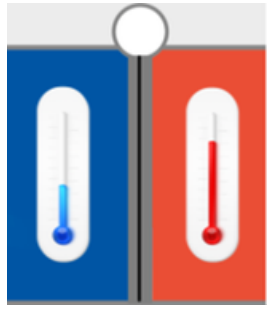
Client – what does the Client want?

- High quality, reliability & durability (doors often operate under tough conditions)
- Cost of ownership & level of after-service (doors are often “business critical ”)
- High speed, low noise level, aesthetics & transparency
- Security , safety and in compliance with all standards that apply
- Competitive pricing
- Easy (quick) installation- first time success
- Self reset, self repair,
- No hard edges



Universal Values & Future Trends

- Safety
- Reduce Energy Loss / Carbon Footprint / “Green” Solutions
- ROI
- Added Value
- Convenience
- Sustainability
- Easy to Use / Install / Maintain “value”
- Cost of Ownership
- After Service Offers / CBIB
- Industry 4.0
- IoT

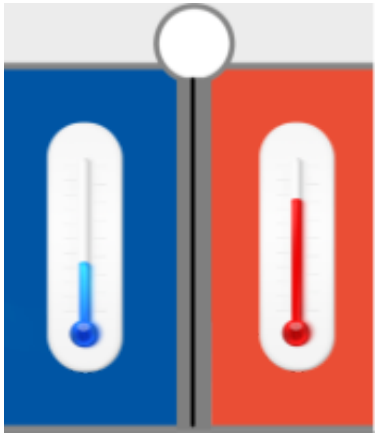


Benefits und savings of High Speed Doors

Optimization and or control of indoor climate

Draft control

Sealing/hygiene, avoiding pollution



Maintaining temperature differences

Noise reduction

Benefits und savings of High Speed Doors

Fast opening and closing cycles

Optimized processes

Fast material flow

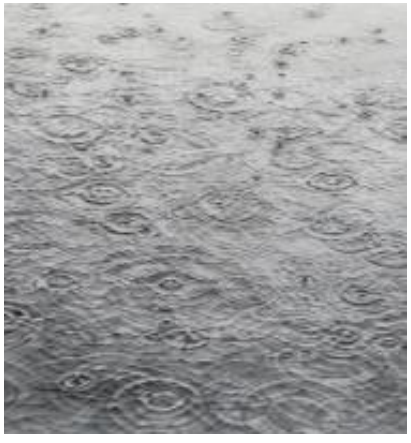
Minimized waiting times

Energy savings

Benefits und savings of High Speed Doors

Protection against weather and environmental influences

Wind



Rain

Extreme temperatures

Dust/dirt

Content

1. **Technology overview**
2. “Gravity” driven doors
3. “Power Drive” driven doors
4. Applications/Options
5. Technical Summaries
6. Sales

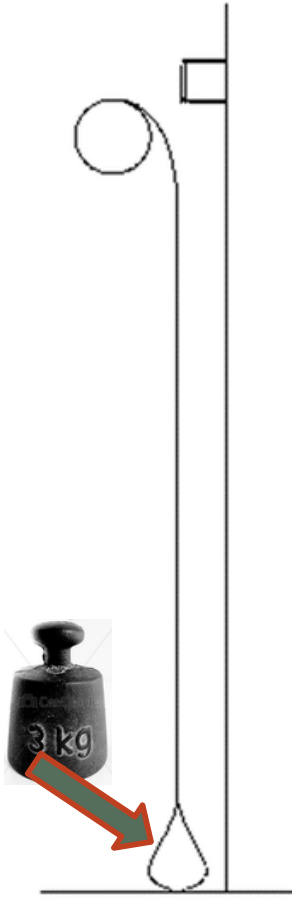
TECHNOLOGIES

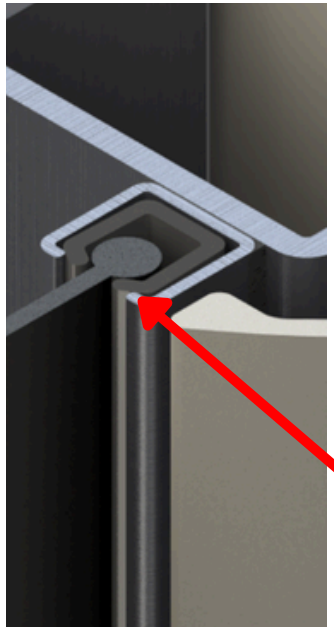
« GRAVITY »

« POWER DRIVE »

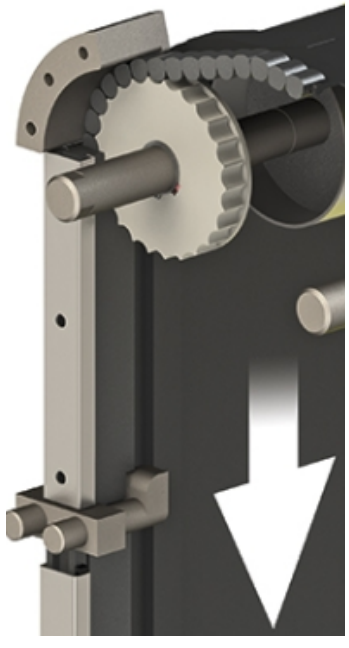
TECHNOLOGIES

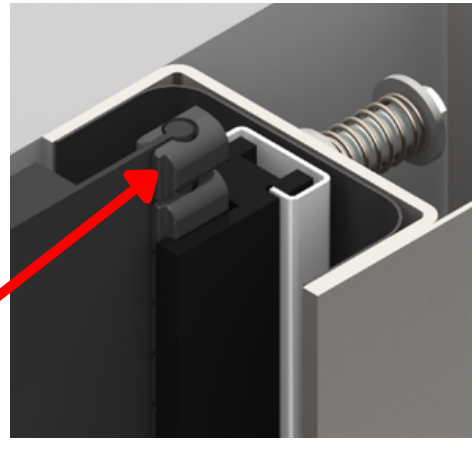
« GRAVITY »





« POWER DRIVE »



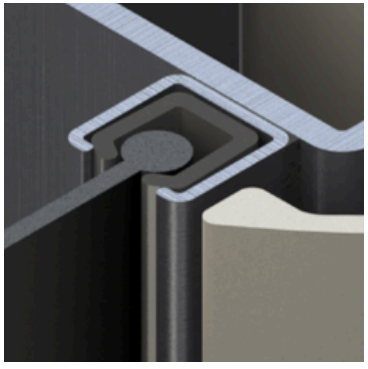


Zipper

TECHNOLOGIES - 5 Platforms

« GRAVITY »

- Roll-up system for opening-cycle
- Closing-cycle achieved through gravity but without the use of rigid elements or heavy ballast
- Inside application only - Not suitable for high wind or pressure applications

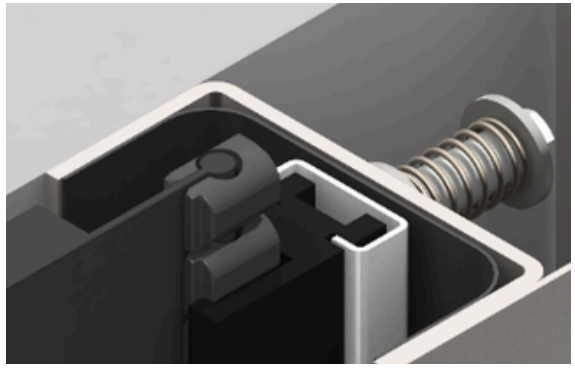


Interior

HS9020 G	HS9030 G
Low sliding friction	Low sliding friction
4000 x 4000	4000 x 4000 Aesthetics

« POWER DRIVE »

- Gear driven system
- No gravity ballast, no rigid bottom elements, offering unrivalled user safety
- Suitable for interior and exterior, high wind or pressure applications



Interior - exterior

HS9010 P	HS9110 P	HS9010 PL
HS8010 P	HS8110 P	HS8010 PL
5500 x 5500m	5500 x 5500m Low noise level	11000 x 5500 9000 x 5500

Door model overview HS-Series

	Gravity			
9	0	20	G	-
				HY
				AT
		30	G	-
				HY
				AT
				EM

	Power Drive			
9	0	10	P	-
				L
				EM
				FR
	1	10	P	-
8	0	10 20	P	- L
	1	10	P	-

8	exterior door			
9	interior door			
	0 1	Generation		
		10 20 30	Product characteristics	
			"P"	Power drive
			"G"	Gravity
			L	Large
			HY	Hygienic
			AT	AirTight
			EM	Emergency
			FR	Freezer

Product name	Aplicatio n	use by
Albany HS9020 G	Interior	Standard
Albany HS9020 GHY	Interior	Food
Albany HS9020 GAT	Interior	Clean
Albany HS9030 G	Interior	Standard
Albany HS9030 GHY	Interior	Food
Albany HS9030 GAT	Interior	Clean
Albany HS9030 GEM	Interior	Emergency
Albany HS9010 P	Interior	Higher Windclass
Albany HS9010 PL	Interior	Max. Windclass
Albany HS9010 PEM	Interior	Emergency
Albany HS9010 PFR	Interior	Freeze
Albany HS9110 P	Interior	Higher Windclass
Albany HS8010 P	Exterior	Higher Windclass
Albany HS8010 PL	Exterior	Max. Windclass
Albany HS8020 P	Exterior	Higher Windclass
Albany HS8020 PL	Exterior	Max. Windclass
Albany HS8110 P	Exterior	Higher Windclass

Product range

Gravity doors

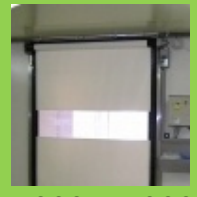
interior

4000 x 4000

HS9020 G

4000 x 4000

HS9020 GAT



4000 x 4000

HS9020 GHY

4000 x 4000

HS9030 G

HS9030 GHY

4000 x 4000

HS9030 GAT

4000 x 4000

HS9030 GEM

Wind Class 1

Power drive doors

interior

5000x5000

HS9010 P

5000x5000

HS9010 PEM

4000x5500

HS9010 PFR

5500x5500

HS9110 P

11000x5500

HS9010 PL

Wind Class 1

Power drive doors

exterior

3-4

5500x5500

HS8010 P

2-3

5500x5500

HS8110 P

4-5

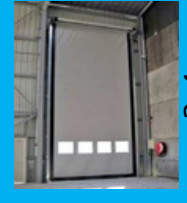
5500x5500

HS8020 P

2-3

9000x5500

HS8010 PL



3-4

8000x5500

HS8020 PL

Wind Class ....

Sample Doors HS9030G / HS8020PL

Content

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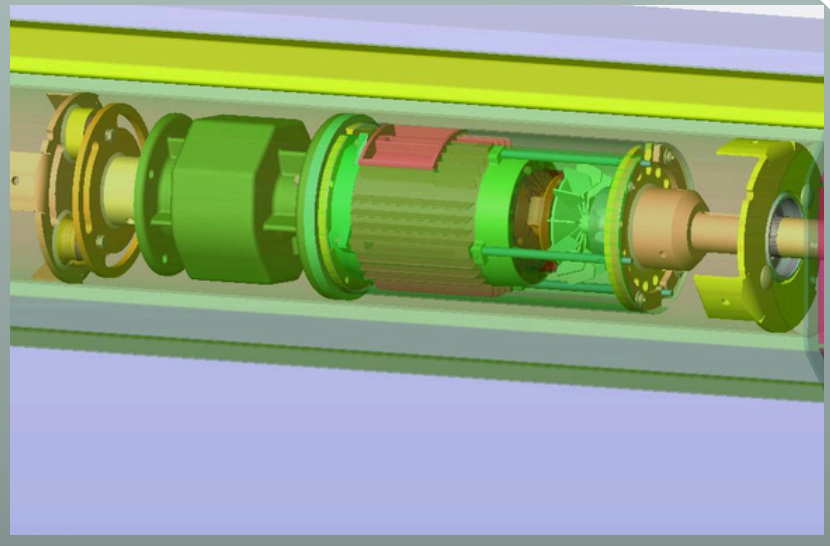
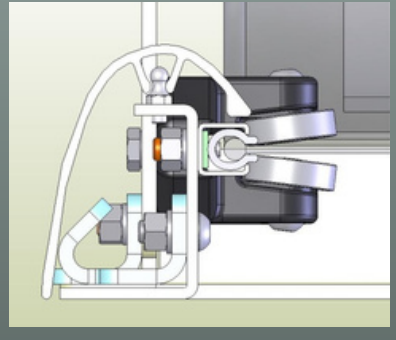
Gravity doors HS90xx Series

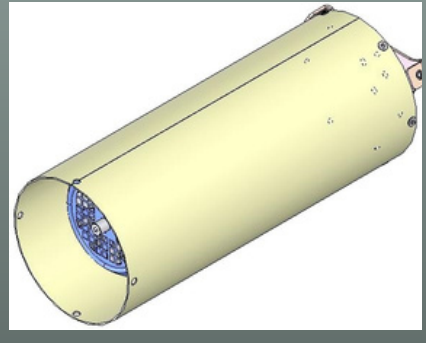
Gravity driven designed for interior use only. Flexible bottom with small pockets of ballast

HS9020 G series

HS9030 G Series







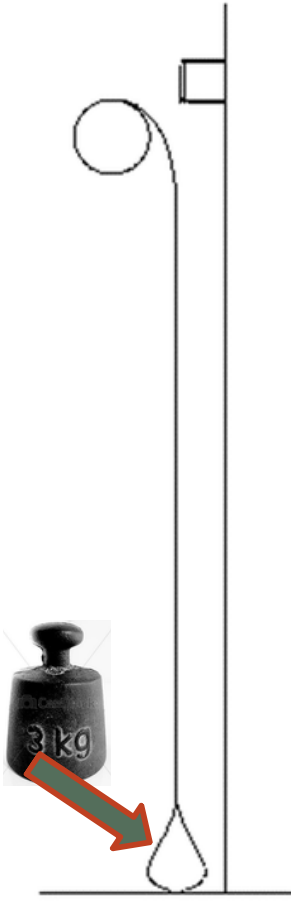
Gravity

HS9020 G
HS9020 GAT
HS9020 GHY

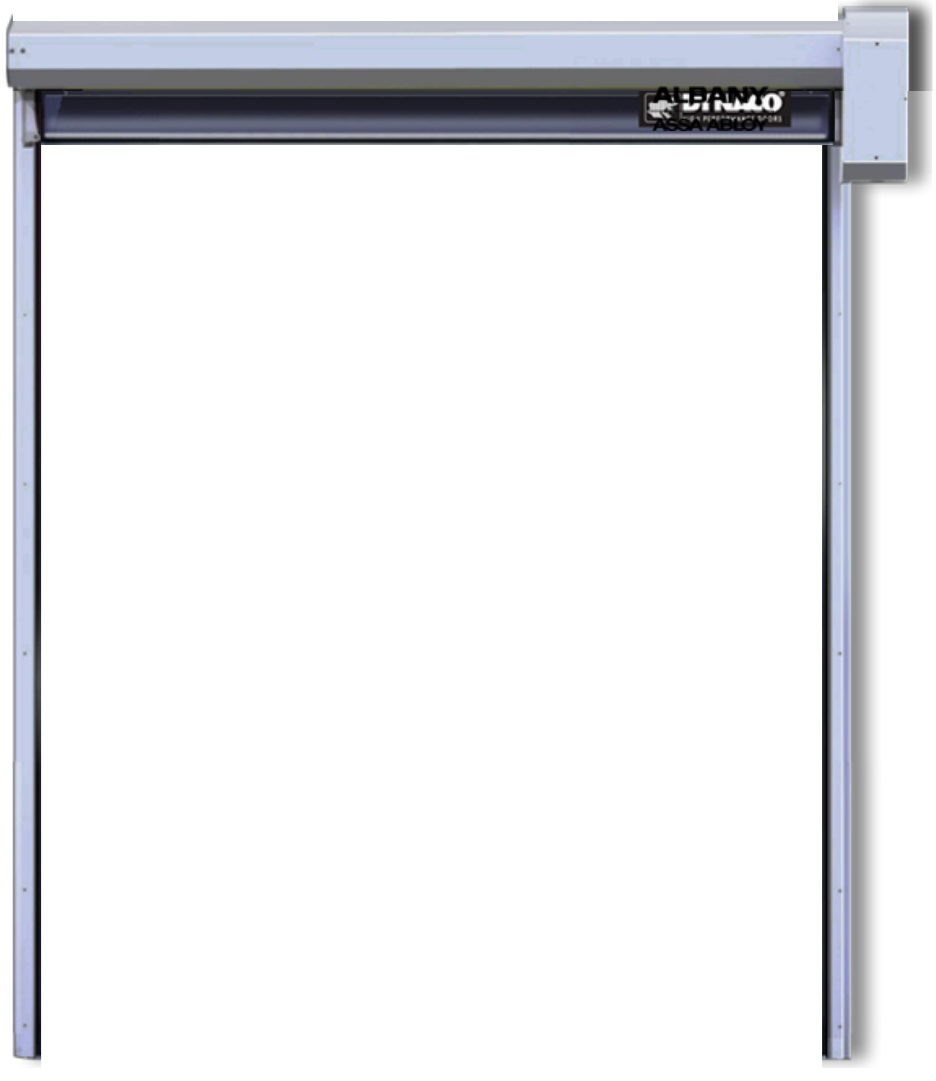
HS9030 G
HS9030 GAT
HS9030 GHY

HS9030 G
HS9030 GEM

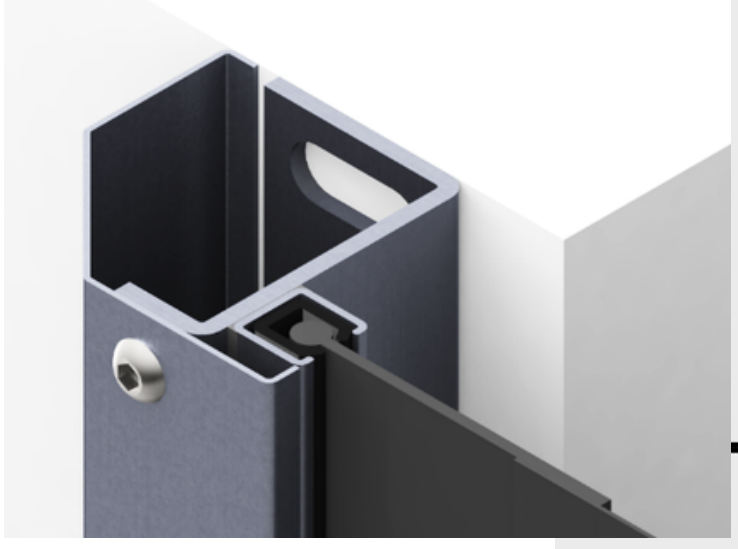
Gravity

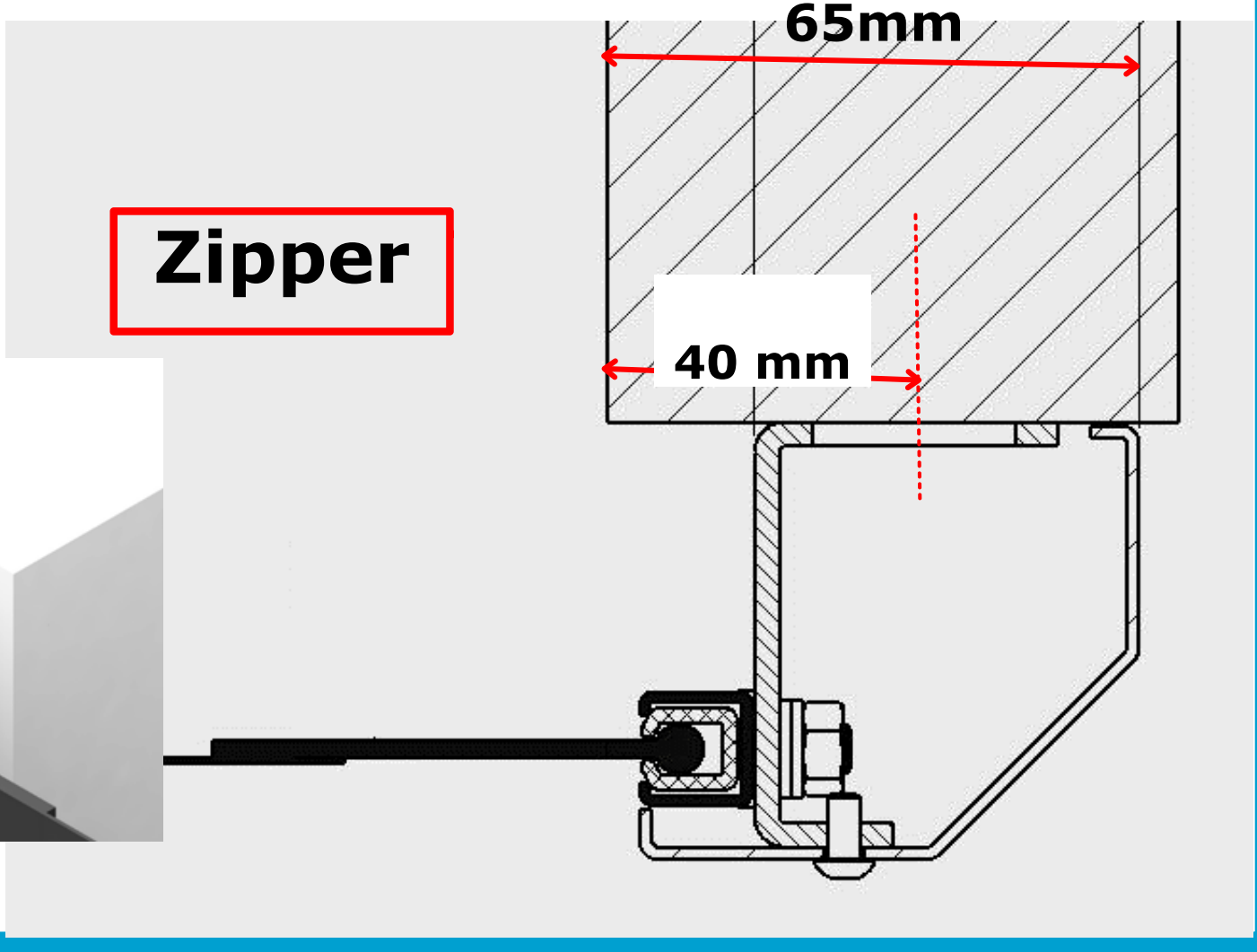


Weight based closing



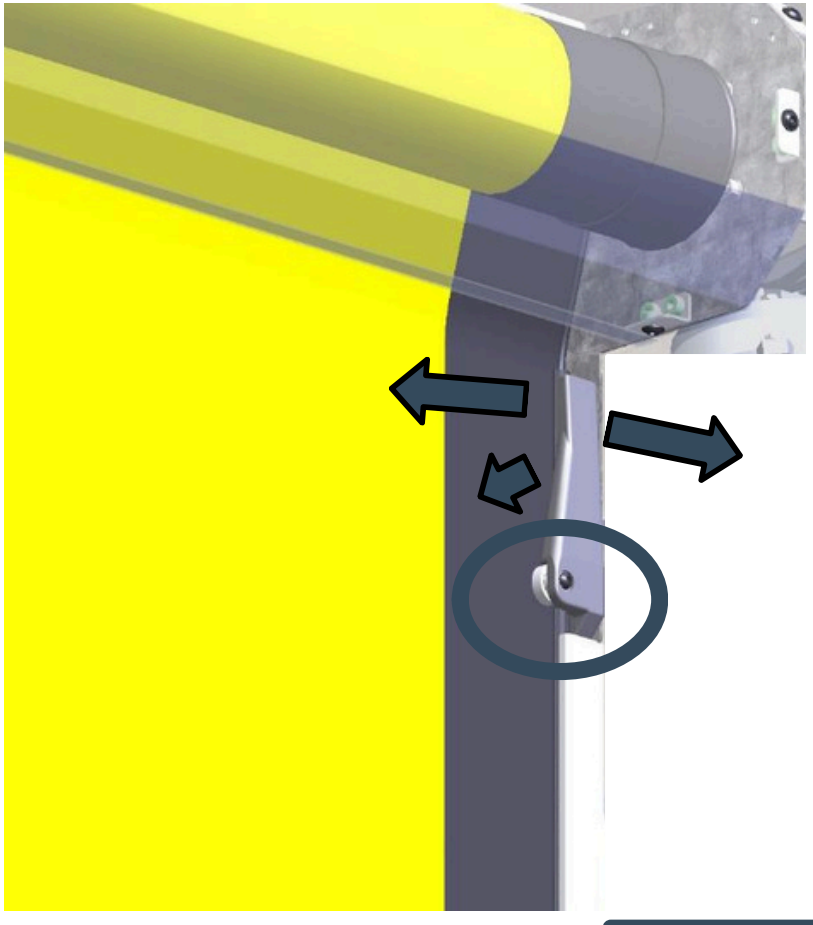
Gravity

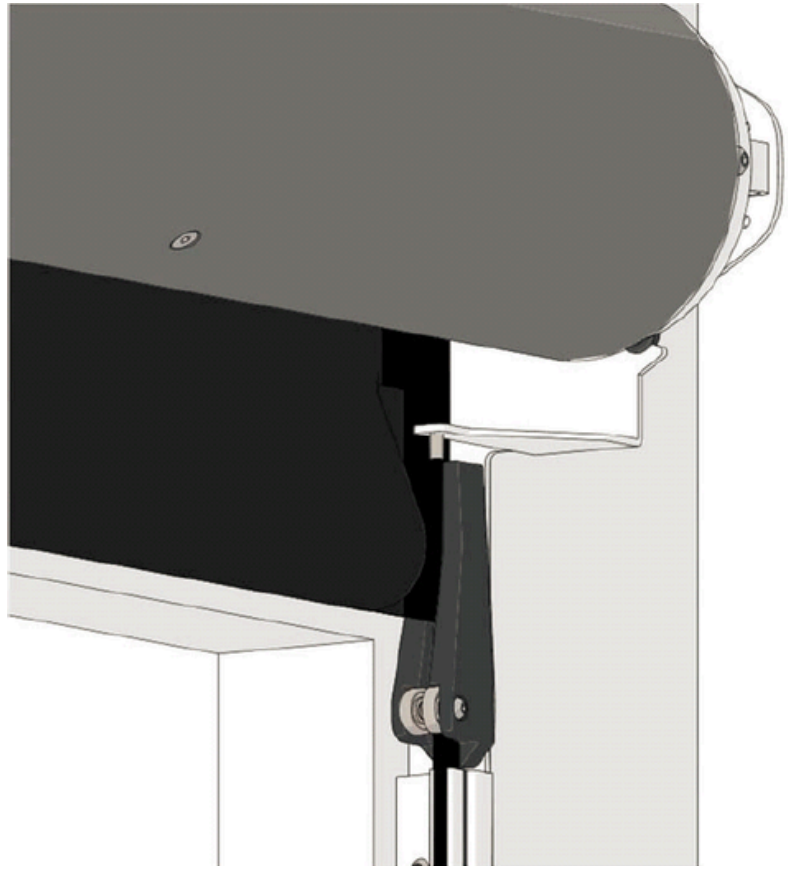




Gravity

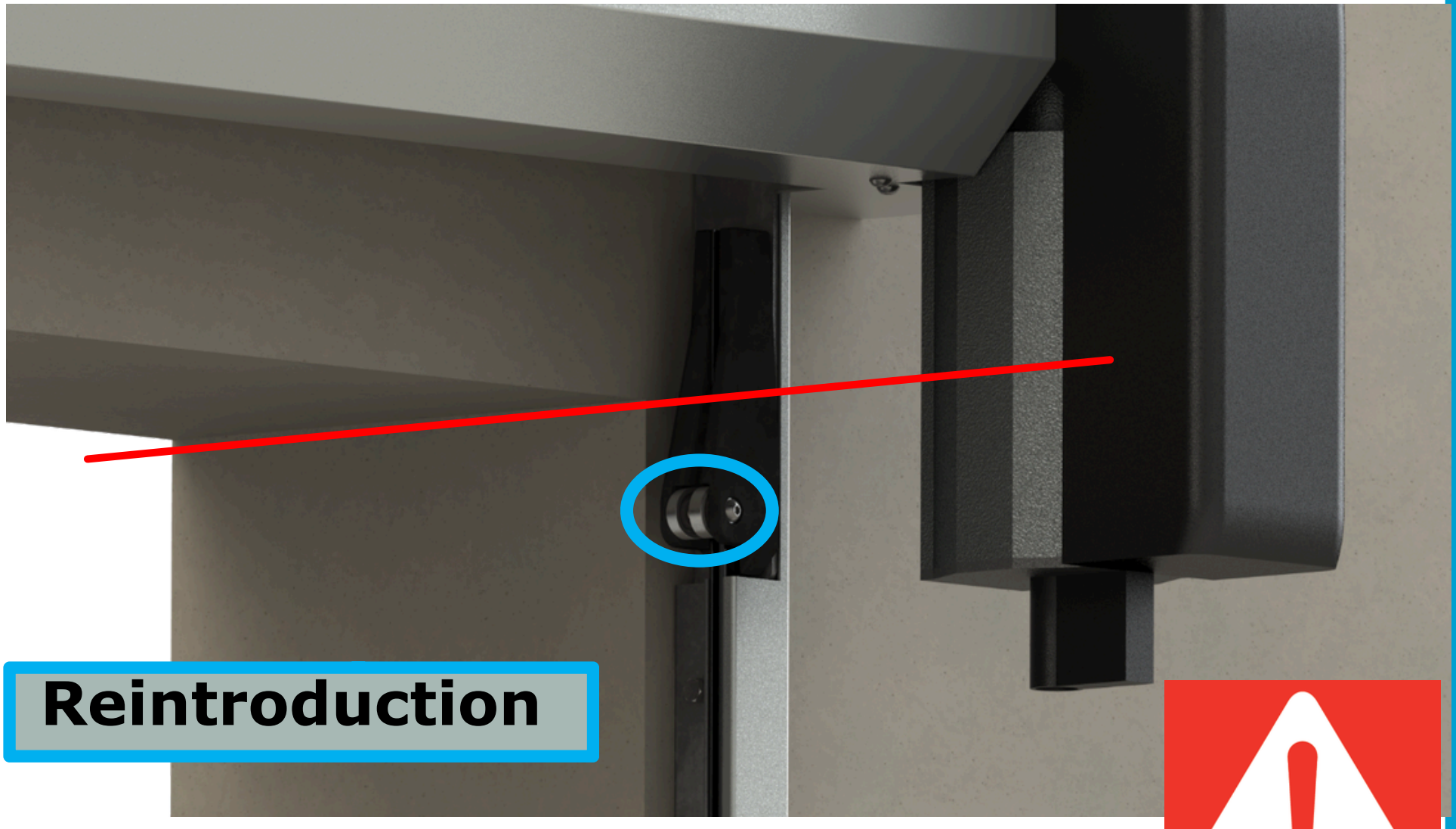
HS90xx





Reintroduction

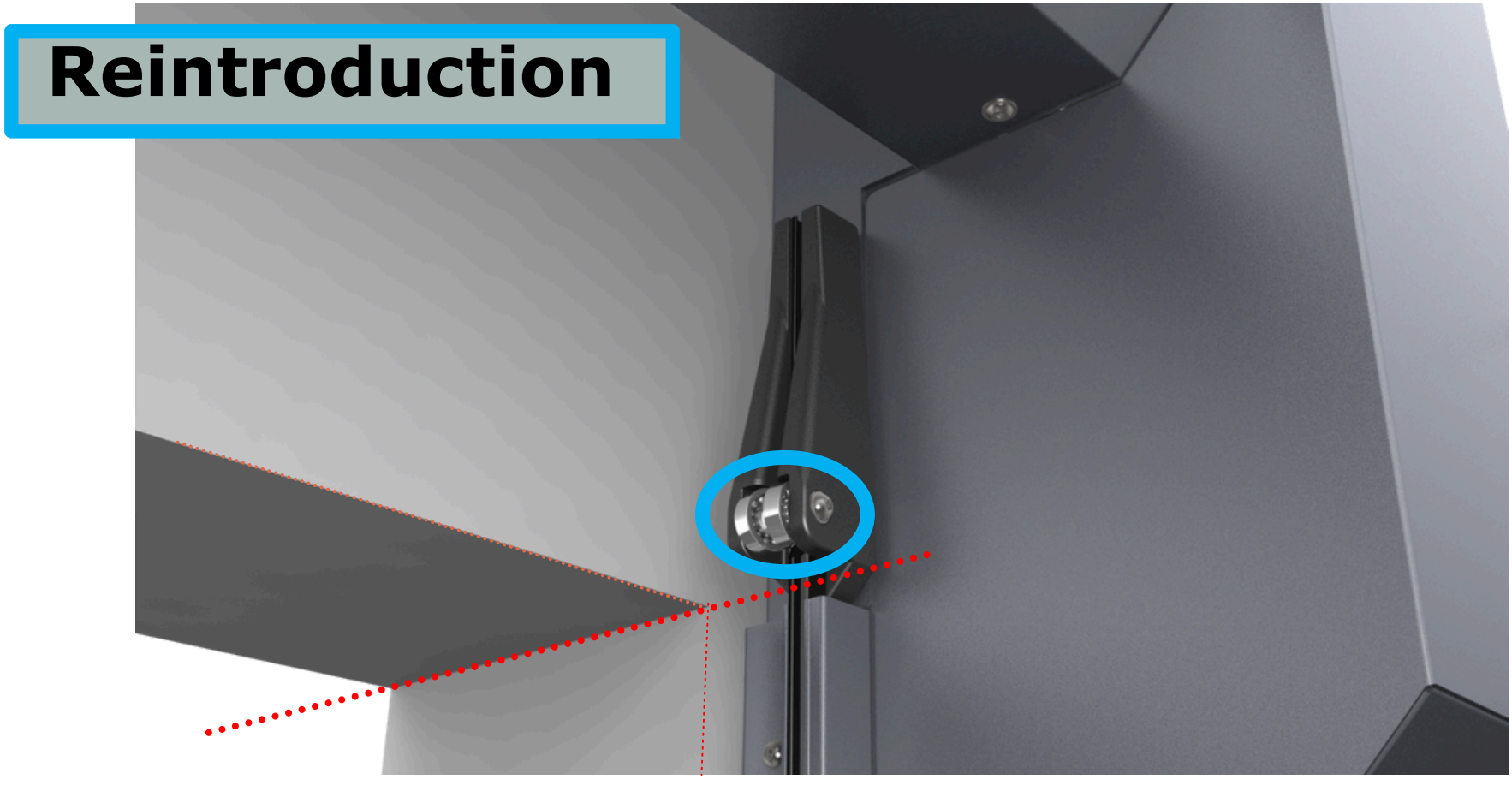
Gravity



Reintroduction point is BELOW lintel. -> risk!

Gravity

Reintroduction point is ABOVE lintel. -> perfect!



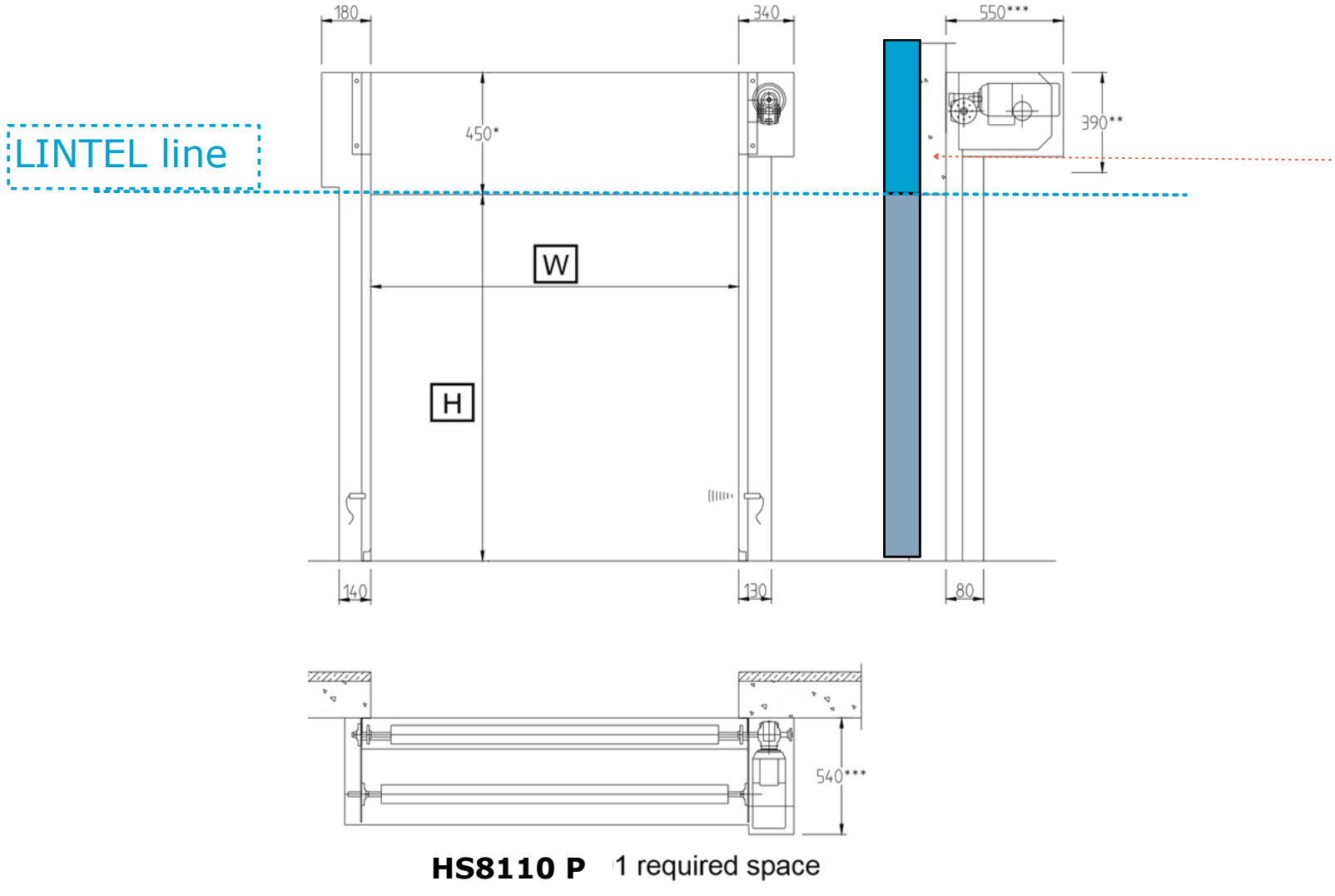
Gravity

Reintroduction

Reintroduction point is BELOW lintel. -> risk!

Gravity

Lintel height



HS8110 P 1 required space

Gravity

A GRAVITY driven door is always an Inside used door

	Gravity			
9	0	20	G	-
				HY
				AT
		30	G	-
				HY
				AT
				EM

	Power Drive			
9	0	10	P	-
				L
				EM
				FR
	1	10	P	-
8	0	10 20	P	- L
	1	10	P	-

Gravity

HS9020 ...

Gravity: HS9020 ...

HS9020 G

HS9020 GAT
Air Tight "Clean"

HS9020 GHY
Hygiene "Food"

Gravity: HS9020 ...

HS9020 G

Used for Openings:

Basic economic door solution for building applications with galvanized framework

Up to 4x4

No Wind (WC 1)

Slim Footprint

Gravity: HS9020 ...

HS9020 GAT cleanroom

<u>Used for Openings:</u>

"AIR TIGHT"

Basic economic door solution

with galvanized framework and extra

sealings (stainless steel optional)

Up to 4x4

No Wind (pressure ok)

Slim Footprint

Sealed bottom profile

Gravity: HS9020 ...

HS9020 GHY
Hygiene/Food

<u>Used for Openings:</u>

"HYGIENE"

Basic economic door solution

with stainless steel frame and

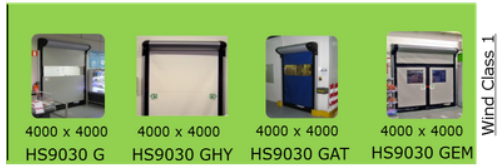
covers for wash down areas

Up to 4x4

No Wind (WC 1)

Slim Footprint

Gravity: HS9030 ...



HS9030 G	HS9030 GAT Air Tight "Clean"	HS9030 GHY Hygiene "Food"	HS9030 GEM Emergency

Gravity: HS9030 ...

HS9030 G

Used for Openings:

"TUBULAR MOTOR"

"AESTHIC COMPOSITE DESIGN"

"NO RH/LH MOTOR SIDING"

Next level of economic door solution

All covers made of composite material (PES resin glass fiber) for high corrosion resistance

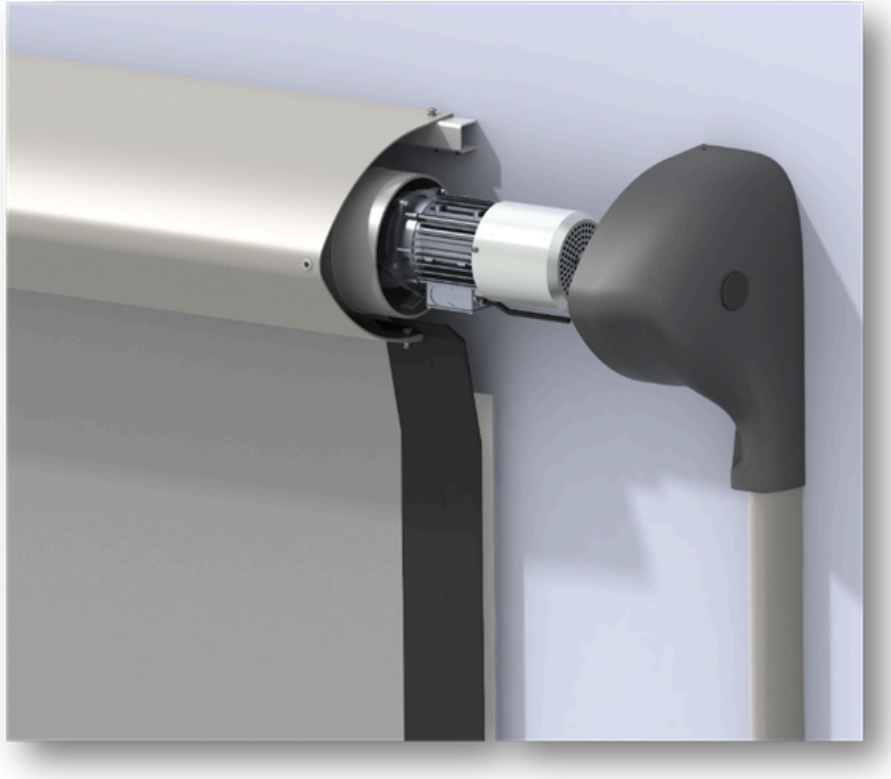
Inner framework galvanized steel

Up to 4x4

No Wind (WC 1)

Gravity: HS9030 ...

HS9030 G



TUBULAR MOTOR = NO RH/LH MOTOR SIDING,

Motor always in the RH side of the barrel

Gravity: HS9030 ...

HS9030 GHY
Hygenic/Cleanroom

Used for Openings:

“Hygiene”

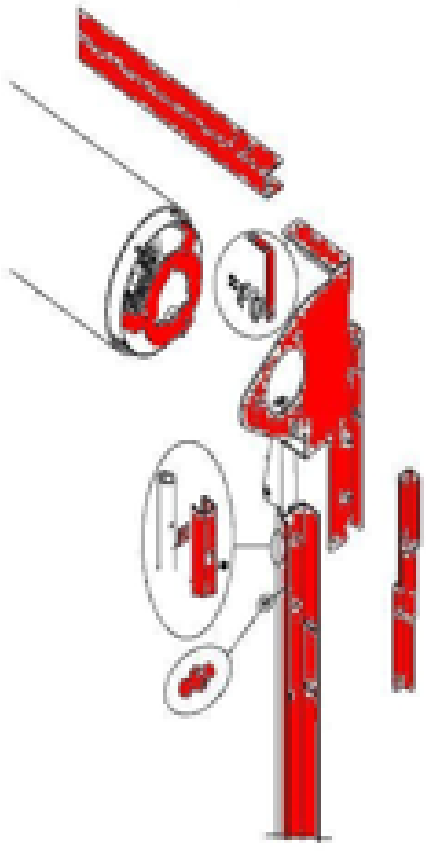
All covers composite material (PES resin glass fiber) for high corrosion resistance

Inner framework stainless steel

Air Leakage Class 2 (at 50 Pa)

Up to 4x4

No Wind (WC 1)



Gravity: HS9030 ...

HS9030 GAT
Air Tight "clean"

Used for Openings:

"AIR TIGHT"

Extra Sealings for air tightness:

Class 3 under/ 4 overpressure (at 50 Pa)

All covers composite material

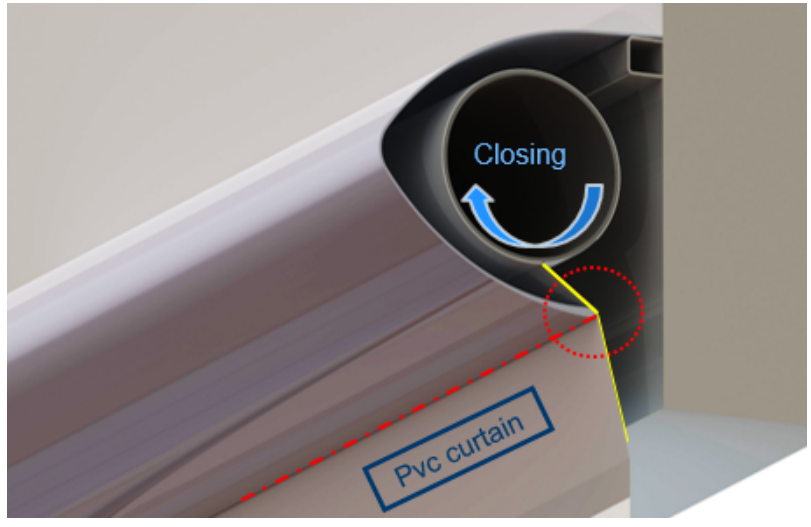
(PES resin glass fiber) for high

corrosion resistance

Inner framework galvanized steel

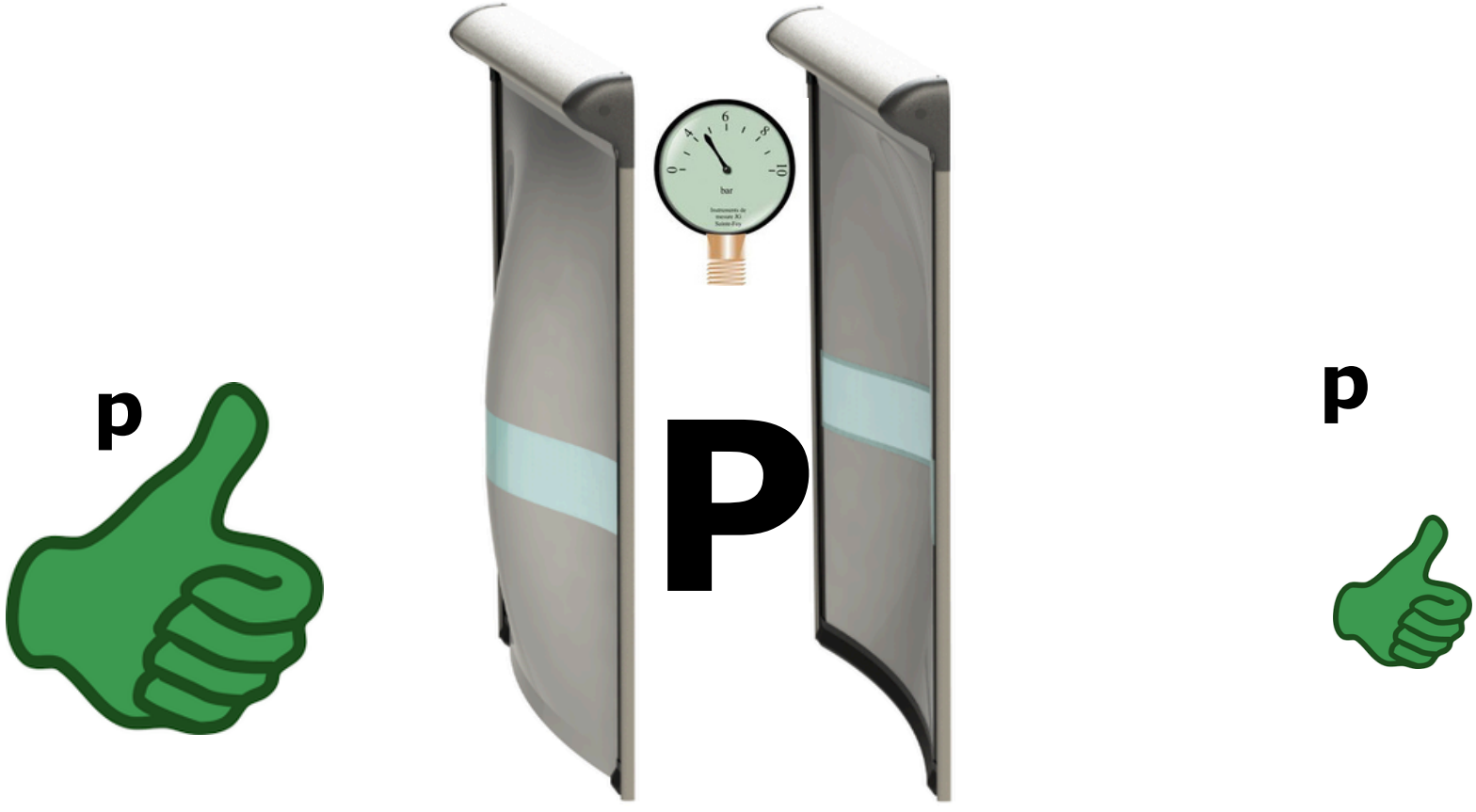
Up to 4x4

No Wind (WC 1)



Gravity: HS9030 ...

HS9030 GAT
Air Tight



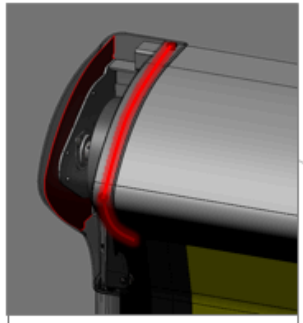
overpressure

underpressure

Gravity: HS9030

HS9030 AT
Air Tight

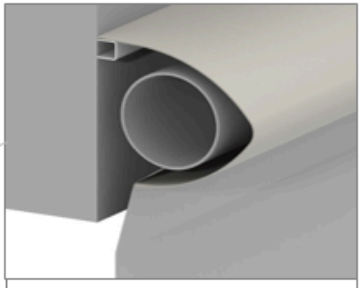
Cleanroom sealing overview



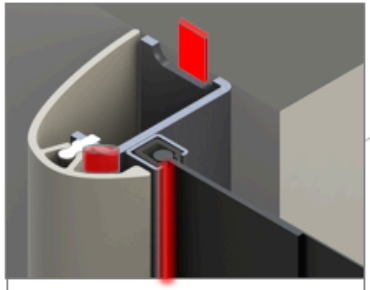
EPDM strip between the drum cover and lateral covers

Sealing plate of plastic inside the lateral covers

Silicon on the periphery of the door



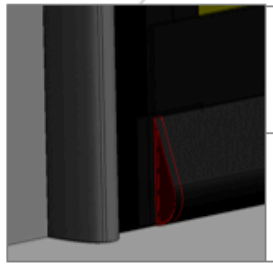
Pressure contact between the curtain and drum cover



EPDM profile inside the door post covers

Continuous side sealing with BEAD technology

EPDM strip behind door post



Bottom bag sealing with drop

Bottom bead straight finish

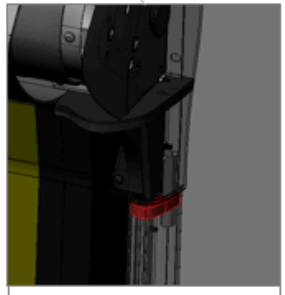
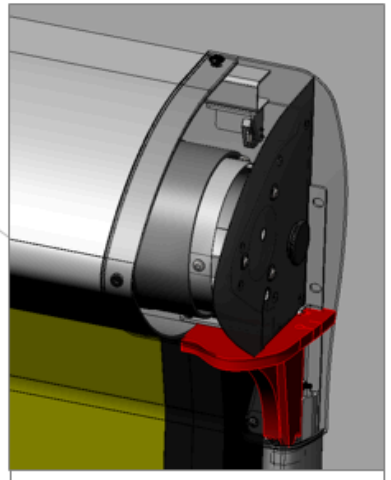


Plate of PE rubber inside the door post cover top



Kit of PE rubber inside the side covers

PE rubber behind reintroduction block

Gravity: HS9030 ...

HS9030 GEM Emergency

For use in emergency exits
Up to 4 x 4 m

Manual push through construction

Gravity: HS9030 ...

HS9030 GEM Emergency

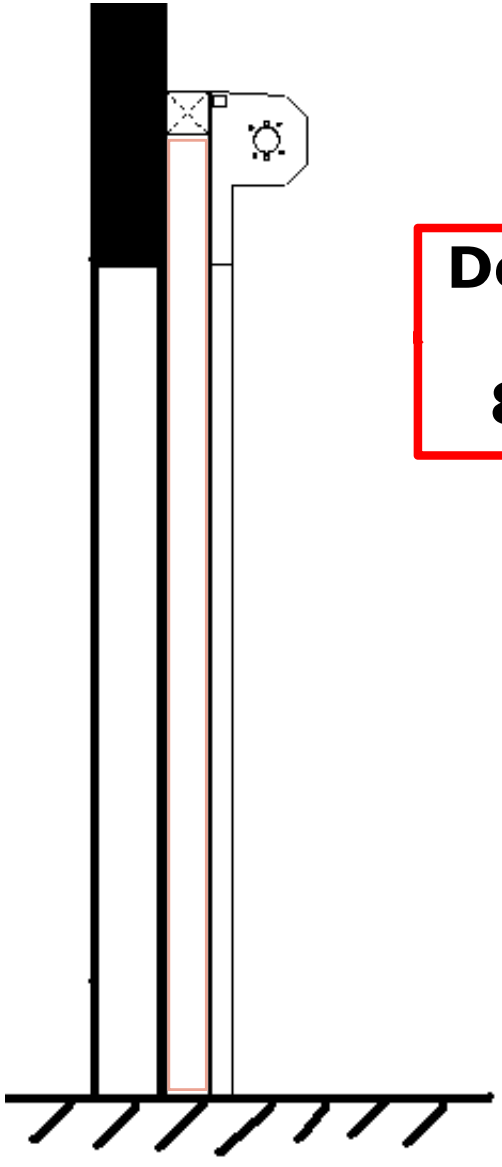
Photocell+ reflector!

Photocell => detects **<u>open zip</u>**

Attention: photo cell ≠ entry detector

Gravity: HS9030 ...

HS9030 GEM Emergency

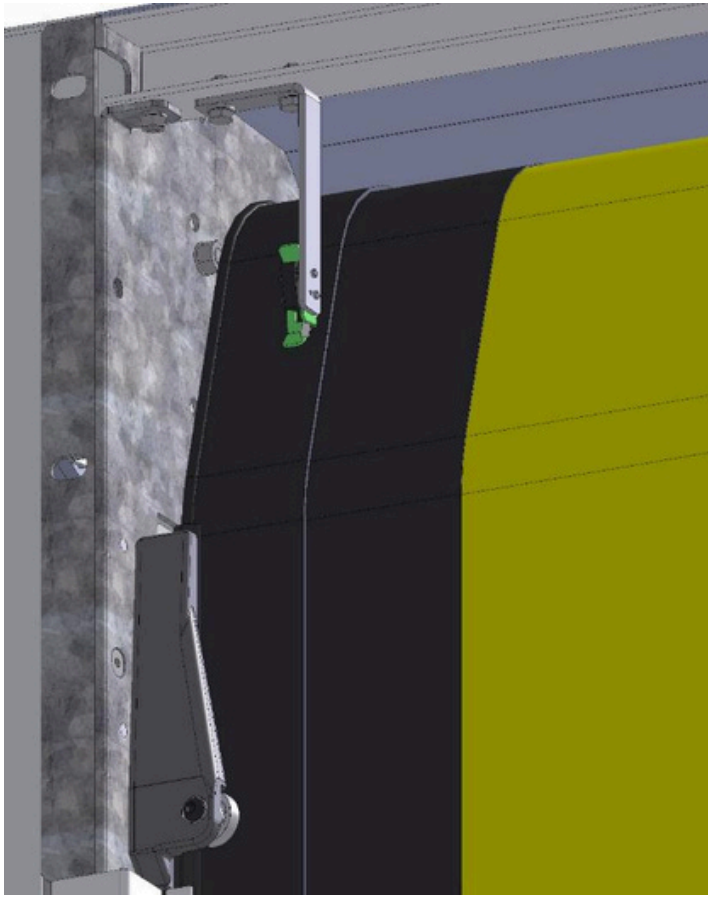


Delivered with Foreframe 80 x 80 mm

Gravity

HS90xx Gxx

Curtain



TRM : Top Roll Monitoring

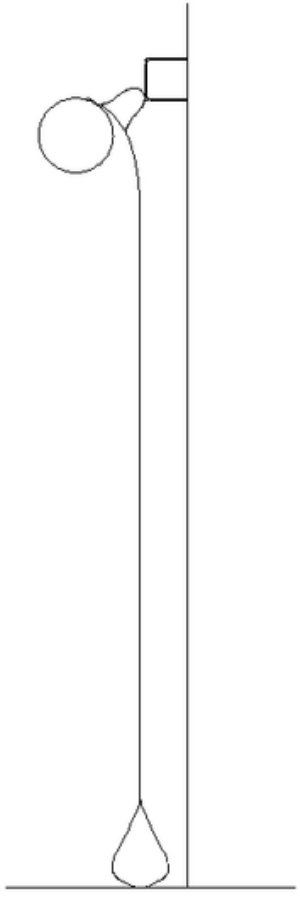
- **Switch to detect curtain deformation**
- **To protect the curtain**
- **Installed motor side**
- **In addition to**
  - **photocell (all doors)**
  - **and WDD (HS9020 G + HS9030 G)**

Gravity: models features

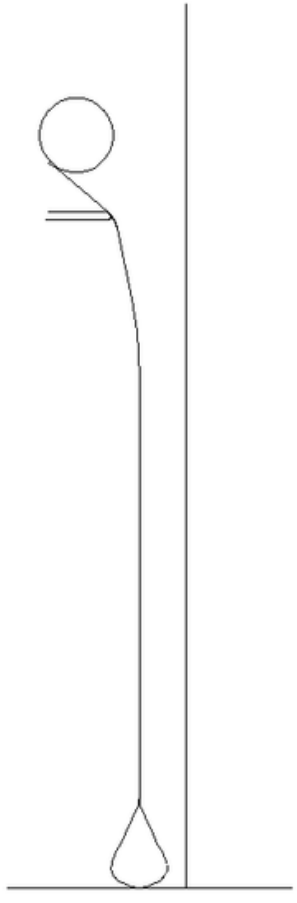
HS90xx Gxx

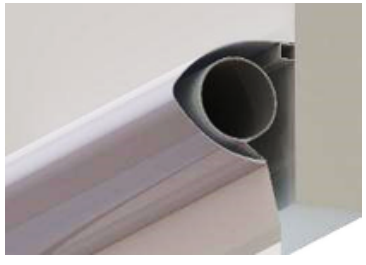
sealing

HS9020 G



HS9030 G, HS9020 GHY
Cleanroom





Gravity

	Max dimension		Top speed		Wind
	Height	Width	UP m/s	DOWN m/s	class
HS9020 G	4000	4000	1,2 Option 2	0,5	1
HS9020 G GHY/GAT	4000	4000	1,2 Option 2	0,5	1
HS9030 G	4000	4000	2,7	0,5	1
HS9030 G GHY/GAT	4000	4000	2,7	0,5	1
HS9030 GEM (Emergency)	3500	3500	2	0,5	1

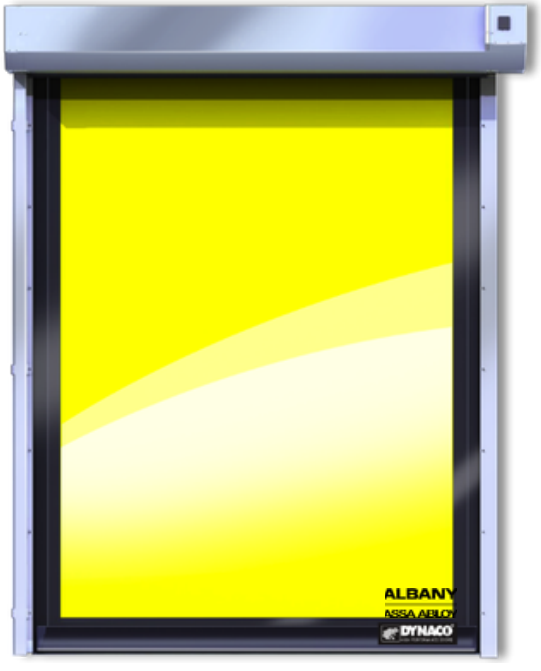
Content

1. Product overview
2. “Gravity” driven doors
3. **“Power Drive” driven doors**
4. Applications/Options
5. Technical summaries
6. Sales

Power Drive

HS9010 P

HS8010 P
HS8020 P



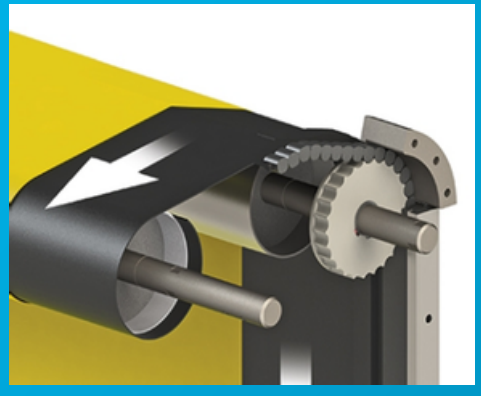
HS9010 PEM
EMERGENCY

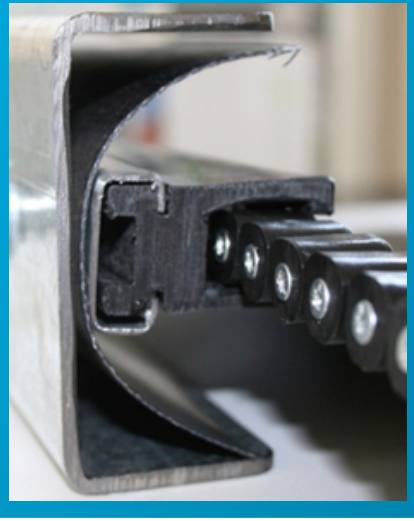
HS9010 PFR
FREEZER

Power drive doors HS80xx Series vs HS81xx

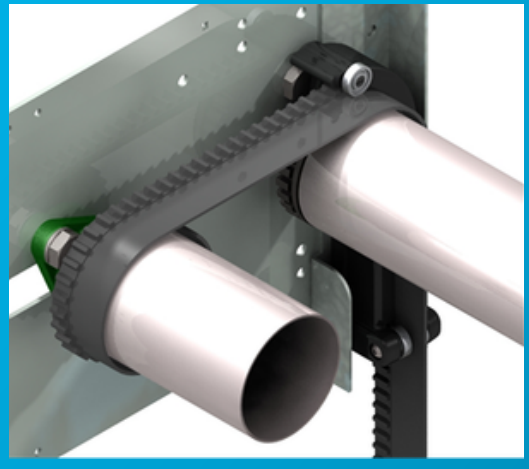
HS80xx with classic Zipper and HS81xx with new “belt” Zipper

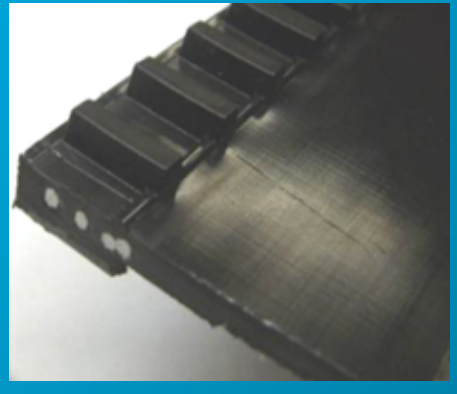
HS8010 Series





HS8110 Series





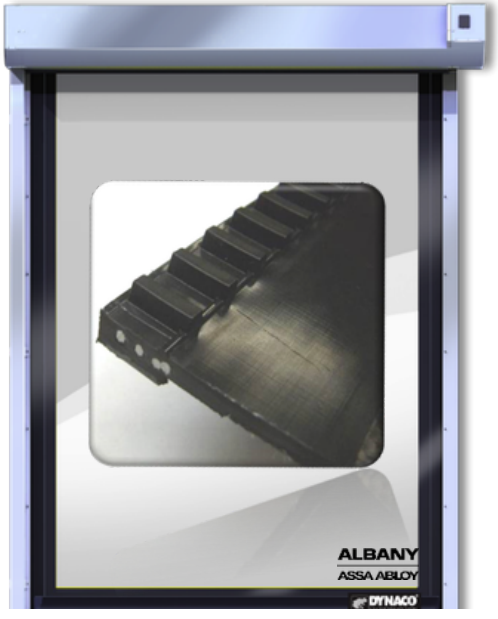
Power Drive

HS9010 PL

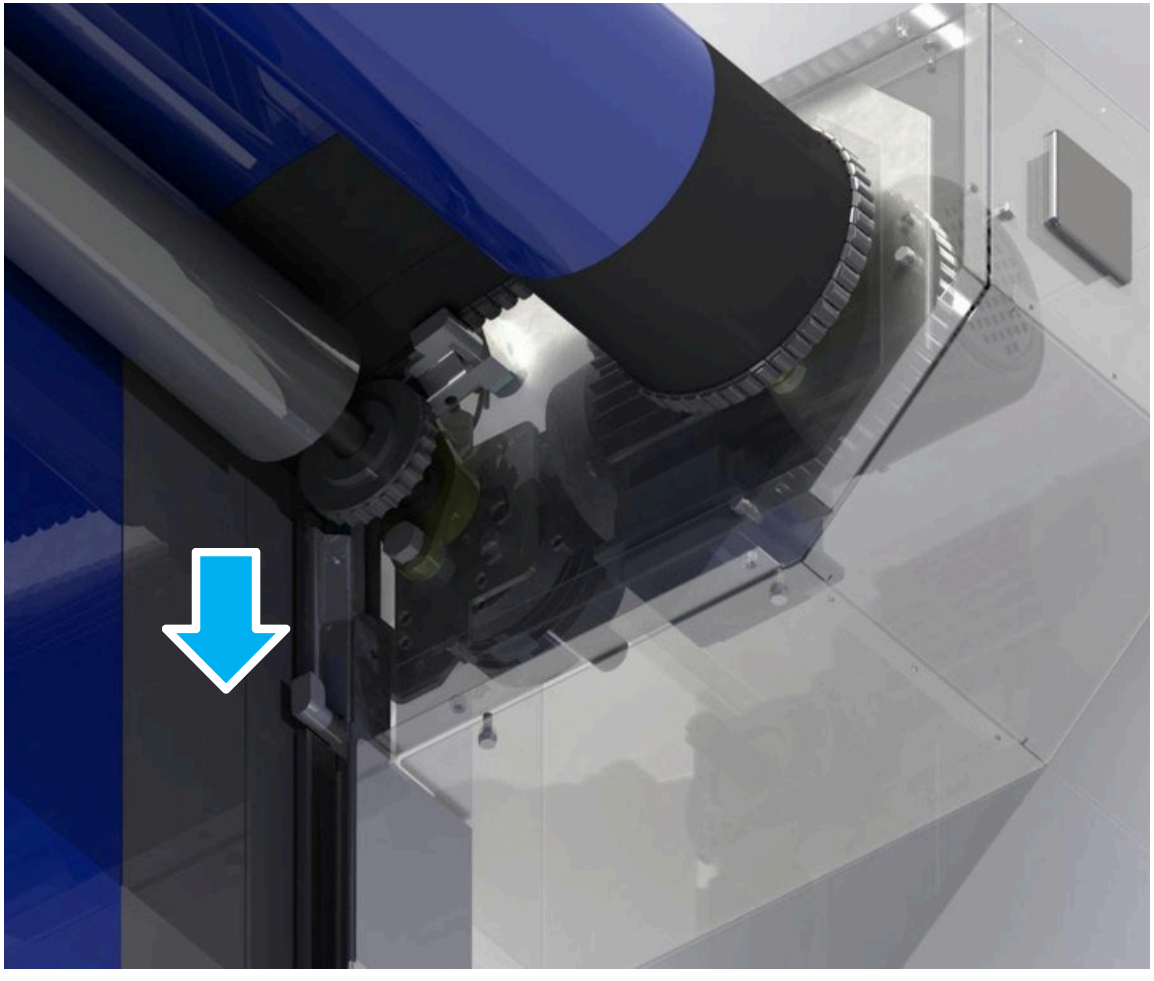
HS8010 PL
HS8020 PL

HS9110 P

HS8110 P



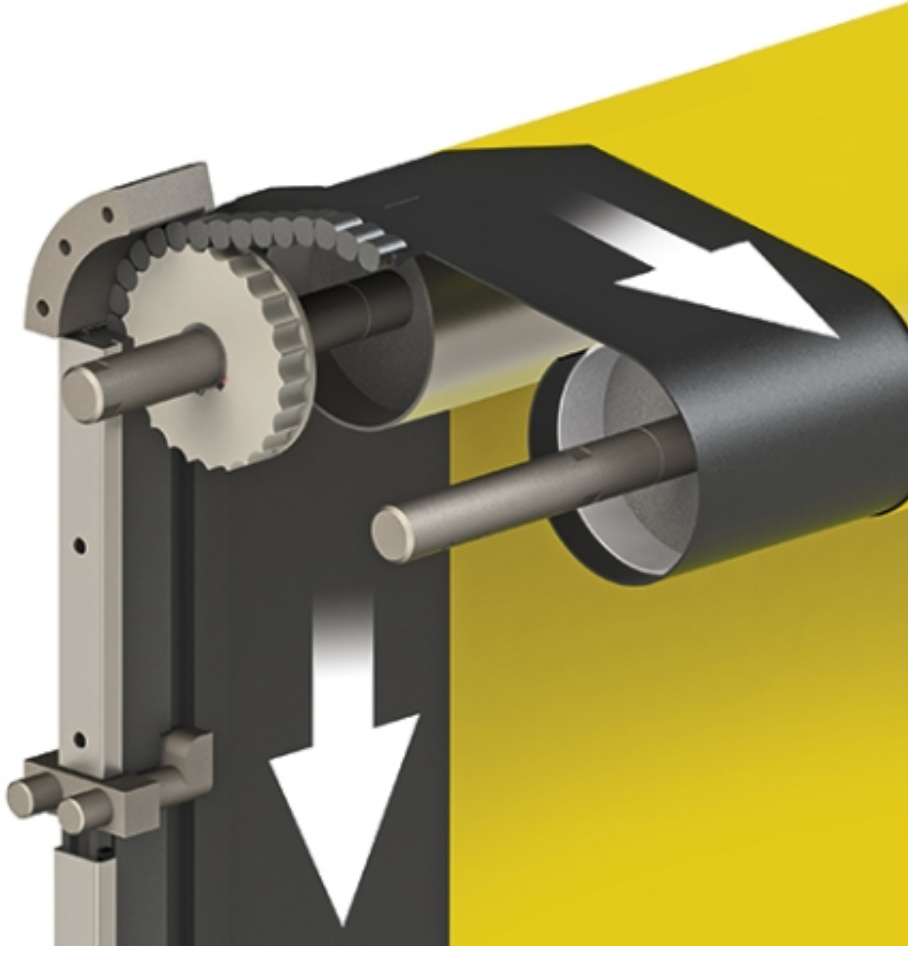
Power Drive



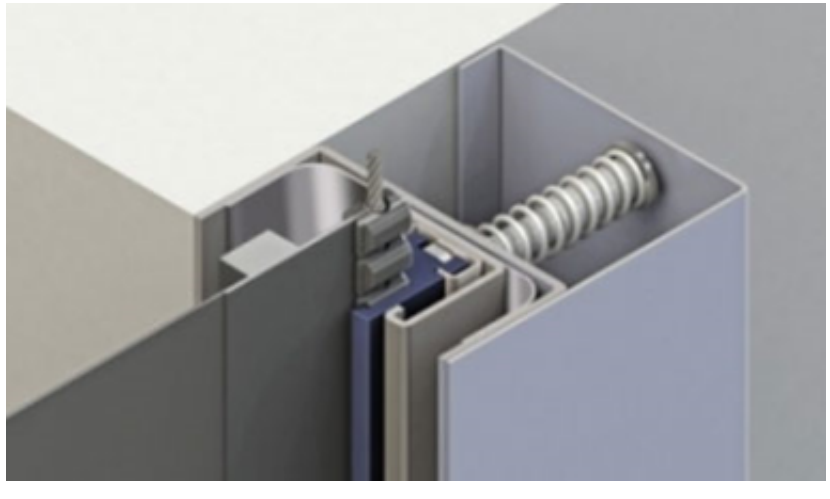
Zipper

Power to close

Power Drive



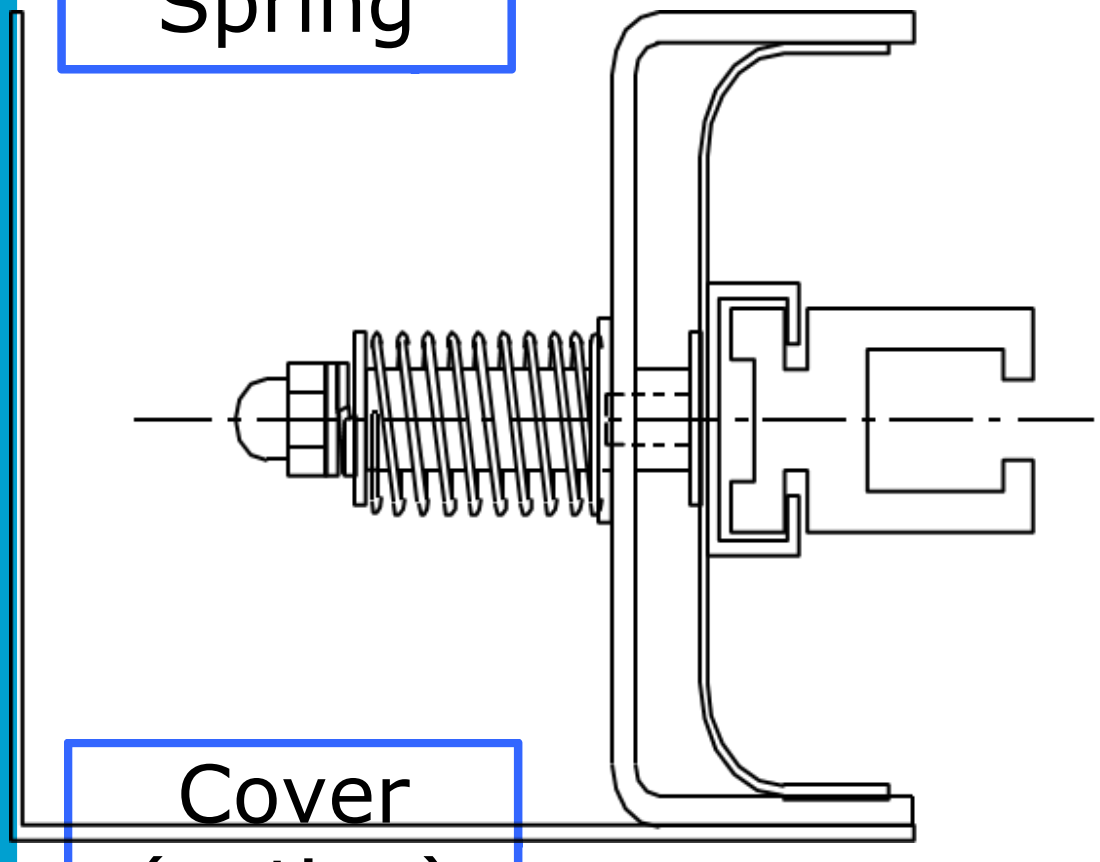
2 drums
Counterweight
Side guides with springs



Power Drive

All P-models

Spring



Cover (option)

Sealing flap

Steel C-rail

PE side guide

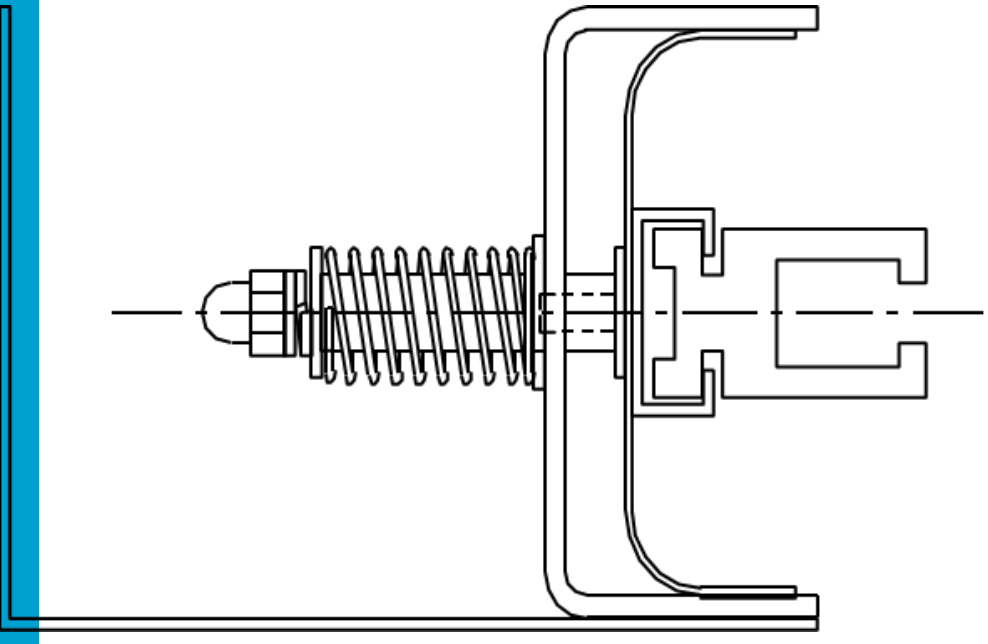
Power Drive

interior

HS9010 P

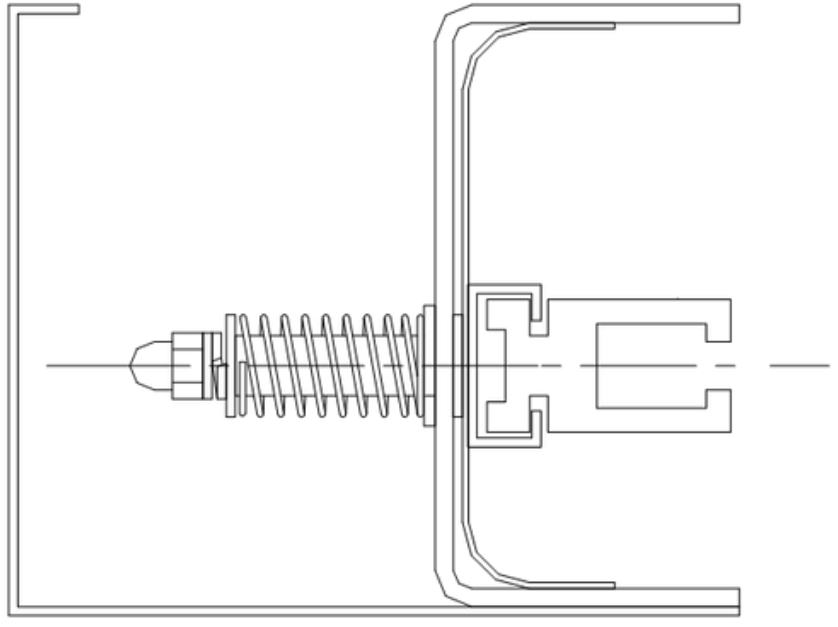
HS9110 P

80x40x3



HS9010 PL

100x50x3



Power Drive

exterior

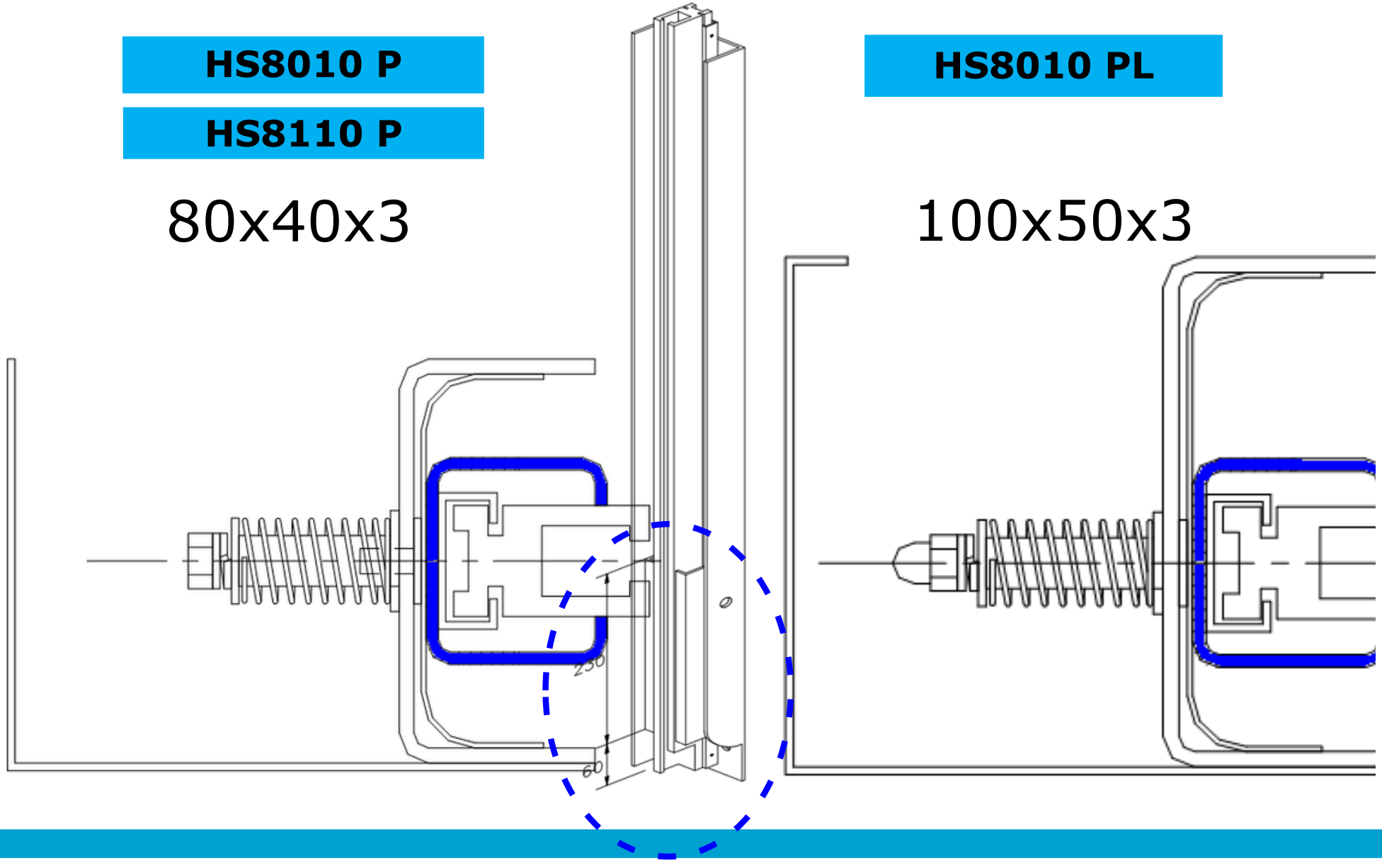
HS8010 P

HS8110 P

80x40x3

HS8010 PL

100x50x3

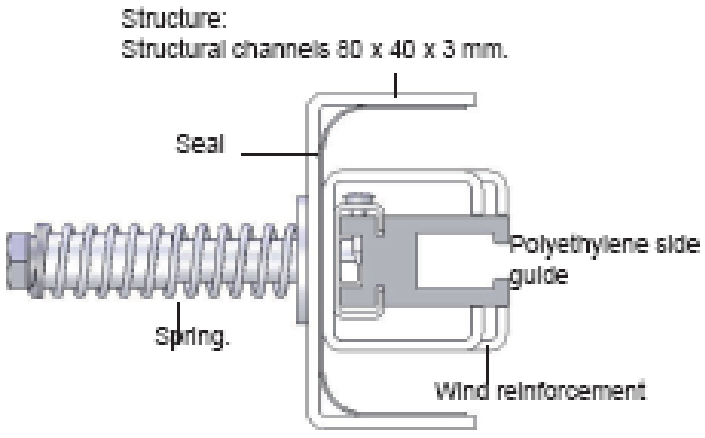


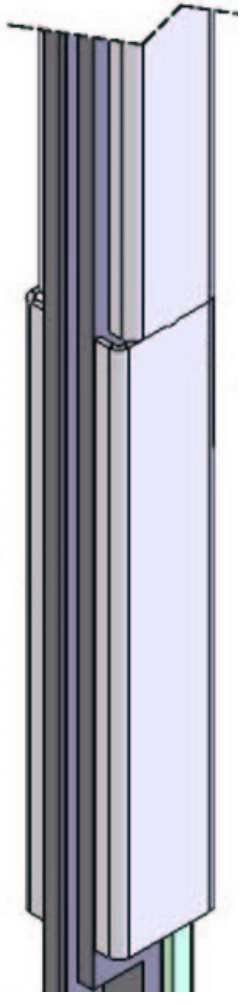
Power Drive

exterior

HS8020 P

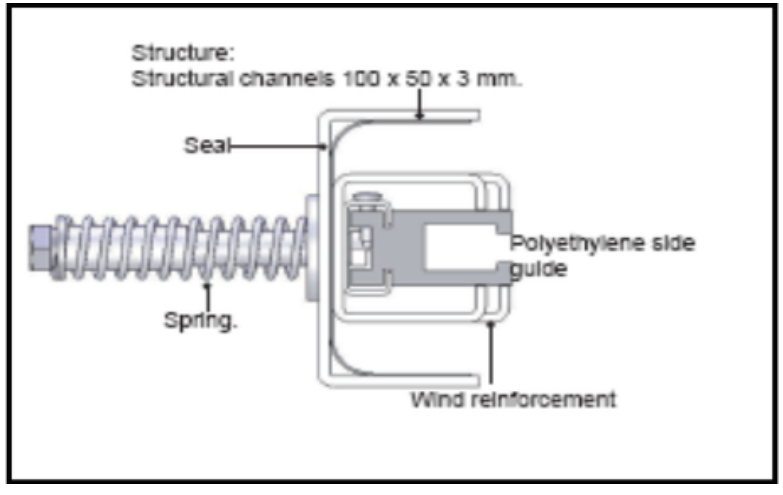
80x40x3





HS8020 PL

100x50x3



Power Drive

HS9010 P

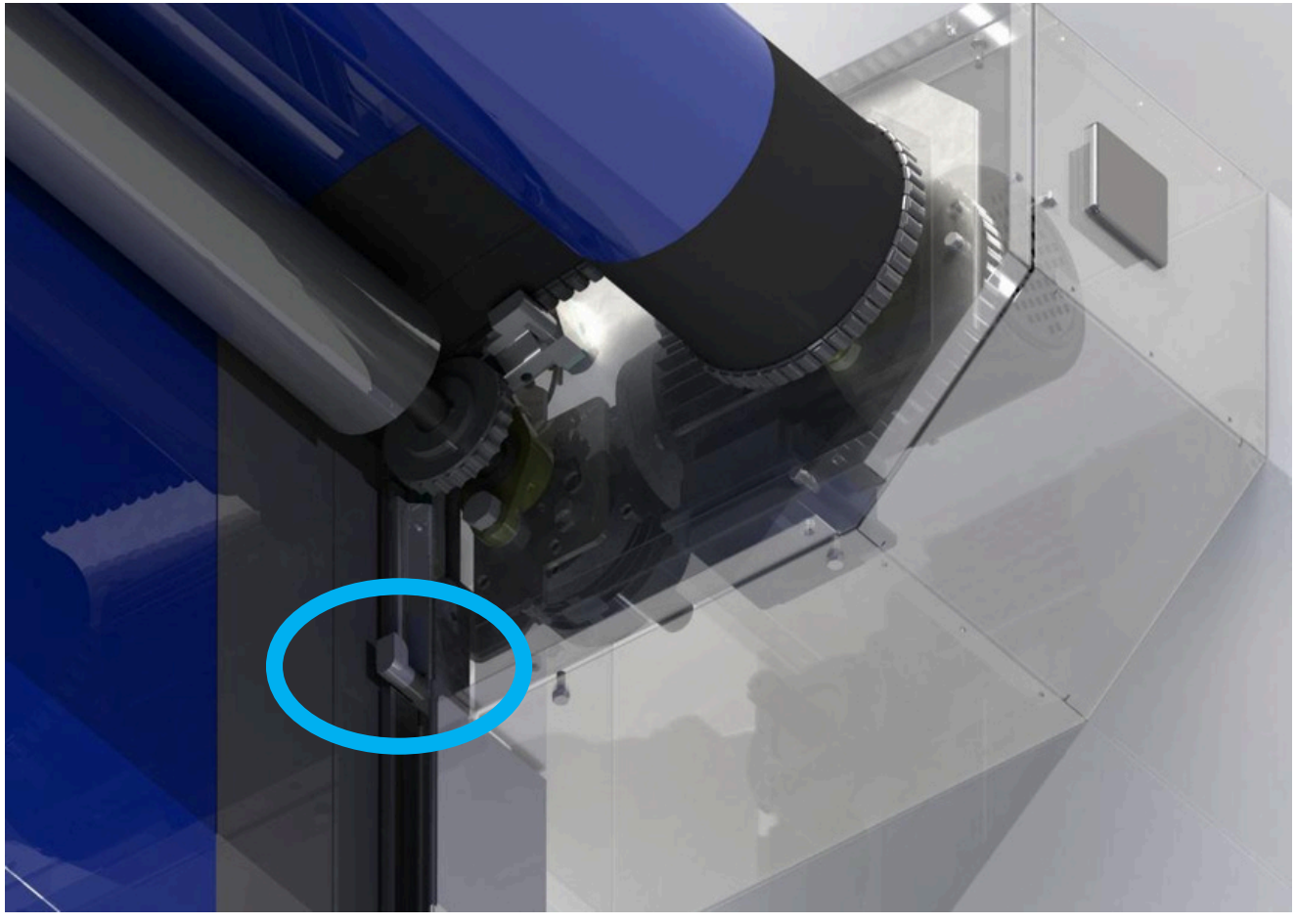
HS8010 P

HS8020 P

Power Drive

HS9010 P

HS8010 P



HS xx10 Zipper

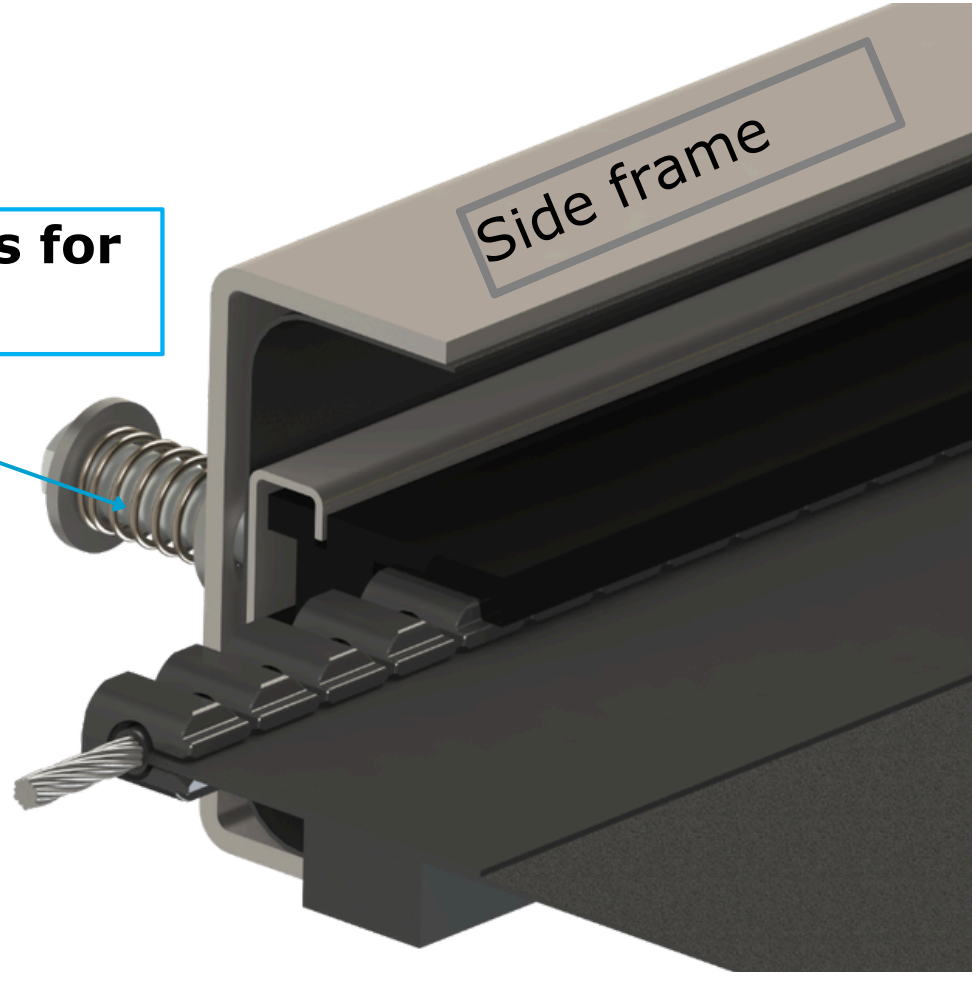
Self resetting

Power Drive

HS9010 P

HS8010 P

Side guides on springs for flexible rail

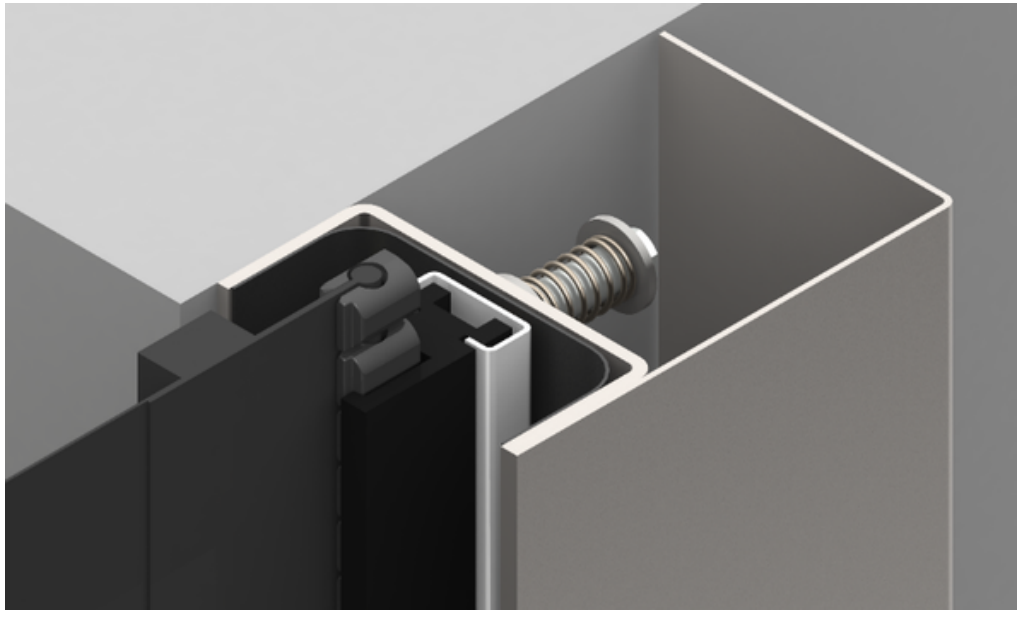


Power Drive

HS9010 P

INSIDE USE

Opening peak STD up to 2,4m/s
Closing 1,2m/s

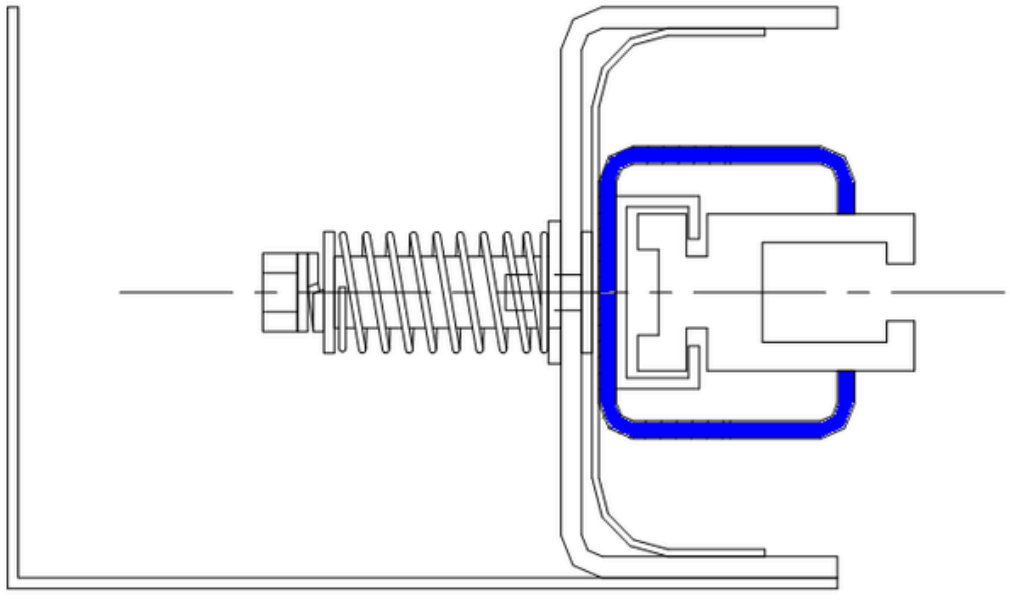


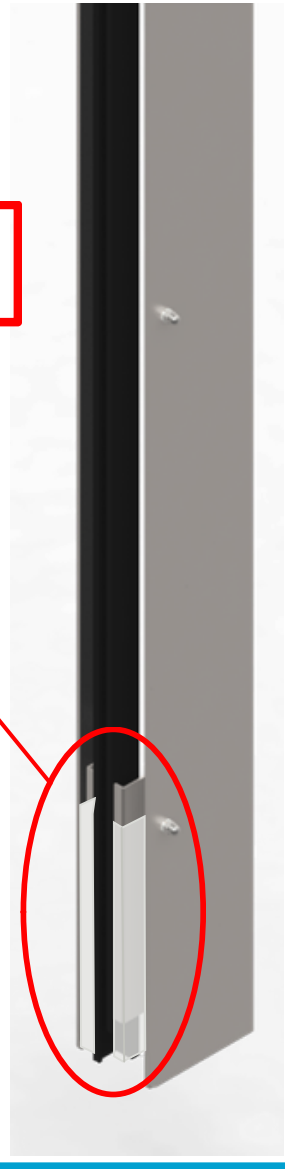
Max dimensions:
W 5500 mm x H 5500 mm
Max windload acc. EN12424 (Door closed)
HS9010 P: Max. Class 1

Power Drive

HS8010 P

OUTSIDE USE



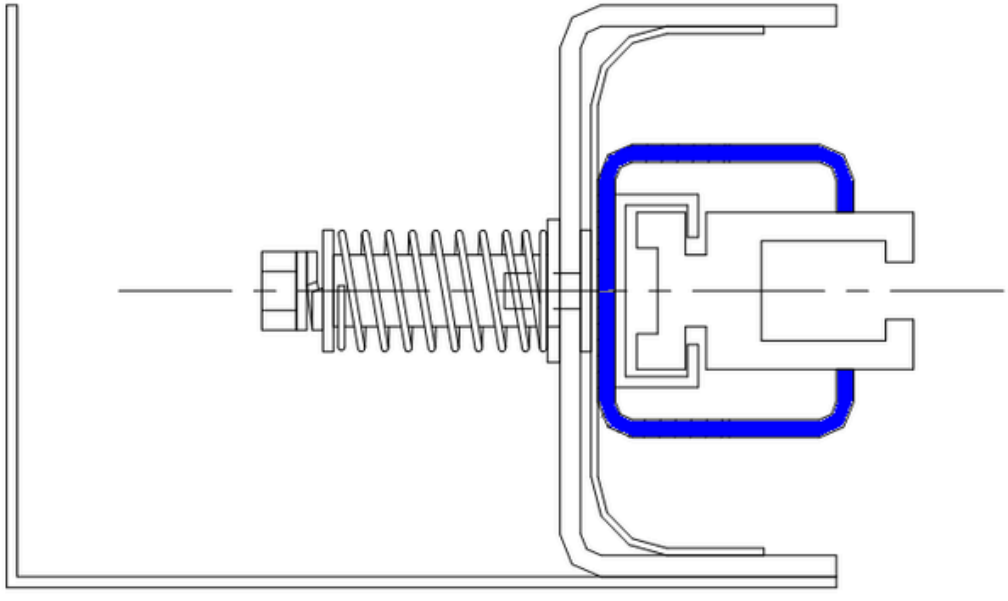


Power Drive

HS8010 P

Outside Door

Opening peak STD 2,4m/s
Closing 1,2m/s

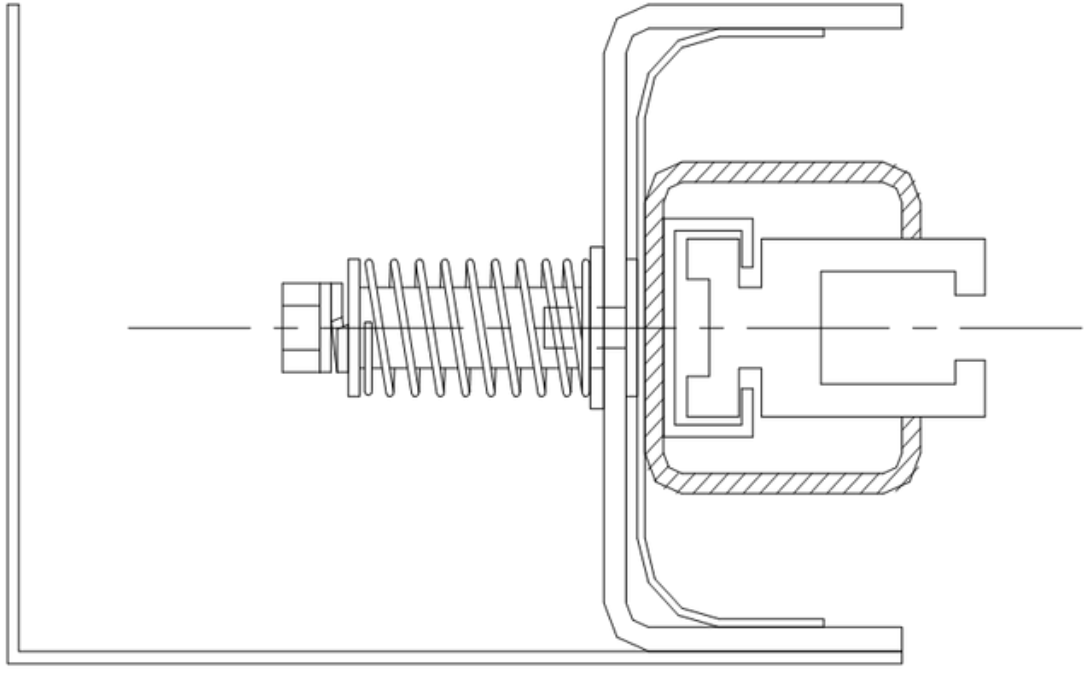


Max dimensions: W 5500 mm x H 5500 mm
Max windload acc. EN12424 (Door closed)
HS8010 P: Max. Class 3

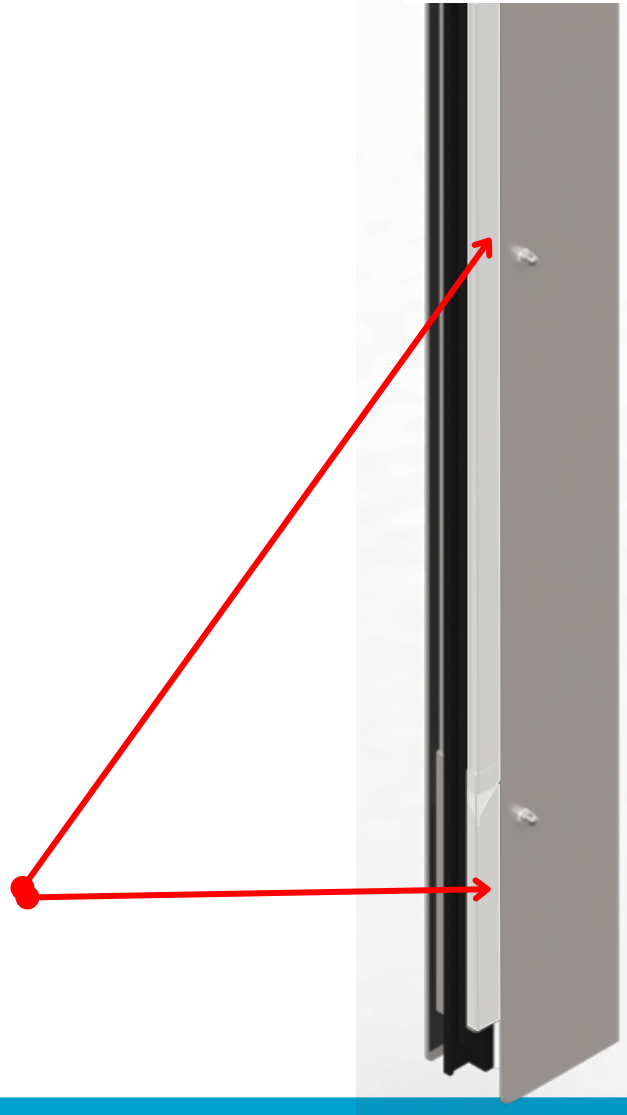
Power Drive
OUTSIDE USE

HS8020 P

Max dim: W 5500 mm x H 5500 mm
Max windload HS8020 P: Max. Class 5
acc. EN12424 (Door closed)



Side guide reinforcement on FULL HEIGHT of the door



Power Drive

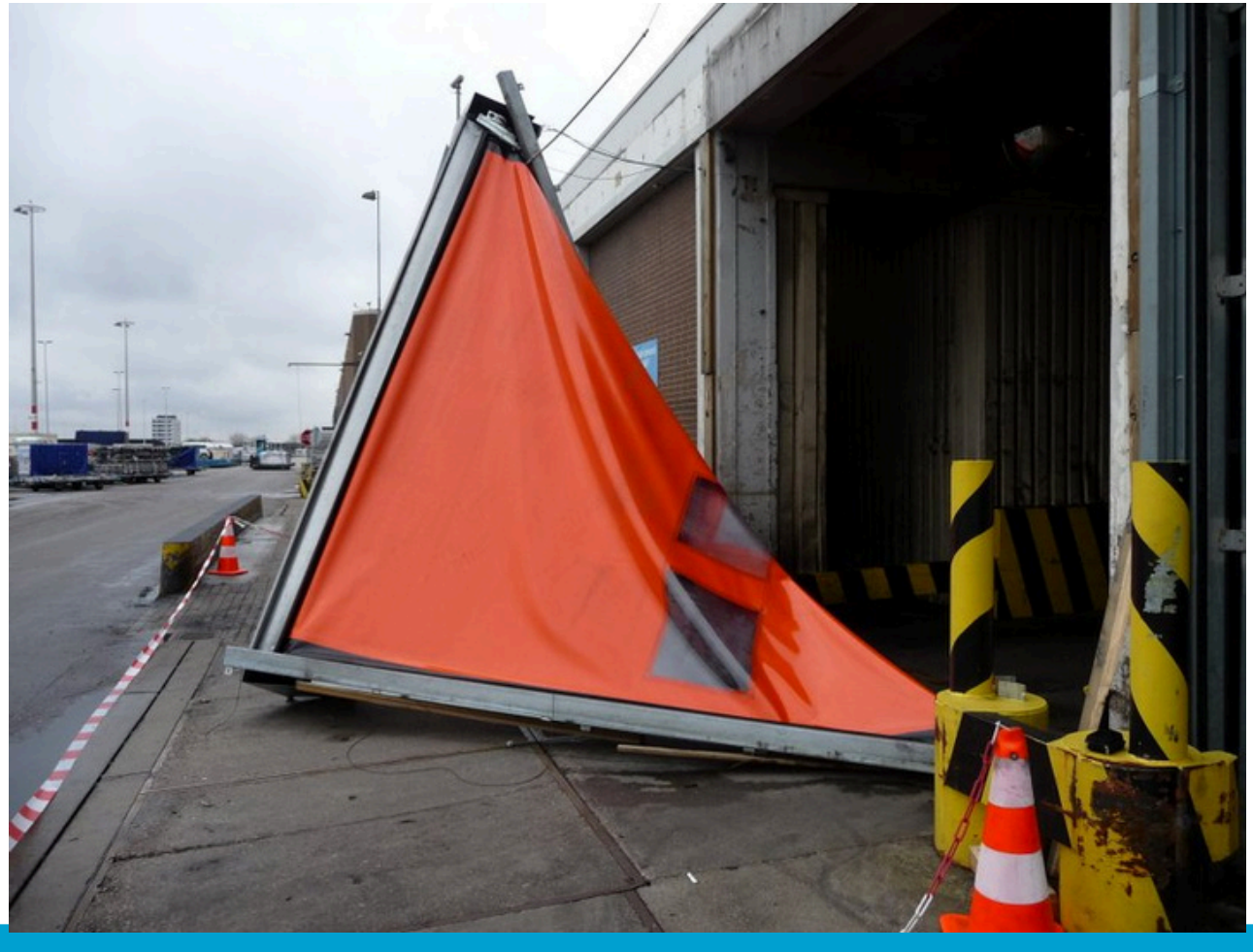
HS8020 P

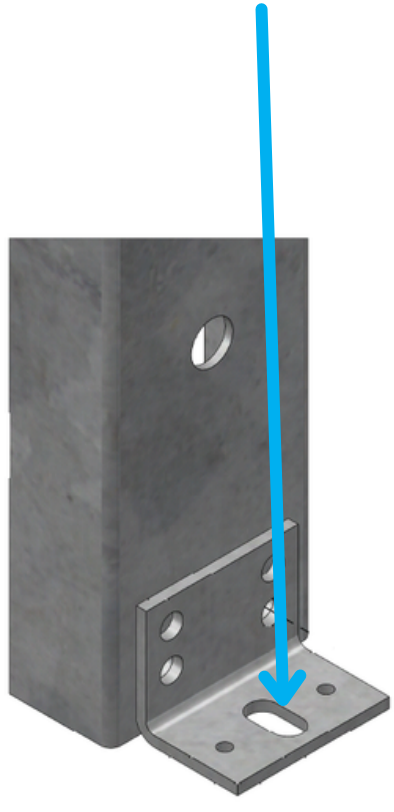
Crashes are accidents!
And must remain unintentional accidents!
Every crash reduces the lifetime of the door!
Watch the 150mm wearing part at the top of the side guide.

Power Drive

HS8020 P

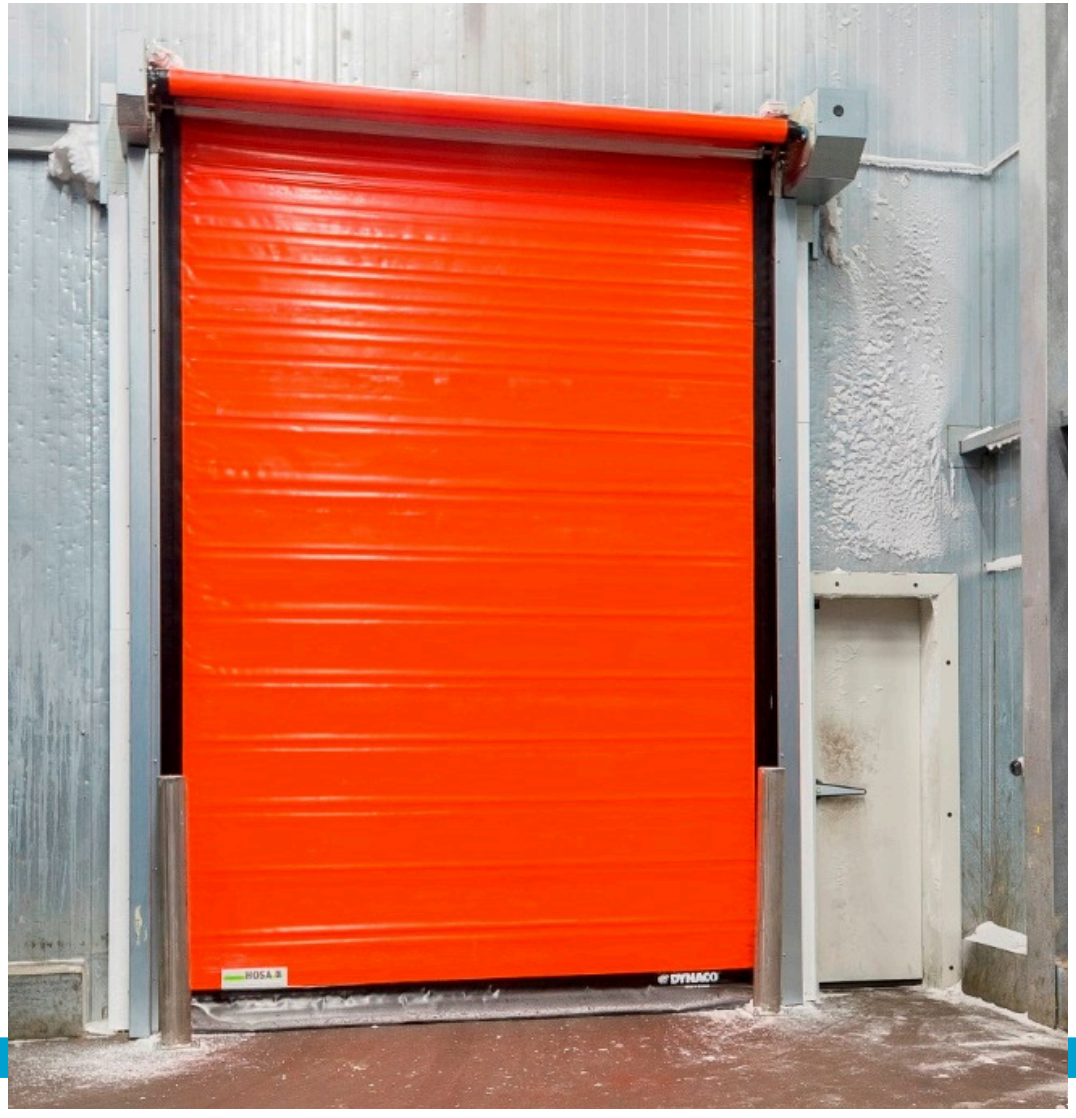
Huge forces involved / wind pressure! Watch the fixations!!!





Power Drive

HS9010 PFR

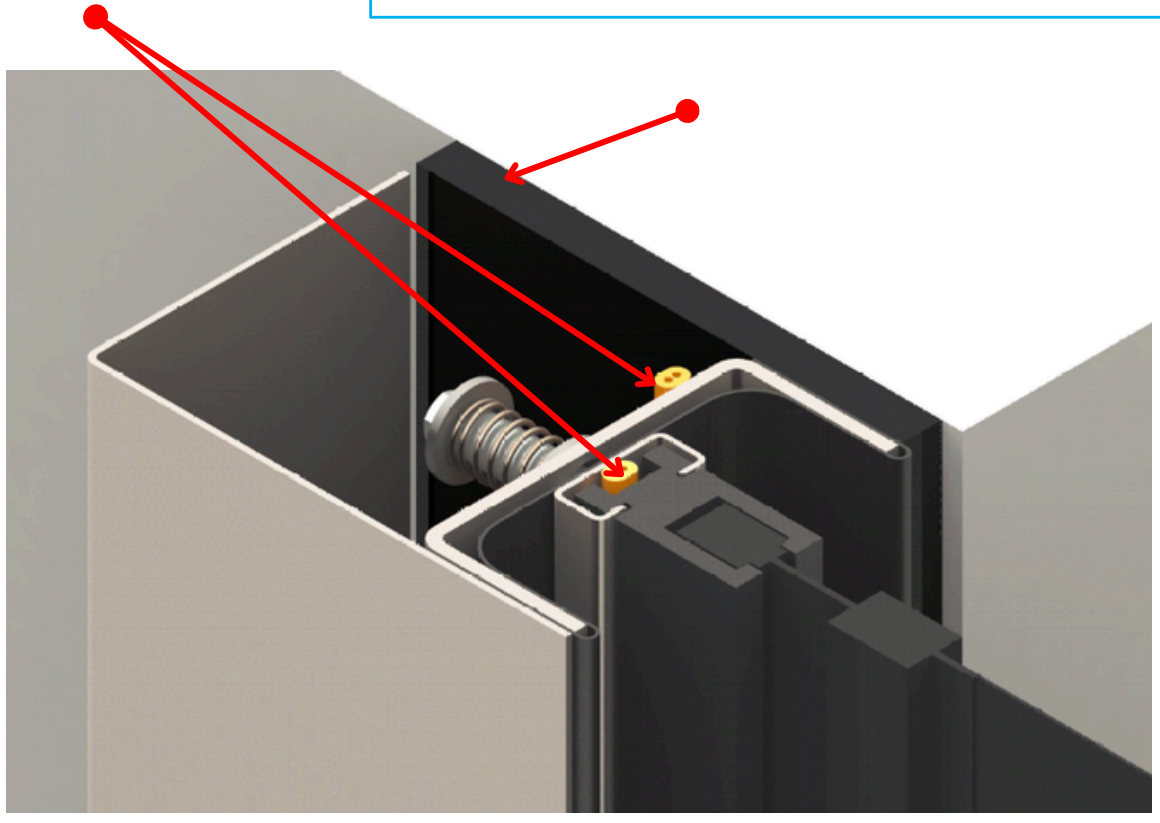


Power Drive

HS9010 PFR

1. Heated side parts

2. Fixed on PE insulation



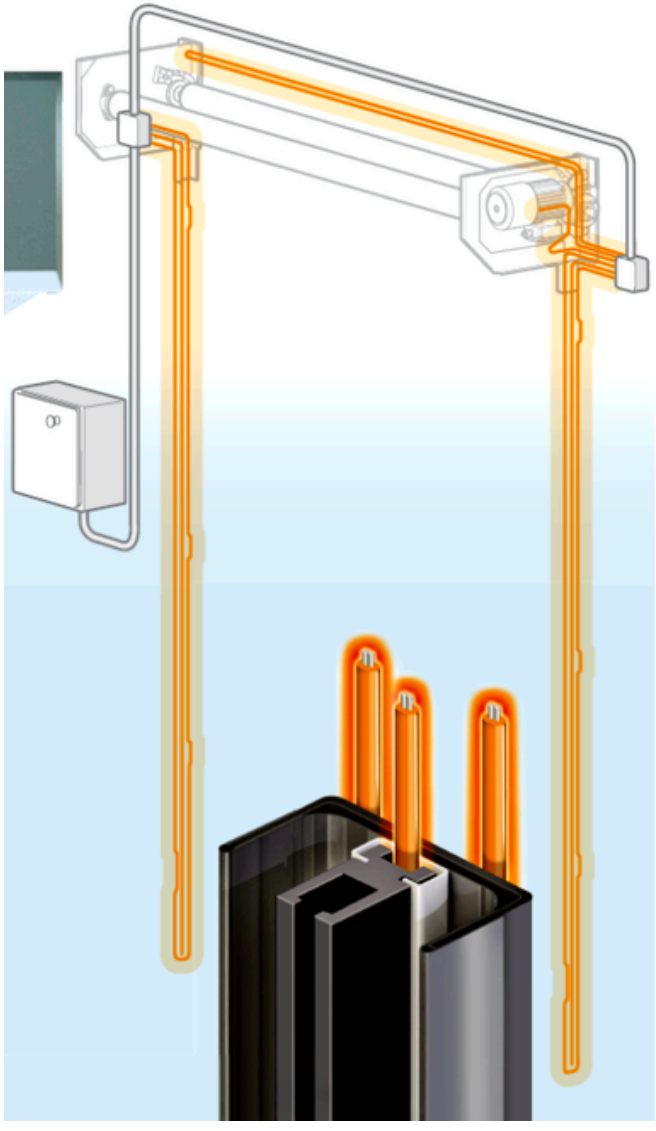
3. Automatic opening

4. weekly deicing required

DO NOT LOCK THE DOOR,
DO NOT SWITCH POWER OFF!

Power Drive

HS9010 PFR

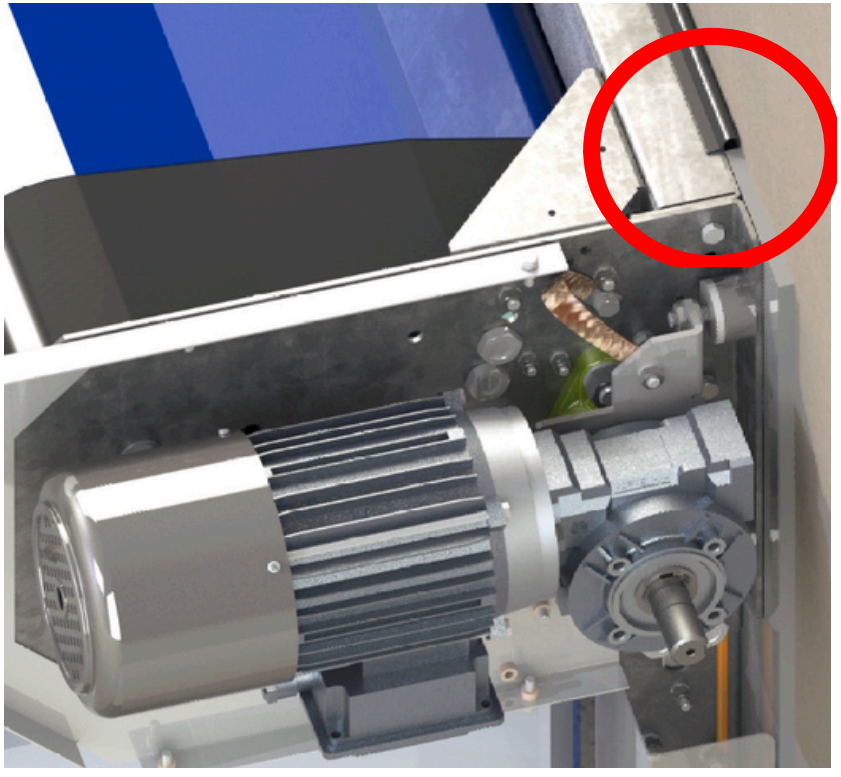


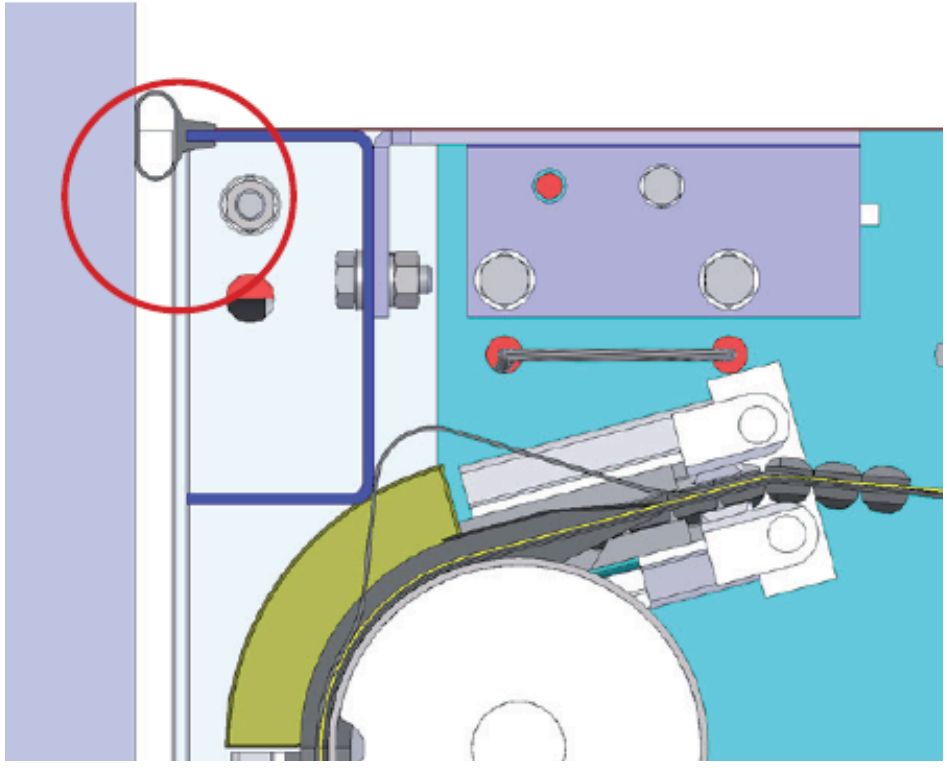
Heating cables through connection boxes

Power Drive

HS9010 PFR

Improved top sealing with rubber profile

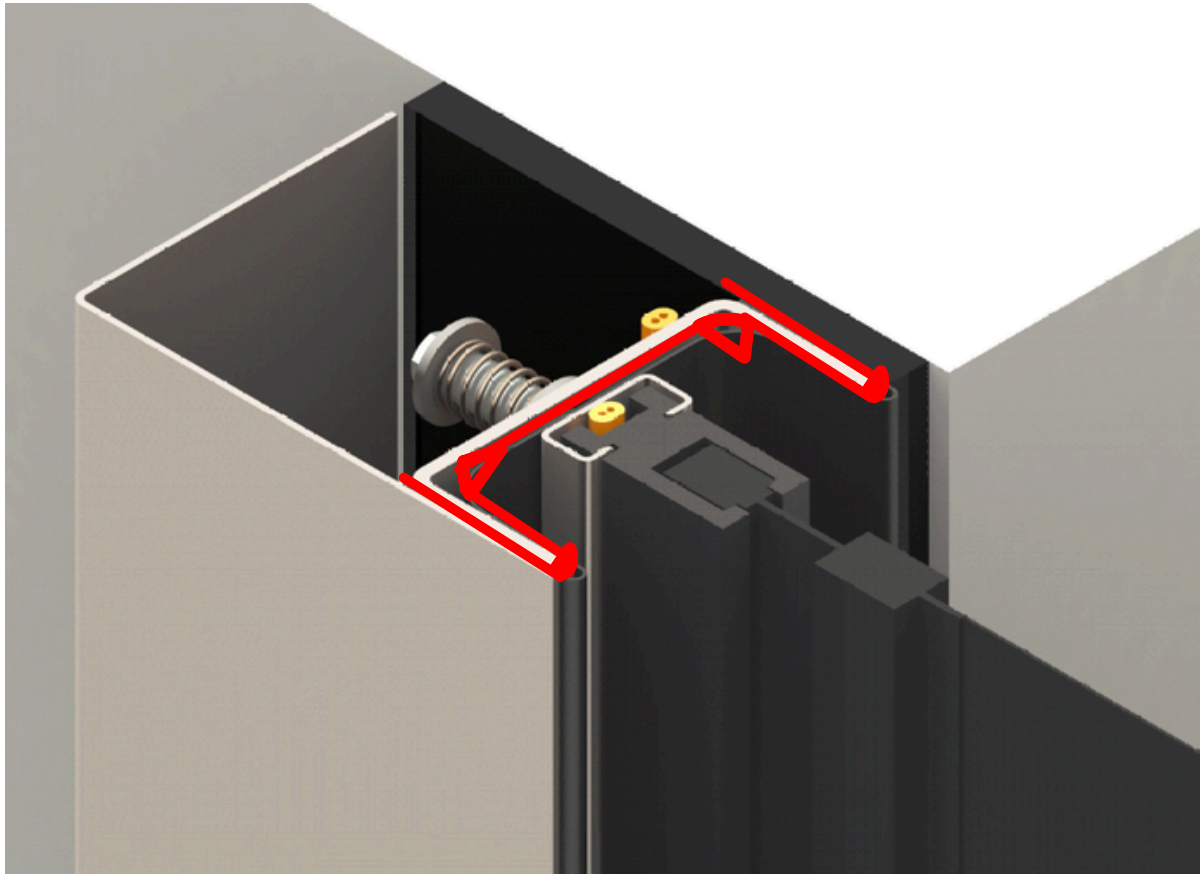




Power Drive

HS9010 PFR

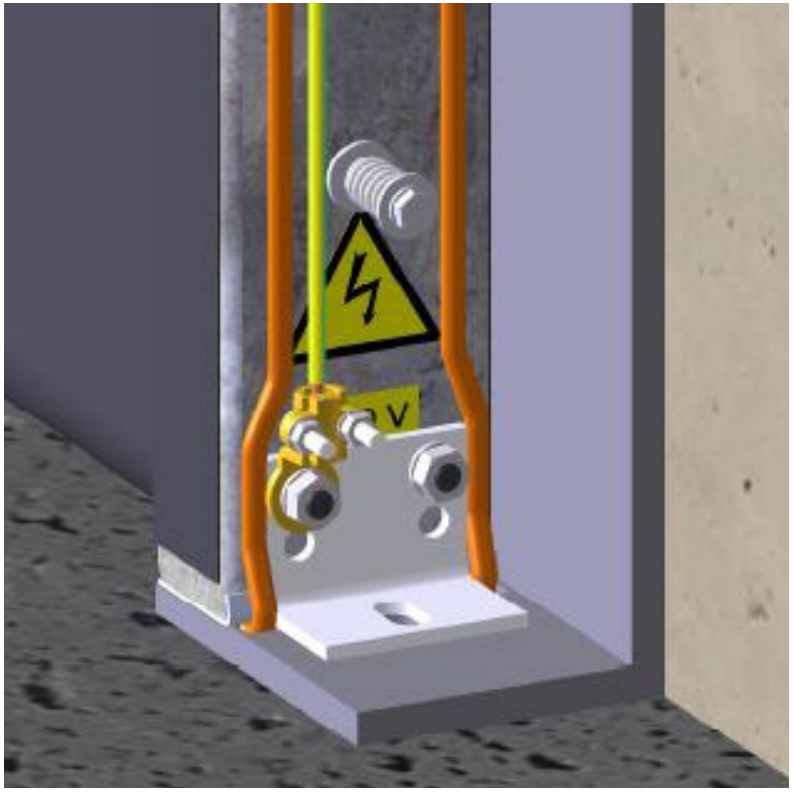
Improved side guide sealing prevents condensation/ice



Power Drive

HS9010 PFR

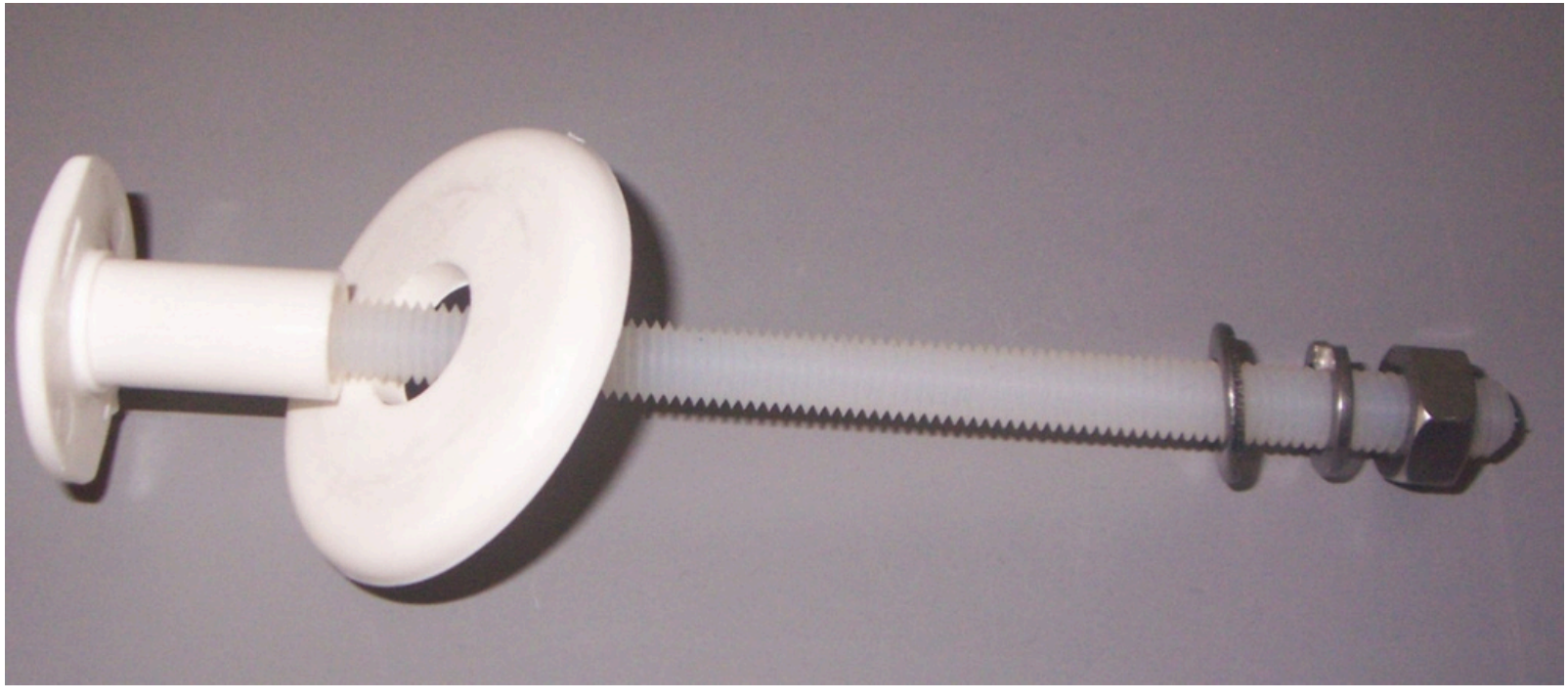
Ground/Earthing Kit



Power Drive

HS9010 PFR

Fi 34 plastic Fixation kit



Power Drive

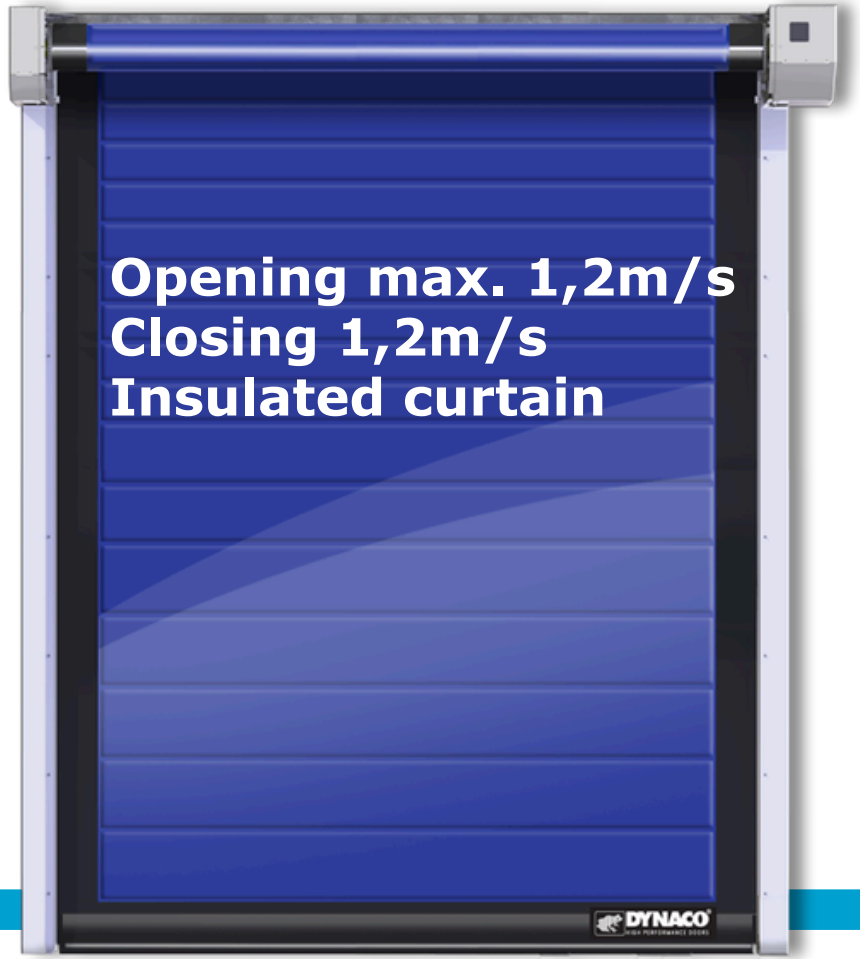
HS9010 PFR

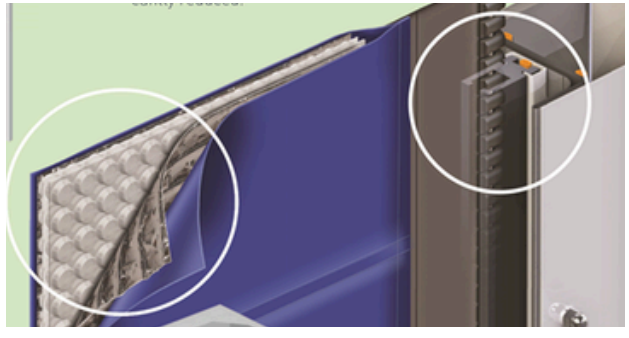
Good sealing only when correct <u>installed!!</u>

Power Drive

HS9010 PFR

Good sealing only when correctly <u>installed!!</u>





HS 9010PFR - Points of attention

- Always order 3 phase power-supply (400V - 16A) not in SSS, order with remark
- Order/use optional Fixations kit plastic Fi34 (thermal insulated threaded rod) for installation of foam-panels
- There is an option: heated CU-box, just in case this is mounted on the cold side
- Special "cold-resistant" window possible as an option, however, is not useful due to frozen fog you will have limited sight.
- Double installation-time needed
- Always install protection/sealing sheet (supplied standard with FR doors)
- Extra attention for shielding (heating) cables going though holes with sharp edges. If damaged heating-cables will burn.
- Best is to go all the way up and place conn. boxes above the door.
- All openings/seams between wall and columns to be sealed with sealing-compound from a dispenser
- check installation and operation after 2 weeks running
- Always in combination with thick sliding-door. (automatically) Close sliding-door after 5 min of no traffic
- The screws in zipper-nodes are standard in stainless steel.

Power Drive

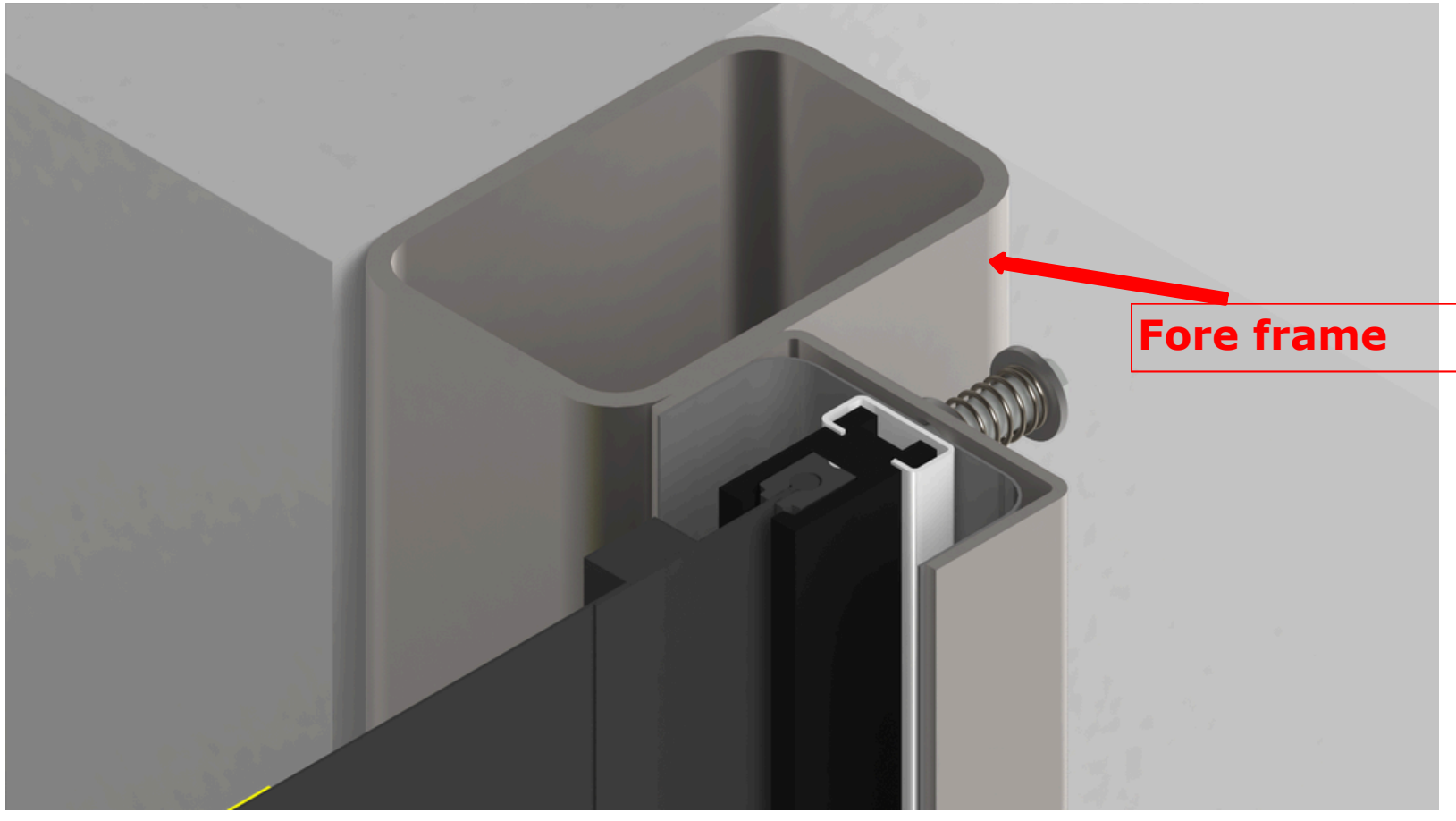
HS9010 PEM

INSIDE USE!!

Photo cell => detects **open zip**

Power Drive

HS9010 PEM



Power Drive

HS9010 PEM

INSIDE USE!!

Opening peak 1,2m/s
Closing 0,6m/s

Power Drive:

HS9010 Pxx

HS8010 P

HS8020 P

	HS9010 P	**HS9010 PFR**	**HS9010 PEM**	**HS8010 P**	**HS8020 P**
Enviroment	Inside	Inside	Inside	**outside**	**outside**
Wind class	1	1	1	4	5
Max width	5500	**4000**	5500	5500	5500
Max Height	5500	5500	5500	5500	5500
Closing speed	1,2m/s	1,2m/s	**0,6m/s**	1,2m/s	1,2m/s
Open topspeed	2,4m/s	**2,4m/s**	**1,2m/s**	2,4m/s	2,4m/s

Power Drive

HS9010 PL

HS8010 PL

HS8020 PL

Power Drive

HS9010 PL

HS8010 PL

HS8020 PL

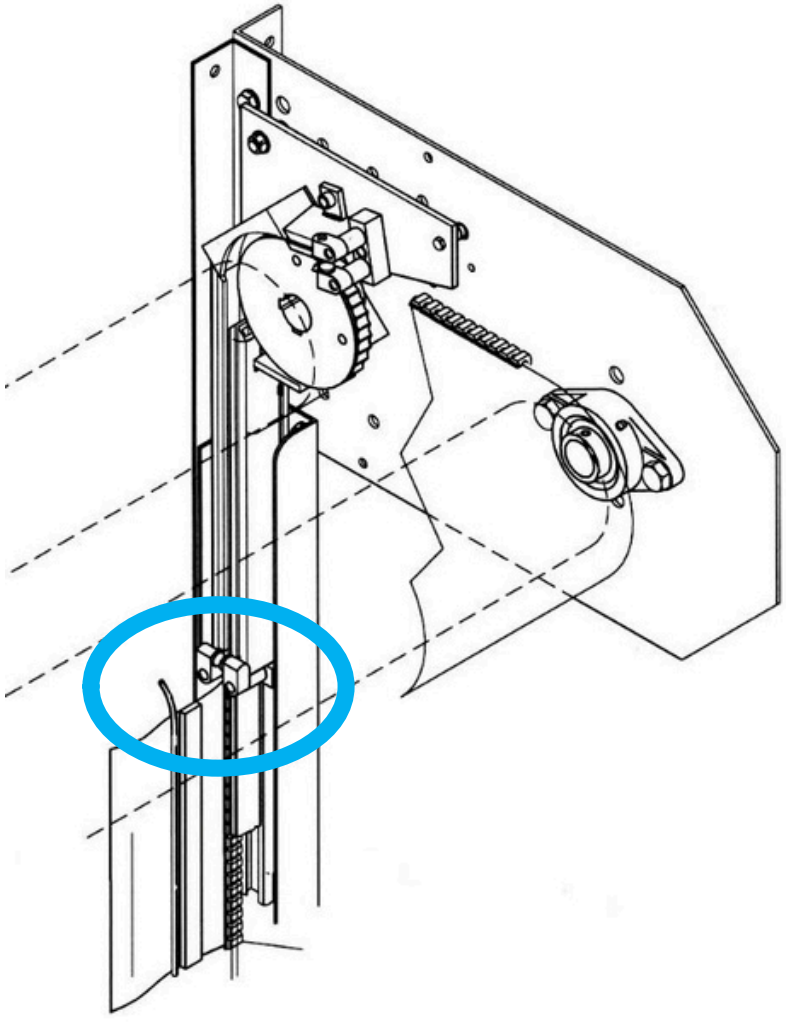
Big sizes up to 60 m²:

- **PUSH to close**
- **PULL to open**
- **2 drums**
- **NO BALLAST**
- **Side guides on lateral spring**
- **Not available in stainless steel**

Power Drive

HSxxxx PL

HSxxxx PL



Zipper

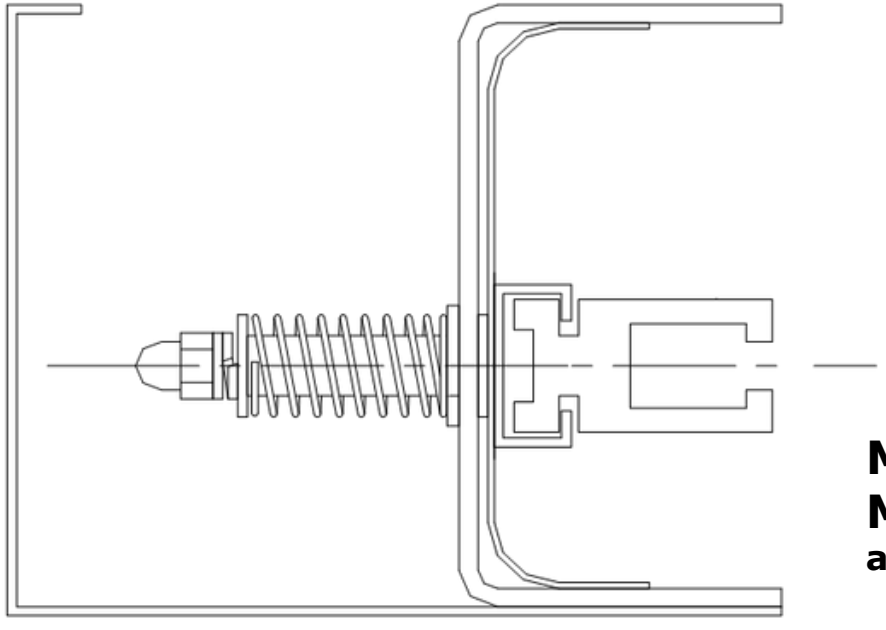
Reintroduction

Power Drive

HS9010 PL

INSIDE USE

100x50x3



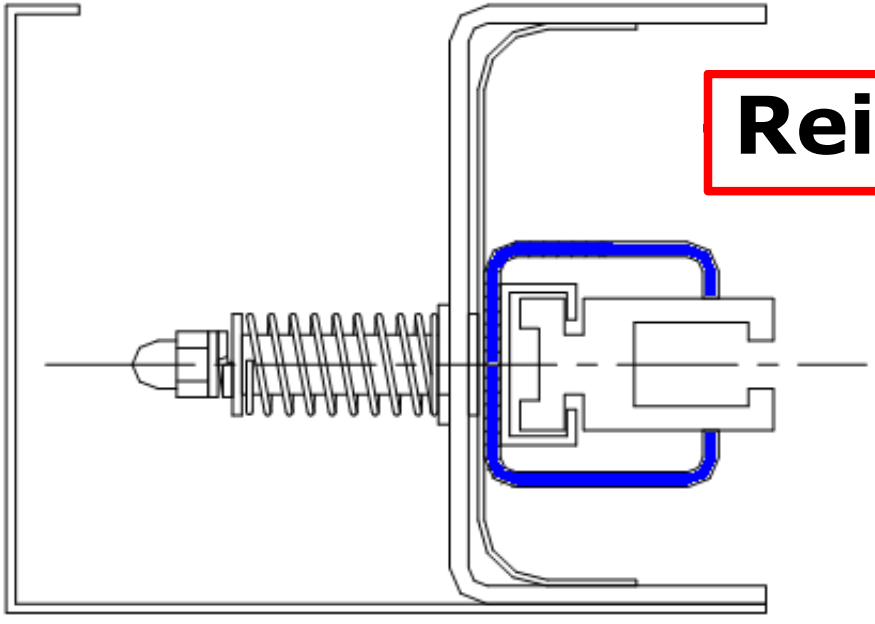
Max dim: W 11000 mm x H 5500 mm
Max windload HS9010 PL: Max. Class 1
acc. EN12424 (Door closed)

Opening max. 1,4/s
Closing 0,8m/s

Power Drive

HS8010 PL

OUTSIDE USE



Reinforcement

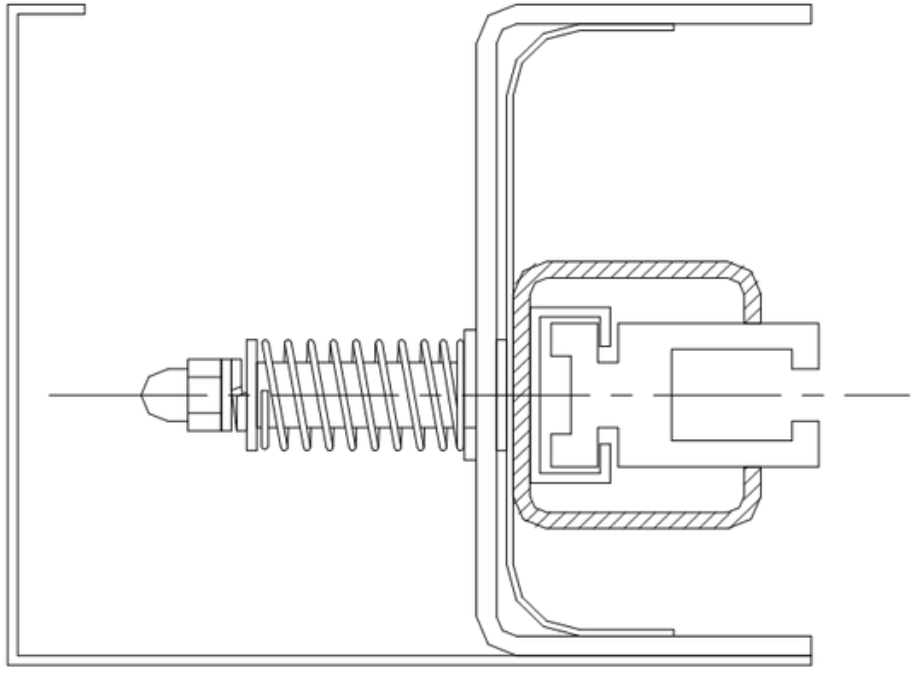
Maximum dim: W 9000 mm x H 5500 mm
Maximum windload HS8010 PL: Max. Class 3
acc. EN12424 (Door closed)

Opening peak 1,4m/s
Closing 0,6m/s

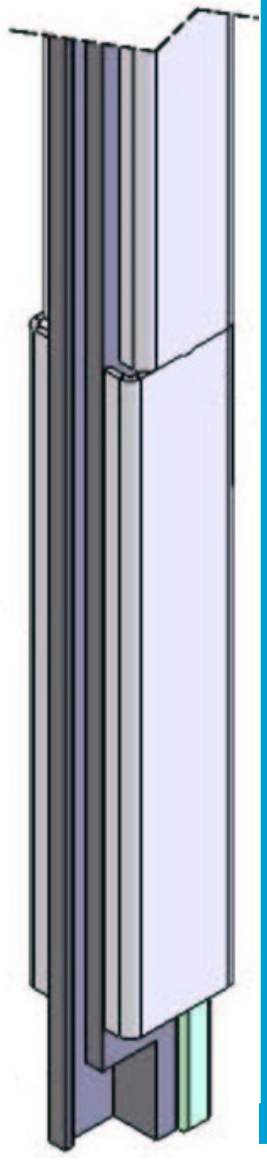
Power Drive

HS8020 PL

OUTSIDE USE



Side guide reinforcement on FULL HEIGHT of the door



Opening peak 0,8m/s
Closing 0,4m/s

Max dim: W 8000 mm x H 5500 mm
Max windload HS8020 PL: Max. Class 4
acc. EN12424 (Door closed)

Power Drive

HSxxxx PL

HSxxxx PL

	HS9010 PL	**HS8010 PL**	**HS8020 PL**
Environment	Inside	**outside**	**outside**
Wind class	1	3	4
Max width	11000	9000	8000
Max Height	5500	5500	5500
Closing speed	0,8m/s	0,8m/s	0,8m/s
Open topspeed	1,4m/s	1,4m/s	1,4m/s

Power Drive

Options – All Doors, except some door types

HS9110 P

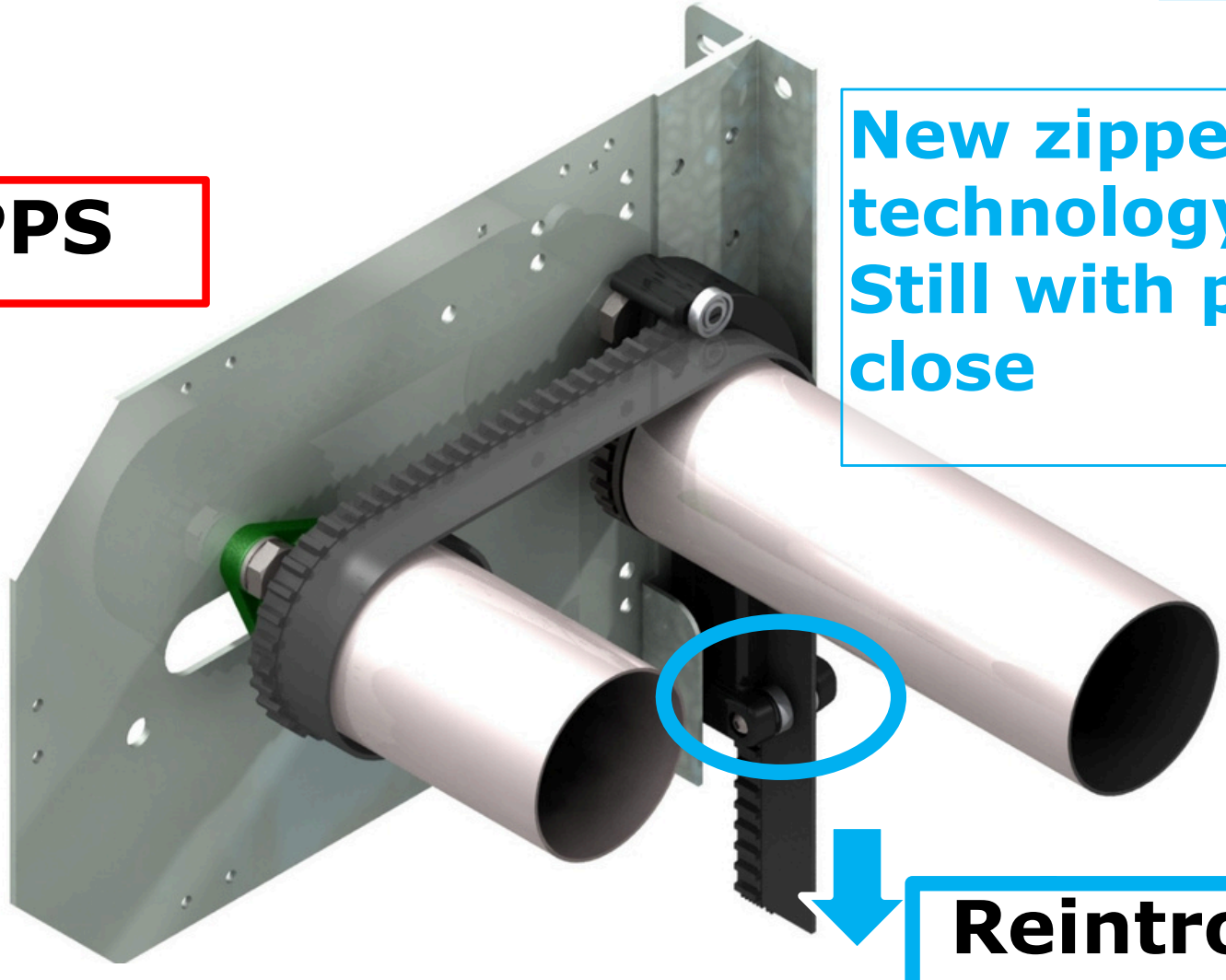
HS8110 P

Power Drive

HS9110 P

HS8110 P

PPS



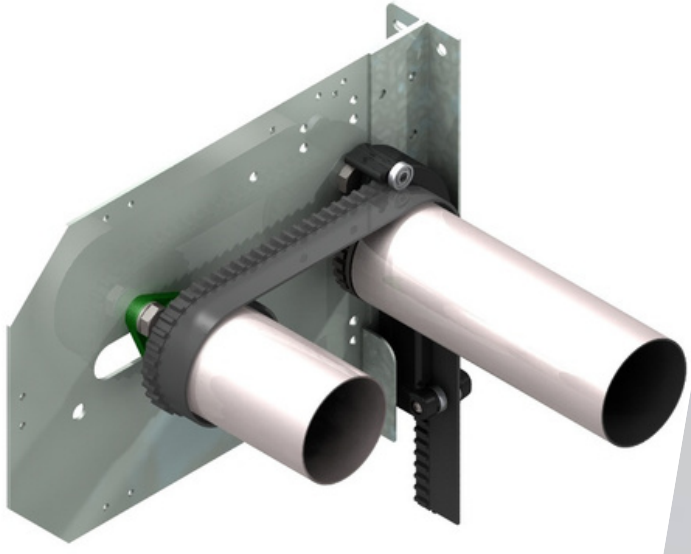
New zipper technology PPS belt Still with power to close

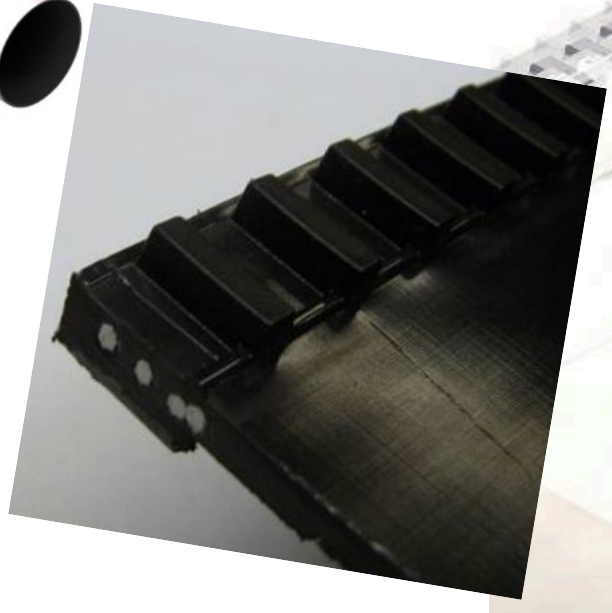
Reintroduction

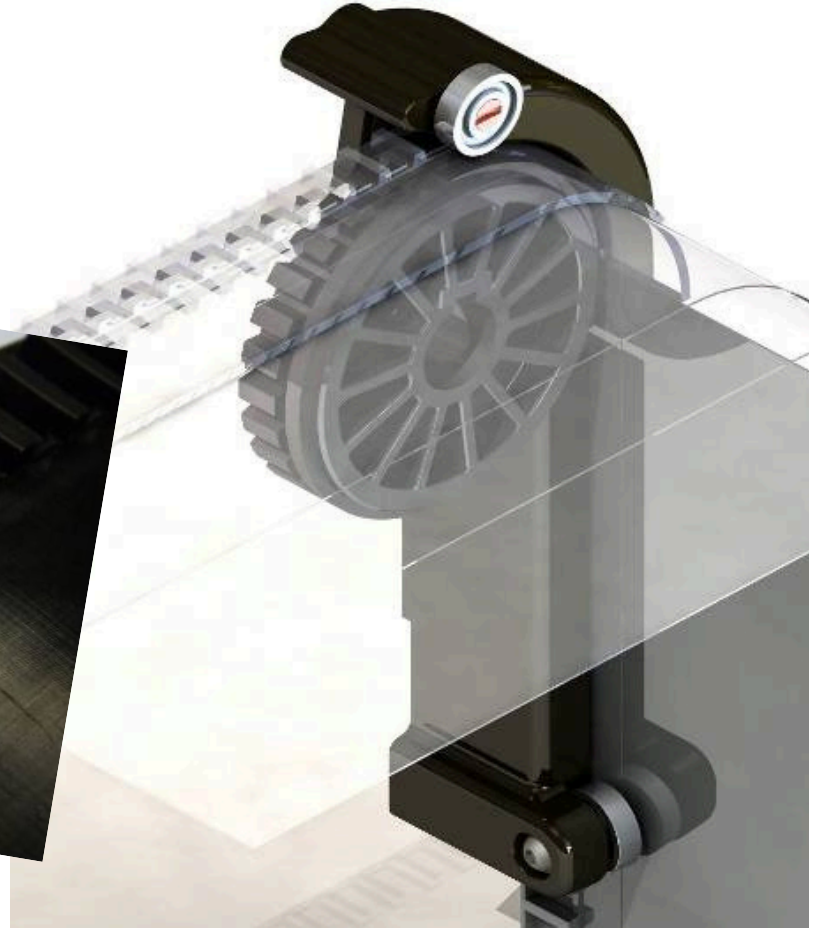
Power Drive

HS9110 P

HS8110 P





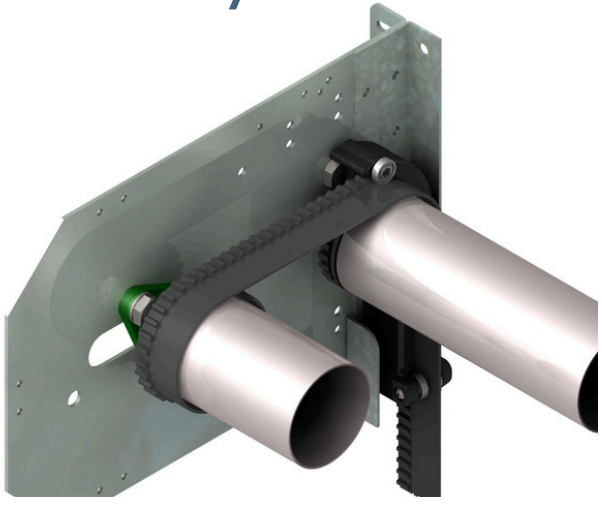


Power Drive

HS9110 P

HS8110 P

Albany HS9110P & HS8110P



Advantages:

Less noise (PPS, spring on plastic washer)

Softer reintroduction (Bearings)

Less wear

Easier installation

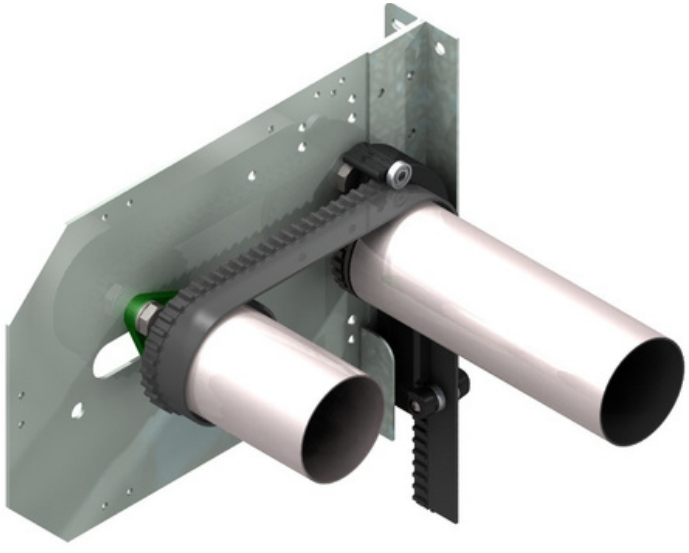
Side Covers included in standard

Lower maintenance

Power Drive

HS9110 P

INSIDE USE



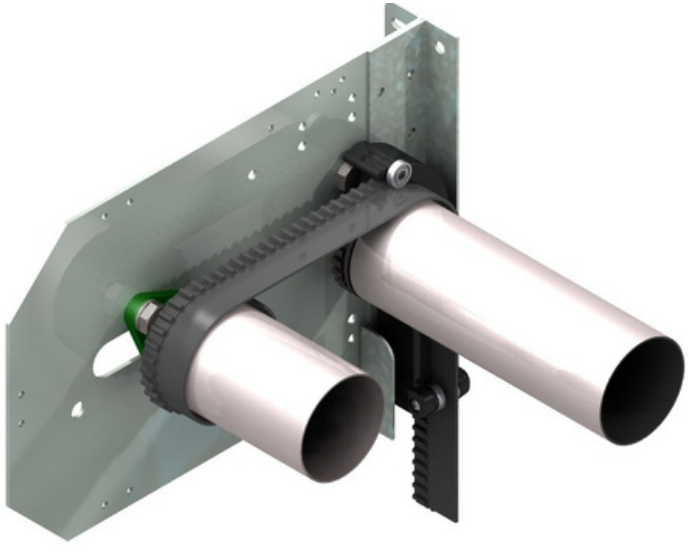
Opening peak STD 2,4m/s
Closing 1,2m/s

Max dim: W 5500 mm x H 5500 mm
Max windload HS9110 P : Class 1
acc. EN12424 (Door closed)

Power Drive

HS8110 P

OUTSIDE USE, no high wind load
Carport, depot, stable, semi open constructions



Opening peak STD 2,4m/s
Closing 1,2m/s

Max dimensions: W 5500 mm x H 5500 mm
Max windload acc. EN12424 (Door closed)
HS8110 P: Max. Class 2

Power Drive

HS9110 P

HS8110 P

	HS9110 P	**HS8110 P**
Enviroment	Inside	**outside**
Wind class	1	2
Max width	5500	5500
Max Height	5500	5500
Closing speed	1,2m/s	1,2m/s
Open topspeed	2,4m/s	2,4m/s

Content

1. Product overview
2. “Gravity” driven doors
3. “Power Drive” driven doors
4. **Applications/Options**
5. Technical details
6. Sales

Applications / Options for all doors

1. Activators
2. Standard Colors
3. Windows
4. Bottom Bag options
5. Support frames
6. Side Frame Covers
7. Where to find information about Application / Options
   Sales Document / Data sheet

Applications / Options

Standard Activators and Safeties

1.5.6 Access and automation

Albany offers a wide range of functions that allow advanced opening and safety control.

1.5.6.1 Basic control functions

Interlocking

Developed for climate control or safety; if door A is open, door B cannot be opened. If door B is open, door A cannot be opened. An interlocked door can remember an up-command, if selected via a micro switch. Optionally an external locked switch can be installed to deactivate it.

- Function Interlock operating (switch interlock ON/OFF delivered with priority door)
- Switch interlock on/off (requires interlock function)
- Switch interlock on/off with key (requires interlock function)

Airlock

Developed for climate control or safety: Other than the interlock, door B will open automatically when door A is closed.
Circuit card installed in control unit. Optionally an external locked switch can be installed to deactivate it.

Reduced opening

When people pass through the door, it may be unnecessary to fully open a door. A manual (pedestrian) command can trigger a reduced door opening, while a radar or magnetic loop still triggers a full door opening.
Pre-fitted micro-switch to be activated in control unit.

- Function Two opening heights ‖ with manual selection (switch included)
- Function Two opening heights ‖ with automatic selection (2 different opening impulses)

1.5.6.2 External control functions

External push button box

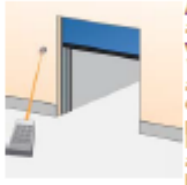
An extra control box is installed outside the building or inside close to the door if the main control unit needs to be installed away from the door opening.
Usually combined with reduced opening.
Installed on the inside or outside wall beside the door.

Pull-rope switch

A pull-rope switch above the door opening can be operated from e.g. a forklift truck. Pulling the rope opens the door.
Installed on the inside construction above the door.

- Pull down switch complete 5 m cord
- Pull switch bracket in galvanised steel – L 3000 mm
- Pull switch bracket in painted steel – L 3000 mm
- Pull switch bracket in stainless steel – L 3000 mm

Remote control



A hand-held radio transmitter allows door operation from a vehicle or any position within 50-100 meters from the receiver and aerial at the door. For closing, the door can be provided with a photocell beam.
Receiver installed in control unit, antenna installed on the wall beside the door.

1.5.6.3 Automatic control functions

- Function Auto/Manual (includes switch on the control box)
- Function Manual closing with opening/closing using a common manual command (e.g. one single pull rope opens and closes the door)
- Function Manual closing with separate impulse (e.g. 2 buttons up and down)

Magnetic loop

A sensor in the floor detects a metal object (usually forklift trucks, pallet trucks) and opens the door automatically. This is an ideal solution for frequent vehicle traffic.
Installed on the outside, inside or both sides of the door in the floor.

Radar

An infrared sensor above the door detects an object (person, vehicle) within a specified distance from the door and opens the door automatically. This is an ideal solution for frequent vehicle or personal traffic. Often combined with automatic closing.
Installed on the inside or outside wall above the door.

Photocell open door

A set of photocells on pillars, on each side of the door. When a person or vehicle passes between the photocells, the beam is interrupted and the door opens.
Photocells installed on pillars, away from the door.

Automatic closing (standard)

A programmable timer that closes the door after a specified time, counted from either the fully open position and/or from passing through the photocell beam.
Usually also a switch on the control unit is used to turn to manual closing.
Adjustable micro switches in control unit.

Applications / Options

Standard Activators and Safeties

1.5.6.4 Safety functions

Wireless edge activator (standard)

All doors are equipped with an edge activator. A wireless edge activator in the bottom seal detects any obstruction under a closing door and reverses the door.
Installed in the bottom edge.

Safety photocells 1-channel (standard)

A set of a photocell transmitter and receiver is installed in the door opening. If the photocell beam is interrupted during closing, the door will stop in less than 30 mm and reverse to the fully open position.
Installed in the door opening 300 mm from floor level.

Warning lights- Red

A red warning light on each side gives information on the current door behaviour. Flashing light seconds before and during door movement.
Installed on the inside and outside wall beside the door.

Acoustic signal

An acoustic signal is given, starting ...ms before the door begins to close and continues until the door is fully closed.
Installed on the inside and outside wall beside the door.

- Acoustic signal 24VAC 80 dB at 1 meter (horn when door is moving)

1.5.6.5 Lights

- Standard function flashing light
- Standard function flashing light with pre-warning before closing and opening

Warning lights- Orange

An orange warning light on each side gives information on the current door behaviour. Flashing light seconds before and during door movement.
Installed on the inside and outside wall beside the door.

Warning lights- Green

A green warning light installed on each side of the door indicating the open position of the door by continuous light signal.

Traffic lights- Red & Green

If traffic through a door needs to be directed; two red and two green traffic lights can be installed to indicate traffic direction. From the side where a vehicle is first detected to approach the door, the green traffic light comes on. The opposing side shows a red traffic light. Traffic from this direction must give way to the other. Usually installed in e.g. parking garages.
Installed on the inside and outside wall beside the door.

1.5.6.6 Additional functions

UPS battery backup

In case of main supply failure or emergency situations, it may be necessary to be able to open the door. The UPS battery stores enough power for one emergency opening cycle.
Installed on the inside wall beside the door.

- Kit UPS Interface, automatic opening in case of power failure
- Kit UPS Interface, semi-automatic in case of power failure

Applications / Options

Standard Colors

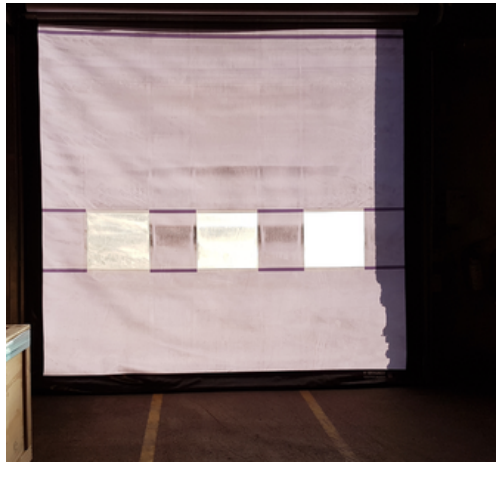
1.2.3 Colours

The Albany HS8010P high speed door is available in 8 standard colors fabric colours and translucent Dynalumin. The RAL-colours are as close as possible to the official RAL HR collection. Max. deviation is 1,0 DE.

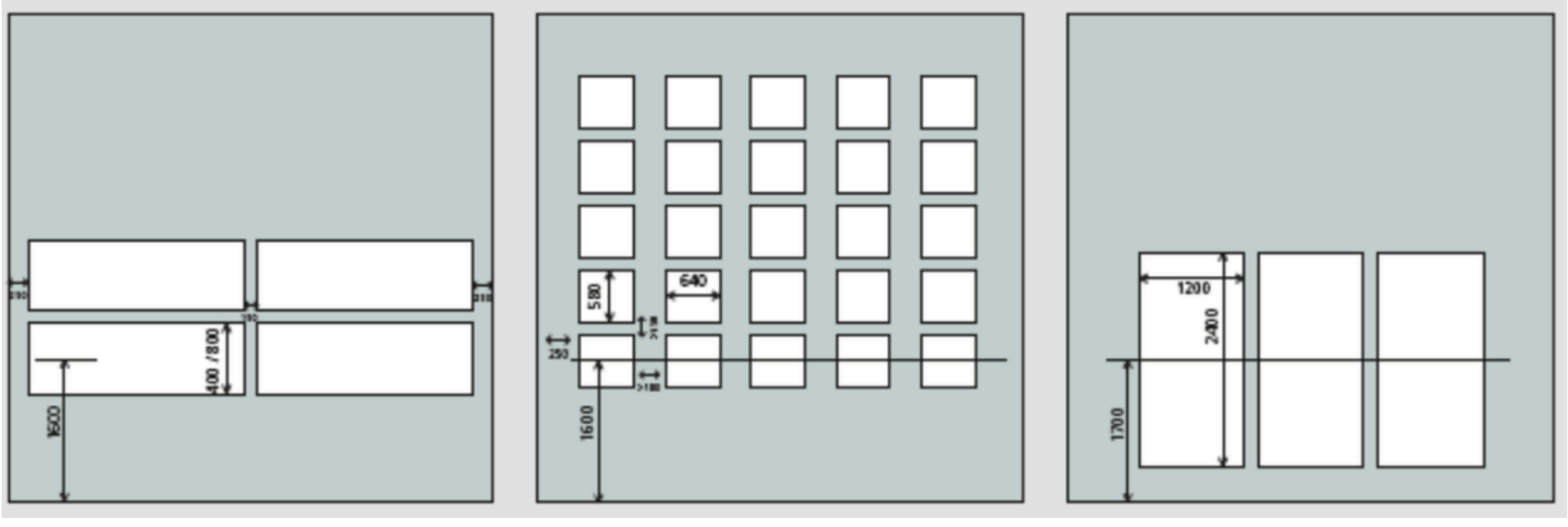
- RAL 1003
- RAL 2004
- RAL 3000
- RAL 5002
- RAL 6005
- RAL 7035
- RAL 9005
- RAL 9010
- Dynalumin, translucent (option)

Translucent:

Curtain to let light in but which is not transparent to still assure privacy.

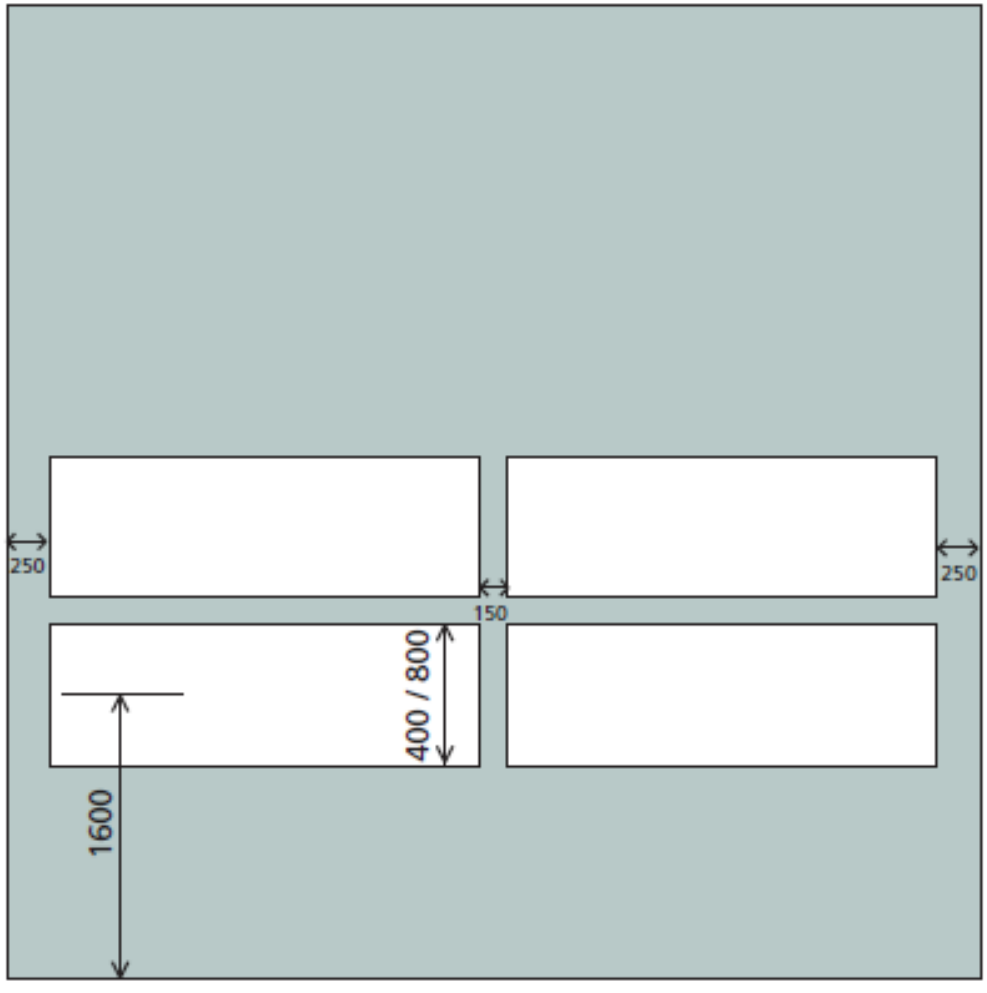


Standard Window and Vision Panels

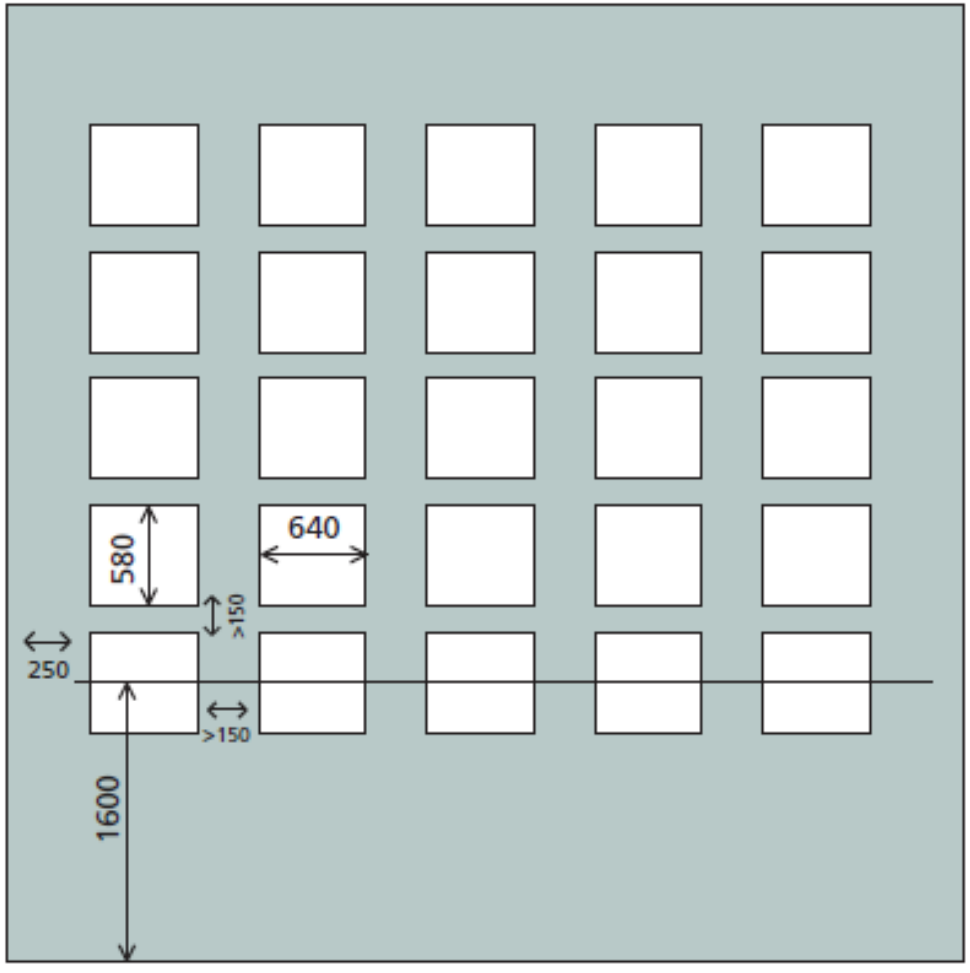


Exact Configuration for each door found in Data Sheet.

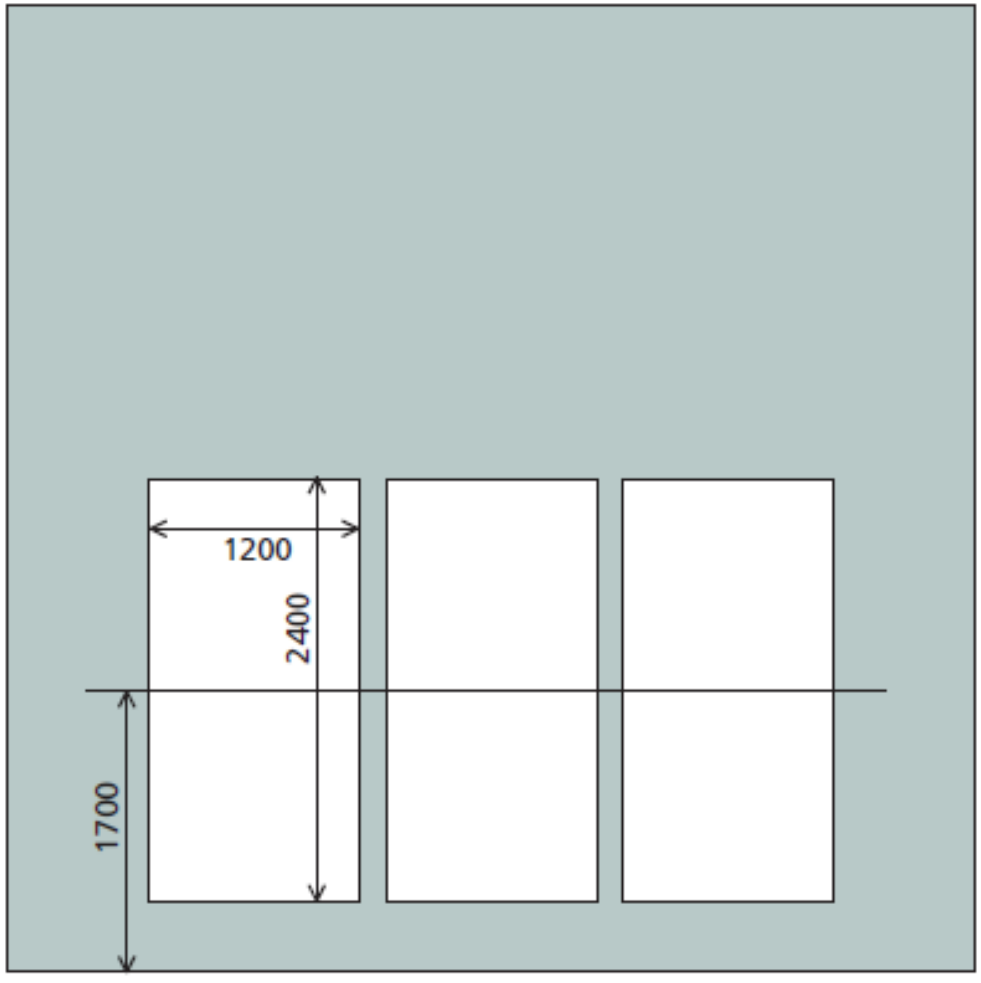
Standard Window and Vision Panels



Standard Window and Vision Panels

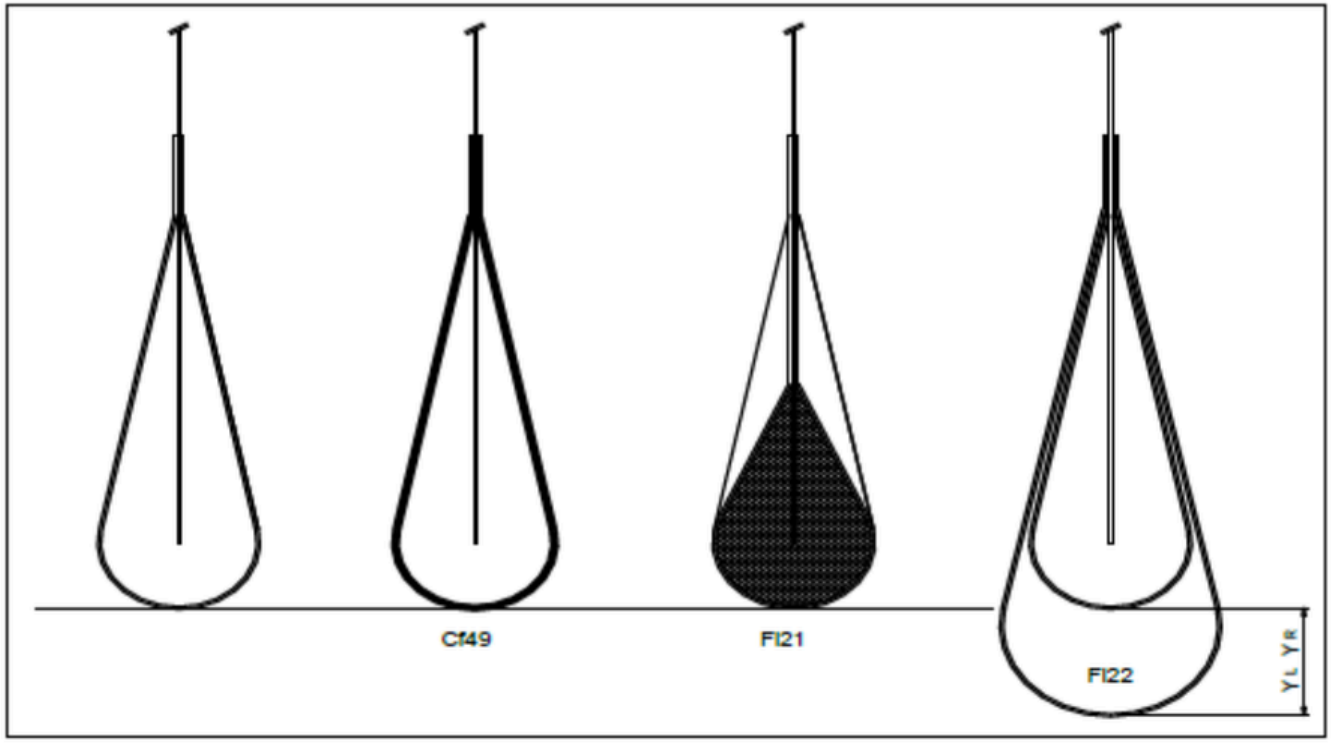


Standard Window and Vision Panels



Applications / Options

Standard Bottom Bag Options



Standard Bottom bag Black	Stronger Bottom bag Grey	Reinforced Bottom bag (End-Plugs)	Additional Bottom bag

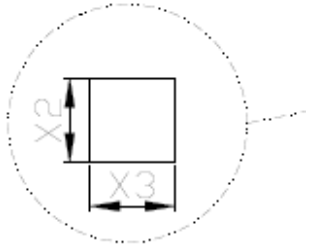
Foreframe “support frame / spacing frame”

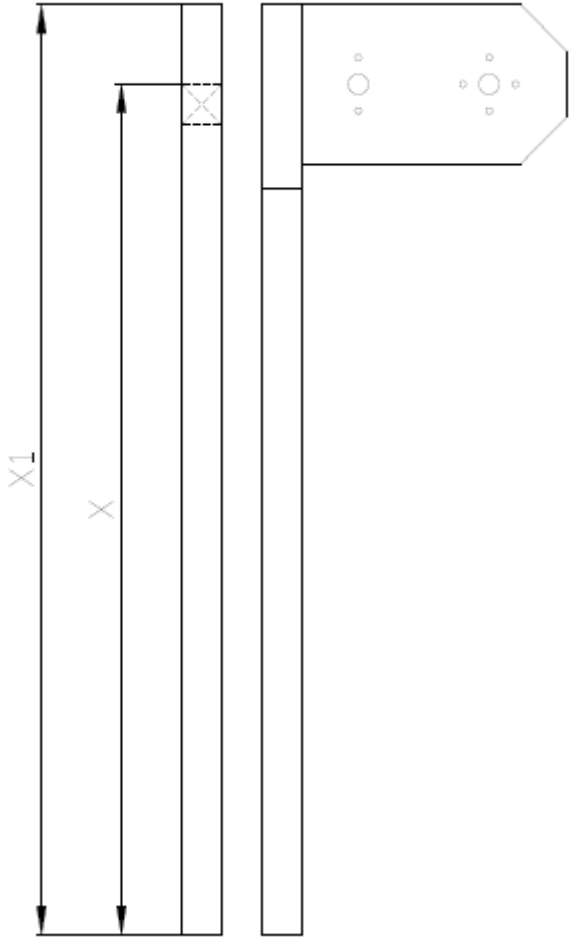
Since HS doors are not self supporting and can deflect under wind pressure, a Foreframe could be necessary when:

1. The wall is not even/strong/stable enough
2. The width is > 4500 mm
3. Emergency doors always come with it

The type of Foreframe is specified in the installation manual per door type

Door type	X	X1	X2	X3
HS9010PL, HS8010PL, HS8020PL	H + 470	H + 640	80	120





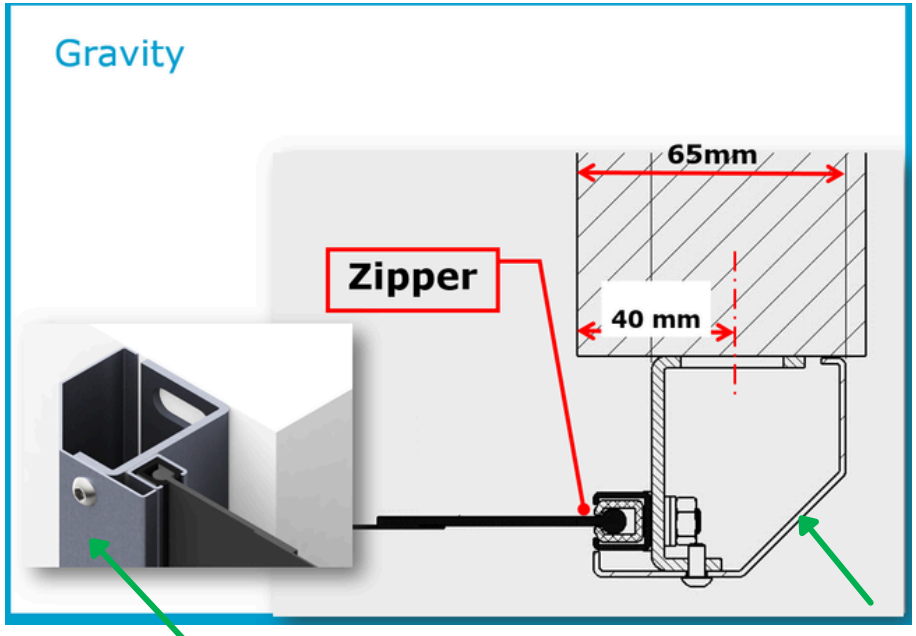
Foreframe “support frame / spacing frame”

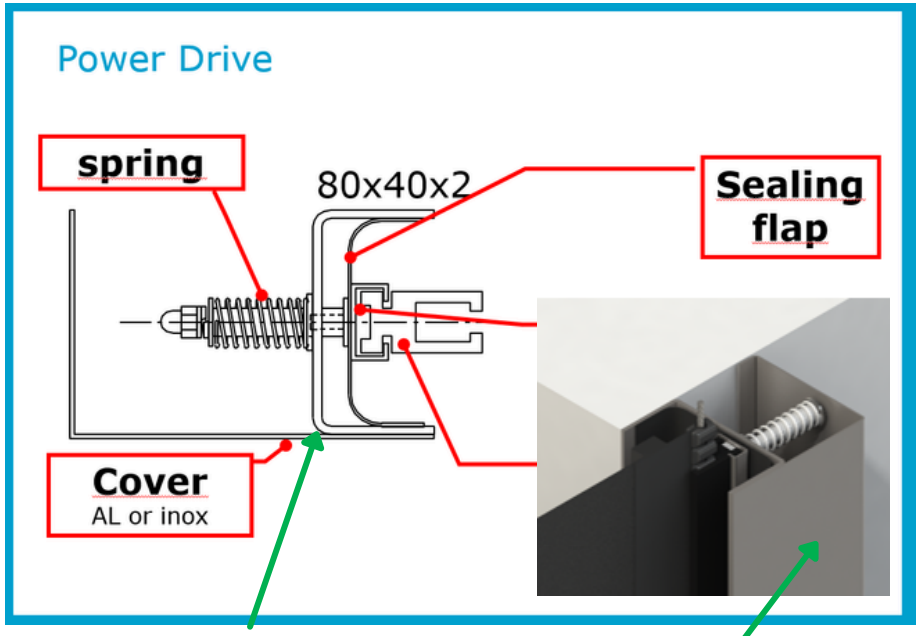
Local Supply Decisions:

- Complete frame in galvanised steel 120x80 for all doors “Power Driven” cut to size and delivered with the door
- Complete frame in galvanised steel 80x80 for all “Gravity driven” doors cut to size and delivered with the door
- Tube (120x80mm) galvanised steel, stock length L= 6000 mm
- Tube (80x80mm) in galvanised steel, stock lenght L= 6000 mm
- Purchase Locally

Door type	X	X1	X2	X3
HS9010PL, HS8010PL, HS8020PL	H + 470	H + 640	80	120

Side Column Covers





HS Doors classically are delivered WITHOUT covers on the side frames:

Gravity: Galvanized steel support / fixation track (frame) is visible

Power Drive: Galvanized Steel support / fixation track (frame) and springs are visible

Exception: HS 9030G always has composite covers and HS8110&9110 Series always has standard galvanized column covers.

Chain winch

- Inside applications
  - HS9010P
  - HS9010PL
  - HS9020G
  - HS9110P
- Outside applications
  - HS8010P
  - HS8110P
- Application doors
  - HS9020GHY
  - HS9030GHY
  - HS9020GAT
  - HS9030GAT
  - HS9030GEM
  - HS9010PEM
  - HS9010PFR

Applications / Options

2 Page Sales Brochure
(short customer sales document)

High speed door
Albany HS8010P

ASSA ABLOY

ASSA ABLOY Entrance Systems

The global leader in door opening solutions

Medium-sized high speed door for outside applications

The main features of the Albany HS8010P are:
- Medium-sized, max. 5500 x 5500 mm
- Heavy-duty operations
- Outside usage
- Galvanised steel construction
- Unique direct door drive system
- Opening speed up to 2,4 m/s

The fast opening and closing speed improves your traffic flow, provides employee comfort, and saves energy.

The unique direct door drive system eliminates the need for weighted bottom bars or tension systems by actively driving the door curtain. An exceptional reliability and smooth operation is ensured by a frequency inverter driven motor for a soft start and stop.

The robust steel, IP54 classified, control box is easy to operate and to program. The mechanical main switch and emergency stop make it foolproof. The clear display gives a wide range of information like an easy to read cycle counter.

Break-away - and Automatic Repair system
Albany high speed doors are equipped with a break-away - and automatic repair system. The curtain reintroduces itself after a crash to reduce downtime and maintenance.

Completely safe
Features that make our doors completely safe.
- The soft bottom edge has no rigid parts, it moulds around any obstruction to prevent injury or damage.
- When an object hits the door, the fabric curtain absorbs the impact and releases itself from its side guides without causing damage or being damaged itself.

Windows and vision panels
To increase the admission of daylight and visibility, the door curtain can be equipped with windows or vision panels. Windows have fixed sizes in a grid. A vision panel uses the full width of the door curtain.

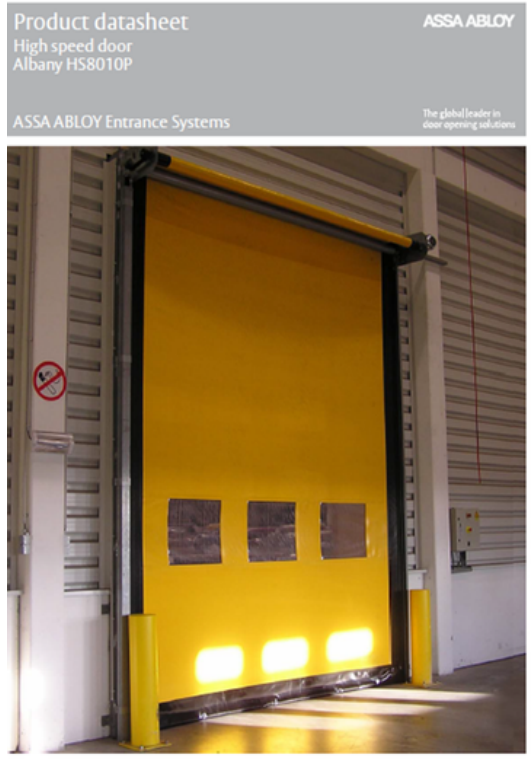
Insulated curtain
An insulated curtain with an increased static insulation of 2,41 W/(m²K) is available as an option.

Technical Data

Min size (W x H)	1000 x 2000 mm
Max size (W x H)[1]	5500 x 5500 mm
Operating speed	opening: up to 2,4 m/s [2] closing: 1,2 m/s
Wind load resistance, EN 12424	class 3 (700 N/m²) class 4 (1000 N/m²) for doors up to 4000 x 5500 mm
Water penetration, EN 12425	class 2 (50 N/m²)
Air permeability, EN 12426	class 1 (24 m³/m²h at 50 Pa)
Thermal transmittance	6,02 W/(m²K)
Lifetime expectations	1.000.000 cycles

1) Other sizes available on request 2) Depending on door size

Detailed Data sheet
(long customer sales document)



Product datasheet
High speed door
Albany HS8010P

ASSA ABLOY

ASSA ABLOY Entrance Systems

The global leader in door opening solutions

Applications / Options

2 Page Sales Brochure (short customer sales document)

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Water penetration, EN 12425	class 2 (50 N/m²)
Air permeability, EN 12426	class 1 (24 m³/m²/h at 50 Pa)
Thermal transmittance	6,02 W/ (m²K)
Lifetime expectations	1.000.000 cycles

1) Other sizes available on request 2) Depending on door size

<u>Content page 1/2:</u>

Max Size of door

Material of door

(galv or stainless steel)

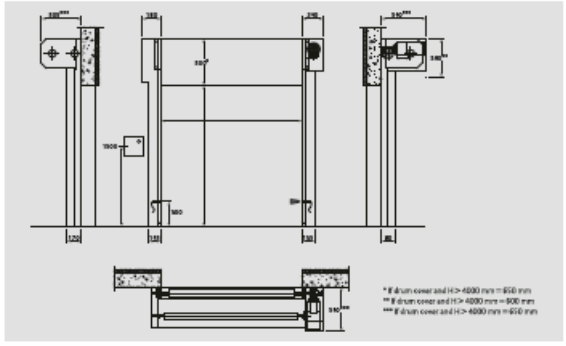
List of Features

Technical Data

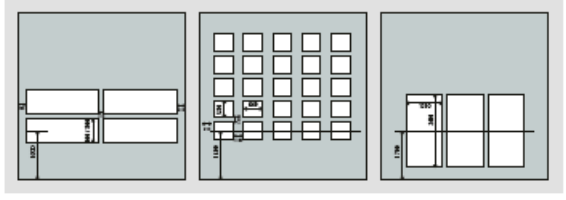
Applications / Options

2 Page Sales Brochure (short customer sales document)

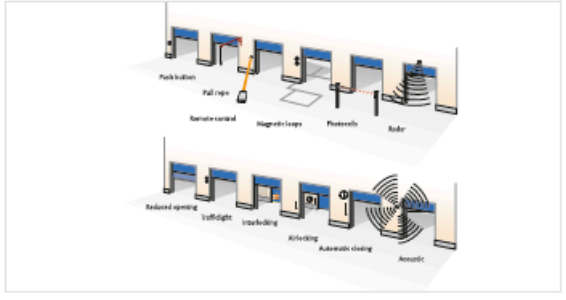
Dimensional specifications



Windows and Vision panels



Options access and automation



Standard available colours

Content page 2/2:

General Drawing

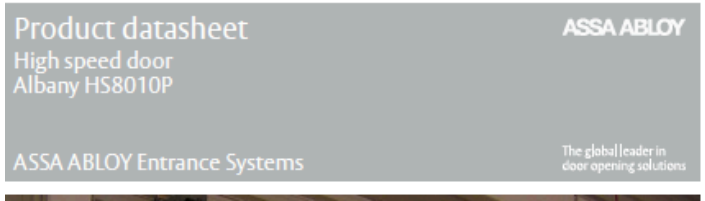
Window specs

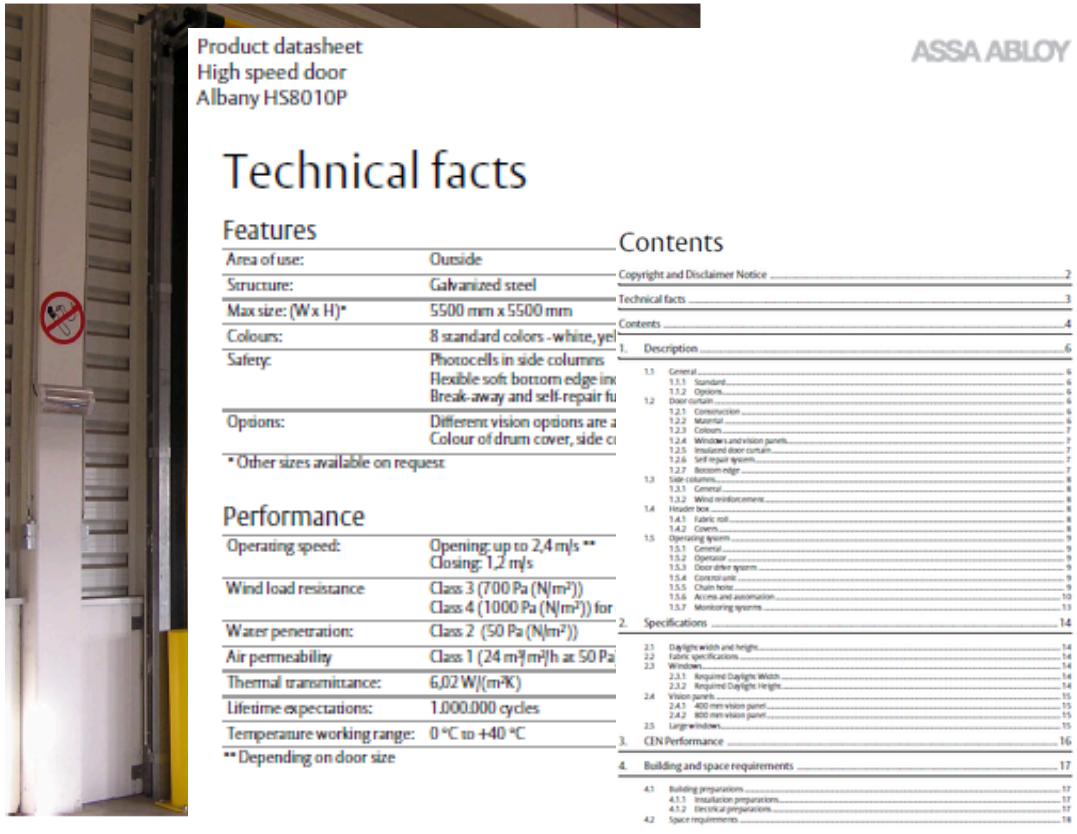
Access and automation

Curtain Colors

Applications / Options

Detailed Data Sheet (long customer sales document)





Content:

Technical Facts

(features/performance)

Contents

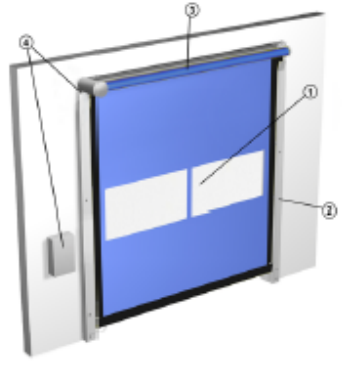
Applications / Options

Detailed Data Sheet (long customer sales document)

1. Description

1.1 General

The Albany HS8010P high speed door is designed for exterior openings in medium-sized heavy-duty operations. It protects your environment against draughts, humidity, dust and dirt. With fast opening and closing speed, the door improves your traffic flow, provides employee comfort, and saves energy.



The Albany HS8010P high speed door has 4 primary parts:

1) Door curtain
2) Side columns
3) Header box
4) Operating system

1.1.1 Standard

The Albany HS8010P high speed door is supplied with the following specifications as standard:

Door curtain:	900 g/m² coloured PVC
Structure:	Galvanized steel
Side column:	Wind reinforcement 230 mm at the bottom
Safety:	Photocells in side columns Flexible soft bottom edge including wireless edge activator Break-away and self-repair function
Operation:	Operator + control unit
Colours:	8 standard colors - white, yellow, green, orange, red, gray, black, blue

1.1.2 Options

Albany provides a wide range of options and accessories to customise the Albany HS8010P high speed door to any customer's needs.

Header box:	Galvanised steel drum cover Galvanised steel operator cover
Side column:	Galvanised steel side column covers
Operation:	Access and Automation Monitoring System
Windows:	Small windows or; Vision panels or; Large windows
Insulated door curtain	Static insulation 2,41 W/m²K (max. 4500 mm x 4500 mm)

1.2 Door curtain

1.2.1 Construction

The door curtain is constructed from one single piece of PVC fabric. The door curtain rolls up above the door opening and requires little space.

Top

The top of the fabric is connected to a fabric roll, located in the header box above the door opening.

Bottom

The flexible bottom edge of the door curtain does not contain any stiffeners, making the door curtain completely safe as the curtain moulds itself in case a person is trapped below the closing door.

Side

The left and right sides of the door curtain are constructed with a patented retaining strap. If the door is hit by a vehicle, a part of this retaining strap is pulled out of the side column. The self repair function acts as a zipper to put the retaining strap back in the side column.

1.2.2 Material

Fabric type

- 900 g/m² colored PVC,
- high resistance

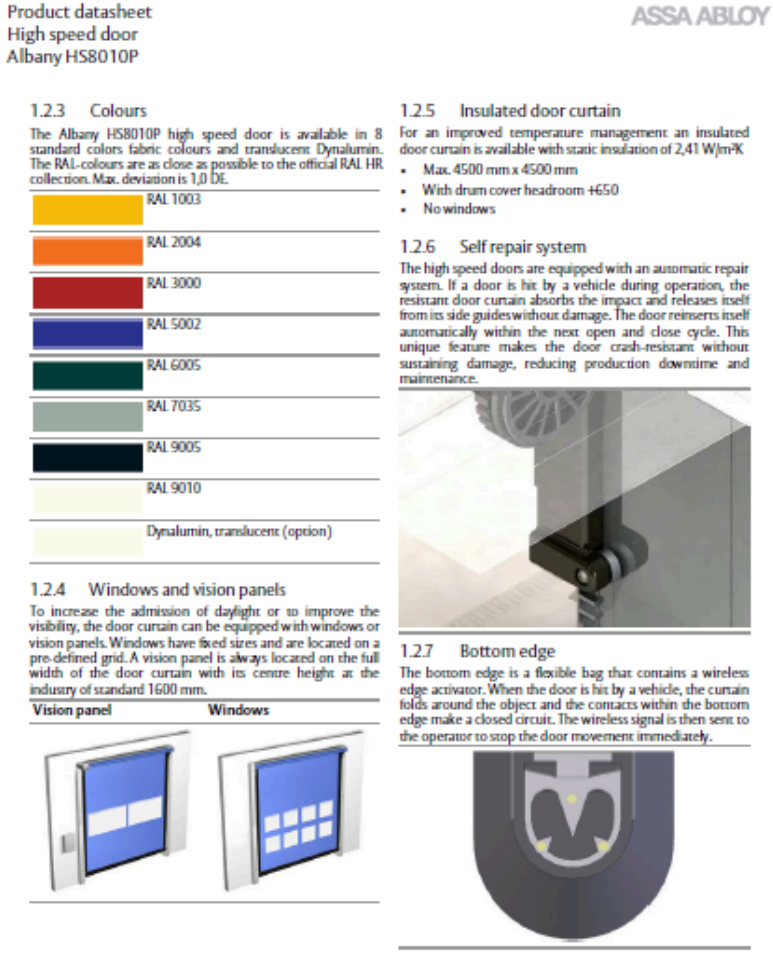
Content:

Component list with material information

Functional description

Applications / Options

Detailed Data Sheet (long customer sales document)



Product datasheet
High speed door
Albany HS8010P

ASSA ABLOY

1.2.3 Colours

The Albany HS8010P high speed door is available in 8 standard colors fabric colours and translucent Dynalumin. The RAL-colours are as close as possible to the official RAL HR collection. Max. deviation is 1,0 DE.

RAL 1003
RAL 2004
RAL 3000
RAL 5002
RAL 6005
RAL 7035
RAL 9005
RAL 9010
Dynalumin, translucent (option)

1.2.4 Windows and vision panels

To increase the admission of daylight or to improve the visibility, the door curtain can be equipped with windows or vision panels. Windows have fixed sizes and are located on a pre-defined grid. A vision panel is always located on the full width of the door curtain with its centre height at the industry of standard 1600 mm.

Vision panel | Windows

1.2.5 Insulated door curtain

For an improved temperature management an insulated door curtain is available with static insulation of 2,41 W/m²K

- Max. 4500 mm x 4500 mm
- With drum cover headroom +650
- No windows

1.2.6 Self repair system

The high speed doors are equipped with an automatic repair system. If a door is hit by a vehicle during operation, the resistant door curtain absorbs the impact and releases itself from its side guides without damage. The door reinserts itself automatically within the next open and close cycle. This unique feature makes the door crash-resistant without sustaining damage, reducing production downtime and maintenance.

1.2.7 Bottom edge

The bottom edge is a flexible bag that contains a wireless edge activator. When the door is hit by a vehicle, the curtain folds around the object and the contacts within the bottom edge make a closed circuit. The wireless signal is then sent to the operator to stop the door movement immediately.

Content:

Available RAL Colors

Windows introduction

Functional Description

Applications / Options

Detailed Data Sheet (long customer sales document)

Product datasheet
High speed door
Albany HS8010P

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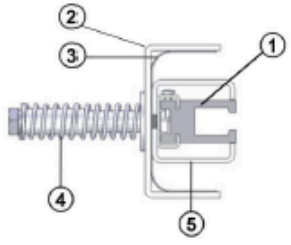
1.3 Side columns

The side columns guide the door curtain up and down. This guidance is a plastic-to-plastic connection, which makes lubrication essential.

1.3.1 General

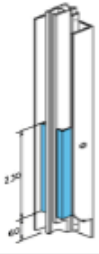
The side columns are part of the frame that also contains the header box. This frame is made of 3 mm thick galvanized steel tubes. The side columns are connected directly to the wall.



1) Low-friction polyethylene side guide inside a steel rail
2) Galvanized steel structure. U-channel 80 mm x 40 mm x 3 mm
3) Flexible rubber seal
4) Tension spring
5) Wind reinforcement (230 mm at bottom)

1.3.2 Wind reinforcement

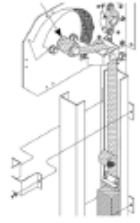
At the bottom of the side columns, a wind reinforcement is installed around the side guide to increase the wind load resistance of the door. The bottom part of the door curtain is the most vulnerable to wind load. This reinforcement prevents that the curtain will be pulled out of the side guides.



1.4 Header box

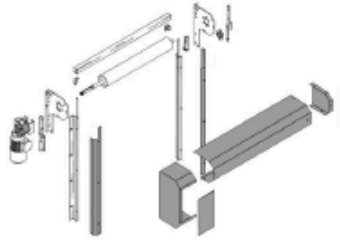
1.4.1 Fabric roll

The fabric roll is installed in the header box above the door curtain. Its function is to roll up the door curtain by means of a counterweight. A gear drive system forces the door curtain up and down the tracks.



1.4.2 Covers

For use in dusty and dirty environments or for aesthetical reasons, an optional header box cover is available to enclose the fabric roll and/or the operator. Also the side columns can be equipped with covers.



Content:

Side Column technical information

Information about covers

Applications / Options

Detailed Data Sheet (long customer sales document)

1.5 Operating system

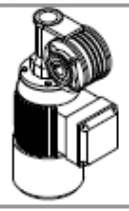
1.5.1 General

The Albany HS8010P high speed door is always operated electrically. The operating system is a combination of an operator and a control unit. The operator opens and closes the door with an electric engine. The operator secures a safe closing speed with a soft start and stop.

1.5.2 Operator

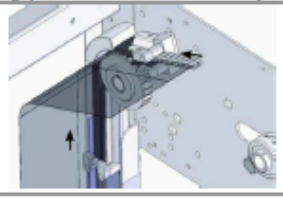
Exceptional reliability and smooth operation is ensured by a motor driven by a frequency inverter. This technology ensures a soft start and stop, which increases the longevity of the motor considerably. It also allows faster opening/closing speed. This motor delivers reliable operations around-the clock. The operator is always combined with a control unit.

The operator drives the fabric roll to open or close the door. In case of a main supply failure, the operator can be disconnected and the door can be opened or closed manually using the hand crank.



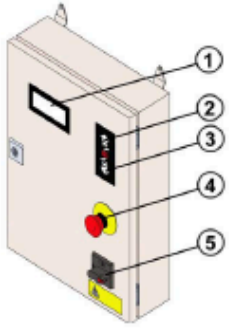
1.5.3 Door drive system

The high speed doors are equipped with a unique door drive system. This system eliminates the need for ballast in the door curtain or tension straps. This gear driven system consists of a pinion on the drive shaft that forces the lateral retaining straps up or down the tracks. This ensures that even under high pressure difference the door will be fully closed.



1.5.4 Control unit

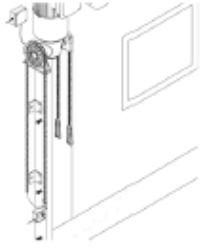
The control unit is installed beside the door. It has impulse UP and DOWN buttons, an emergency stop mushroom button and a mechanical mains switch.



1) Display
2) UP button
3) DOWN button
4) Emergency stop
5) Mechanical main switch

1.5.5 Chain hoist

A chain hoist makes it possible to manually operate the door, e.g. during a power failure. When the electrical operation is in use, the chain hoist is disconnected. When manual operation is required, pulling a string will release the motor brake, allowing the drum to rotate freely, using the chain hoist.



Content:

Operator technology

Control Box information

Chain Hoist information

Applications / Options

Detailed Data Sheet (long customer sales document)

Product datasheet
High speed door
Albany HS8010P

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1.5.6 Access and automation

Albany offers a wide range of functions that allow advanced opening and safety control.

1.5.6.1 Basic control functions

Interlocking

Developed for climate control or safety; If door A is open, door B cannot be opened. If door B is open, door A cannot be opened. An interlocked door can remember an up-command, if selected via a micro switch. Optionally an external locked switch can be installed to deactivate it.

- Function Interlock operating (switch interlock ON/OFF delivered with priority door)
- Switch interlock on/off (requires interlock function)
- Switch interlock on/off with key (requires interlock function)

Airlock

Developed for climate control or safety: Other than the interlock, door B will open automatically when door A is closed. Circuit card installed in control unit. Optionally an external locke switch can be installed to deactivate it.

Reduced opening

When people pass through the door, it may be unnecessary to fully open a door. A manual (pedestrian) command can trigge a reduced door opening, while a radar or magnetic loop still triggers a full door opening. Pre-fitted micro-switch to be activated in control unit.

- Function Two opening heights (Il with manual selectio (switch included)
- Function Two opening heights (Il with automatic selection (2 different opening impulses)

1.5.6.2 External control functions

Product datasheet
High speed door
Albany HS8010P

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1.5.6.3 Automatic control functions

- Function Auto/Manual (includes switch on the control box)
- Function Manual closing with opening/closing using a common manual command (e.g. one single pull rope opens and closes the door)
- Function Manual closing with separate impulse (e.g. 2 buttons up and down)

Magnetic loop

A sensor in the floor detects a metal ol trucks, g the doo ideal sol traffic. Installe both sid

Radar

An infra detects within a the doo automa solution persona with au Installe wall abo

Photocell open door

A set of each sid person the pho interrup Photoce away fr

Automatic closing (standard)

A progr the doo counte position through Usually control manual Adjust control

1.5.6.4 Safety functions

Wireless edge activator (standard)

All doors are equipped with an edge activator. A wireless edge activator in the bottom seal detects any obstruction under a closing door and reverses the door.
Installed in the bottom edge.

Product datasheet
High speed door
Albany HS8010P

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1.5.6.5 Lights

- Standard function flashing light
- Standard function flashing light with pre-warning before closing and opening

Warning lights - Orange

An orange warning light on each side gives information on the current door behaviour. Flashing light seconds before and during door movement.
Installed on the inside and outside wall beside the door.

Warning lights - Green

A green warning light installed on each side of the door indicating the open position of the door by continuous light signal.

Traffic lights - Red & Green

If traffic through a door needs to be directed; two red and two green traffic lights can be installed to indicate traffic direction. From the side where a vehicle is first detected to approach the door, the green traffic light comes on. The opposing side shows a red traffic light. Traffic from this direction must give way to the other. Usually installed in e.g. parking garages.
Installed on the inside and outside wall beside the door.

1.5.6.6 Additional functions

UPS battery backup

In case of main supply failure or emergency situations, it may be necessary to be able to open the door. The UPS battery stores enough power for one emergency opening cycle.
Installed on the inside wall beside the door.

- Kit UPS Interface, automatic opening in case of power failure
- Kit UPS Interface, semi-automatic in case of power failure

Content:

Activators and their functions

Applications / Options

Detailed Data Sheet (long customer sales document)

Product datasheet
High speed door
Albany HS8010P

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1.5.7 Monitoring systems

As an option on all our products, a Monitoring System can be installed. This system helps to ensure efficiency and security in daily operations. All doors or docking stations are connected to the Monitoring System's server, which gives the opportunity to supervise, monitor and report a wide variety of aspects in a facility.

1.5.7.1 Saving energy

A monitoring system reduces energy costs and contributes to a better environment. Energy is lost every time a door is open. If a door is open when no truck is at the bay, even more energy is lost.

A Monitoring System automatically ensures that no door will open unless there is a truck at the bay and even set it to close when there an activity is delayed.

1.5.7.2 Security enhancement

Closing and locking doors is an obvious daily routine. However, checking this manually can be time consuming in a busy facility.

A Monitoring System can automatically ensure that all doors are closed and locked when they need to be. It can also activate all doors and locks from its remote location, and give a real-time overview of the building's situation.

1.5.7.3 Dock management

A good way to increase throughput and thereby efficiency at a logistics facility is to reduce the time of having no truck – or the wrong truck – at a loading bay.

A Monitoring System makes visible – in real-time –which bays are occupied or free, and for how long. It makes it possible to reserve bays for docking activities and to inform drivers via SMS. Since it incorporates information from cameras and other inputs (RFID, card readers, etc.), the system stays updated in real-time.

1.5.7.4 Facility management

The Monitoring System gives a real-time service status for all your door and docking equipment. If an error code occurs, the Albany service organisation is automatically notified, and will respond quickly. Other maintenance information can easily be integrated, further reducing the overall costs.

Content:

Monitoring Systems

Applications / Options

Detailed Data Sheet (long customer sales document)

Product datasheet
High speed door
Albany HS8010P

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2. Specifications

2.1 Daylight width and height

The standard Albany HS8010P high speed door is delivered in the following size range:

Standard door sizes*

	Daylight width	Daylight height
Min.:	1000 mm	2000 mm
Max.:	5500 mm	5500 mm

*Other sizes may be available on request

2.2 Fabric specifications

	Coloured fabric	Vision panel
Material	Reinforced PVC	PVC
Thickness	0,8 mm	2,0 mm
Weight	900 g/m²	2,5 kg / m²
Tensile strength (kN/5cm)	4,01 / 3,5 W	1,6
Tearing resistance	600 N (DIN 53363)	100 N (DIN 53515)

2.3 Windows

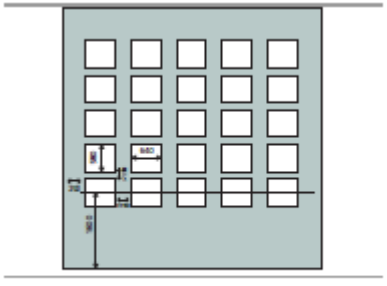
Dimensions: W x H: 640 mm x 580 mm.
Combinations: Any combination of rows is possible.

2.3.1 Required Daylight Width

DLW	Available no. of columns
1140 mm – 1929 mm	1
1930 mm – 2719 mm	2
2720 mm – 3509 mm	3
3510 mm – 4299 mm	4
4300 mm – 5089 mm	5

2.3.2 Required Daylight Height

DLH	Available no. of rows	CC
2040 mm	1	1600 mm
2770 mm	2	2330 mm
3500 mm	3	3060 mm
4230 mm	4	3790 mm
4960 mm	5	4520 mm



Content:

Specifications such as:

Sizes

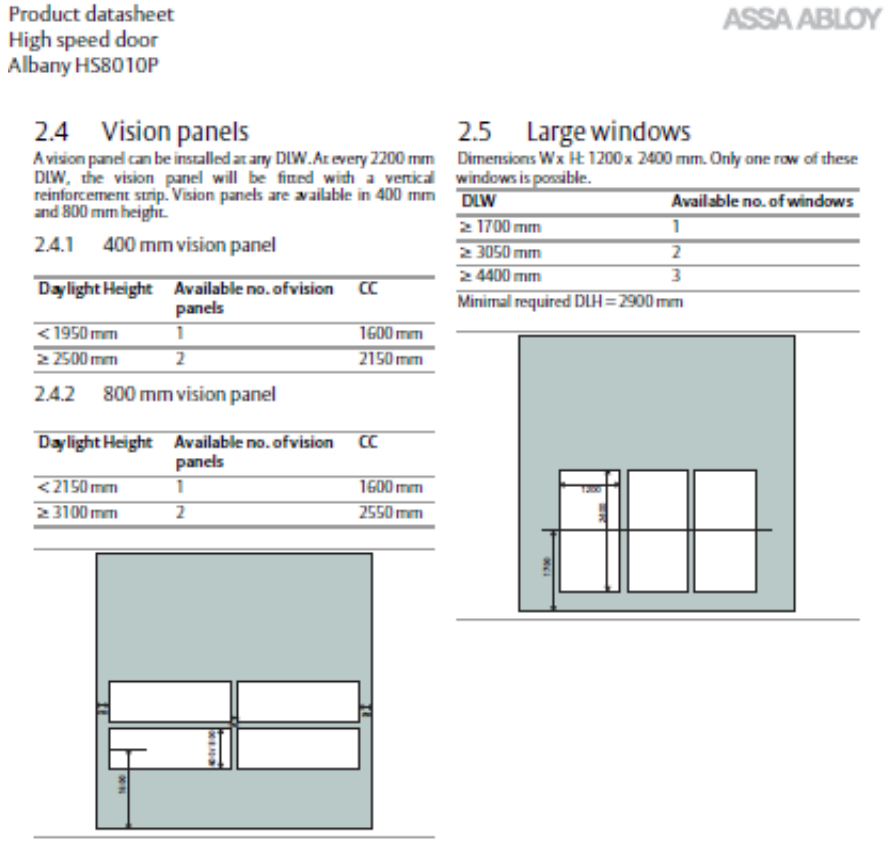
Fabric specifications

Specific Window and

Vision panel allocation

Applications / Options

Detailed Data Sheet (long customer sales document)



Product datasheet
High speed door
Albany HS8010P

ASSA ABLOY

2.4 Vision panels

A vision panel can be installed at any DLW. At every 2200 mm DLW, the vision panel will be fitted with a vertical reinforcement strip. Vision panels are available in 400 mm and 800 mm height.

2.4.1 400 mm vision panel

Daylight Height	Available no. of vision panels	CC
< 1950 mm	1	1600 mm
≥ 2500 mm	2	2150 mm

2.4.2 800 mm vision panel

Daylight Height	Available no. of vision panels	CC
< 2150 mm	1	1600 mm
≥ 3100 mm	2	2550 mm

2.5 Large windows

Dimensions W x H: 1200 x 2400 mm. Only one row of these windows is possible.

DLW	Available no. of windows
≥ 1700 mm	1
≥ 3050 mm	2
≥ 4400 mm	3

Minimal required DLH = 2900 mm

Content:

Vision Panel / Large window allocation

Applications / Options

Detailed Data Sheet (long customer sales document)

Content:

Test Values

Product datasheet
High speed door
Albany HS8010P

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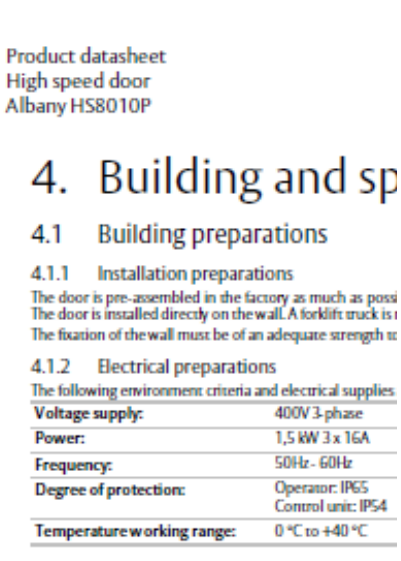
3. CEN Performance

Characteristic	Standard	Test acc.	Result	Value
Wind load	EN 12424	EN 12444	Class 3*	700 Pa (N/m²)
Water permeability	EN 12425	EN 12489	Class 2	50 Pa (N/m²) water spray for 20 minutes
Air permeability	EN 12426	EN 12427	Class 1	24 m³/m²/h at 50 Pa
Safe openings	EN 12453	EN 12445	Pass	
Mechanical resistance	EN 12604	EN 12605	Pass	
Unintended movements	EN 12604	EN 12605	Pass	
Thermal resistance	EN 12428		6,02 W/(m²K)	
Performance (cycles)	EN 12604	EN 12605	1.000.000 cycles	

*Indicated wind-load classification is for maximum dimension. For doors up to W 4000 mm x H 5500 mm: Class 4 (1000 Pa (N/m²))

Applications / Options

Detailed Data Sheet (long customer sales document)


Product datasheet
High speed door
Albany HS8010P

4. Building and sp

4.1 Building preparations

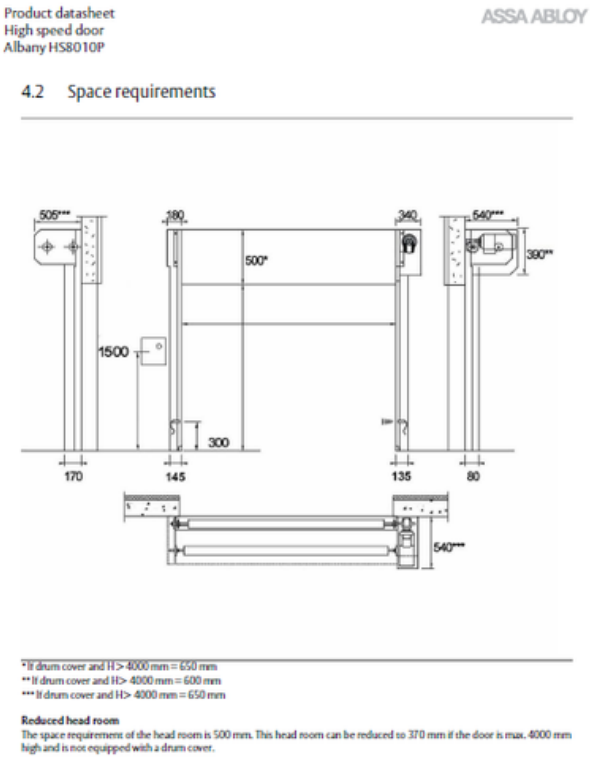
4.1.1 Installation preparations

The door is pre-assembled in the factory as much as possi
The door is installed directly on the wall. A forklift truck is
The fixation of the wall must be of an adequate strength t

4.1.2 Electrical preparations

The following environment criteria and electrical supplies

Voltage supply:	400V 3-phase
Power:	1,5 kW 3 x 16A
Frequency:	50Hz - 60Hz
Degree of protection:	Operator: IP65 Control unit: IP54
Temperature working range:	0 °C to +40 °C



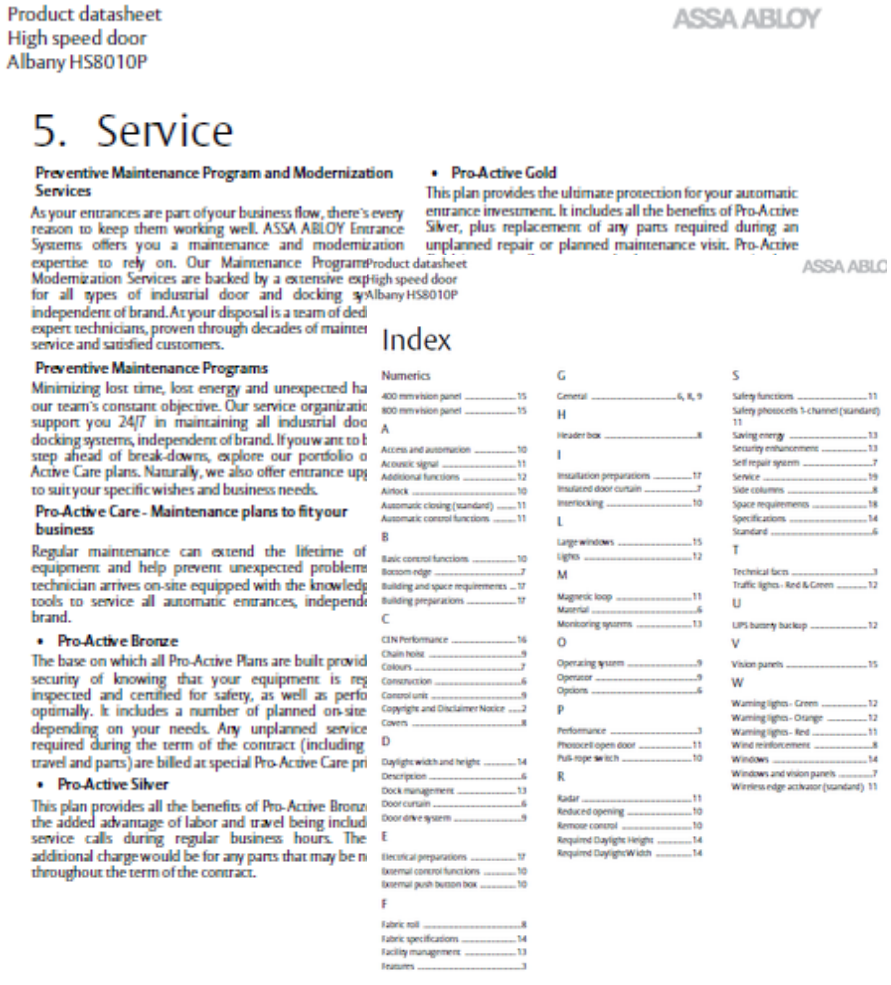
Content:

Building and space requirements

Electrical requirements

Applications / Options

Detailed Data Sheet (long customer sales document)



Product datasheet
High speed door
Albany HS8010P

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5. Service

Preventive Maintenance Program and Modernization Services

As your entrances are part of your business flow, there's every reason to keep them working well. ASSA ABLOY Entrance Systems offers you a maintenance and modernization expertise to rely on. Our Maintenance Program/ Modernization Services are backed by a extensive exp for all types of industrial door and docking sy independent of brand. At your disposal is a team of ded expert technicians, proven through decades of maintei service and satisfied customers.

Preventive Maintenance Programs

Minimizing lost time, lost energy and unexpected ha our team's constant objective. Our service organizatic support you 24/7 in maintaining all industrial doo docking systems, independent of brand. If you want to b step ahead of break-downs, explore our portfolio o Active Care plans. Naturally, we also offer entrance up to suit your specific wishes and business needs.

Pro-Active Care - Maintenance plans to fit your business

Regular maintenance can extend the lifetime of equipment and help prevent unexpected problems technician arrives on-site equipped with the knowledg tools to service all automatic entrances, independe brand.

- **Pro-Active Bronze**

The base on which all Pro-Active Plans are built provid security of knowing that your equipment is reg inspected and certified for safety, as well as perfo optimally. It includes a number of planned on-site depending on your needs. Any unplanned service required during the term of the contract (including travel and parts) are billed at special Pro-Active Care pri

- **Pro-Active Silver**

This plan provides all the benefits of Pro-Active Bronz the added advantage of labor and travel being includ service calls during regular business hours. The additional charge would be for any parts that may be n throughout the term of the contract.

- **Pro-Active Gold**

This plan provides the ultimate protection for your automatic entrance investment. It includes all the benefits of Pro-Active Silver, plus replacement of any parts required during an unplanned repair or planned maintenance visit. Pro-Active

Product datasheet
High speed door
Albany HS8010P

ASSA ABLO

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Service Information

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Noteworthy details for daily handling

- The contact strip on all HS Series doors in not self monitoring as this is not per norm necessary. The closing force is by all doors lower then then the norm making a contact strip unnecessary. Due to the market expectation and usefulness that a door reverses when coming in contact with an object all HS Series doors are equipped with a contact strip. All contact strips are monitored by a wireless radio frequency detection device (WDD)
- Standard self monitored photocell at approx. 300 mm above FFL (as per norm also on RR doors)
- CEE Plug should be ordered separately
- Recommended Voltage
  Single phase power supply
  Fuse rating 20A K characteristic

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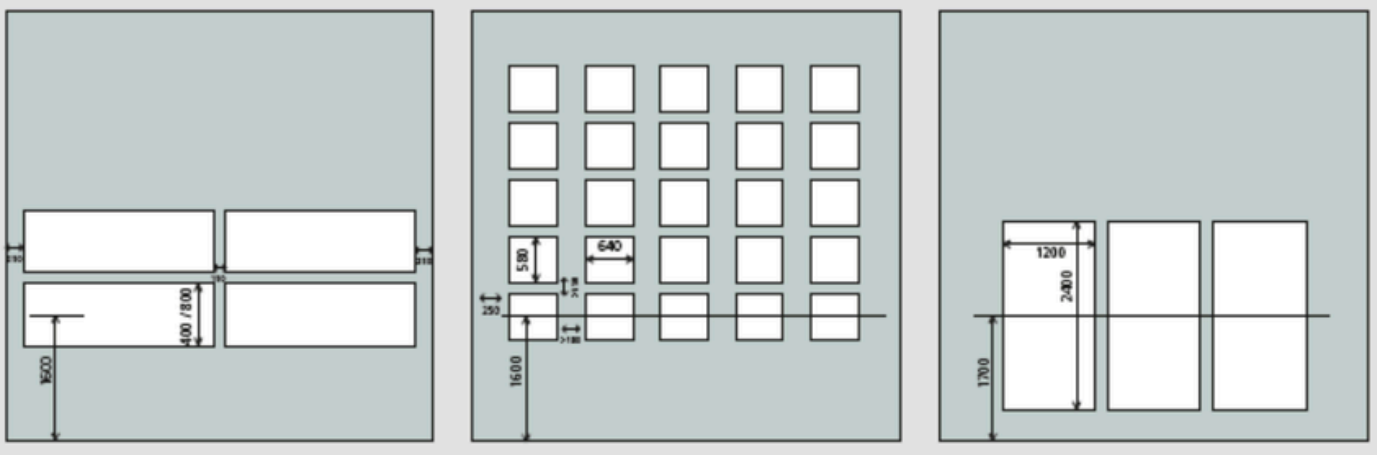
2. “Gravity” driven doors
3. “Power Drive” driven doors
4. Applications/Options
5. **Technical summaries**
6. Sales

Properties Summary

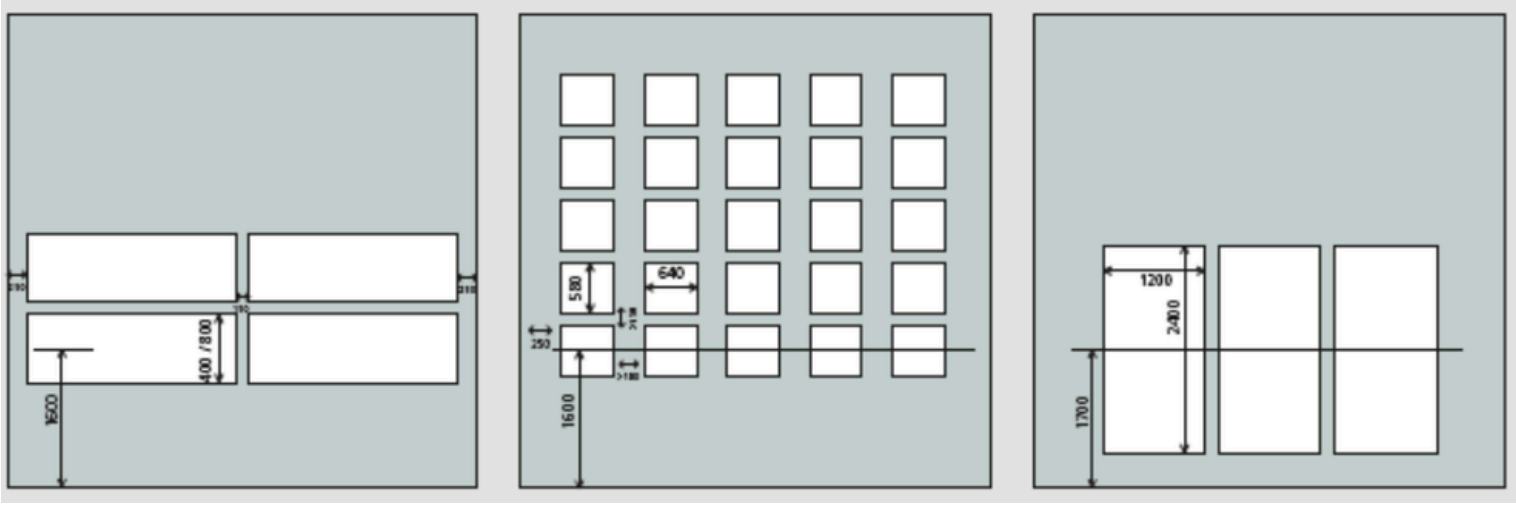
	HS9020 G	HS9020 GAT	HS9030 G	HS9010 P HS8010 P	HS9010 PFR	HS8010 PL HS8020 PL	HS9110 P HS8110 P
Technology	**Gravity**	**Gravity**	**Gravity**	**Power drive**	**Power drive**	**Power drive**	**Power drive**
Frequency inverter	Yes	Yes	Yes	Yes	Yes	Yes	Yes
encoder	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Parking brake	With High speed option	With High speed option	Yes	No	No	No	No
covers	Opt.	full	full	Opt.	Part.	Opt.	part.
WDD	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Photocells	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Windload	+	+	+	+++	++	+++	++
Sealing	++	+++	+++	+	++	+	+

Transparency Inside Applications



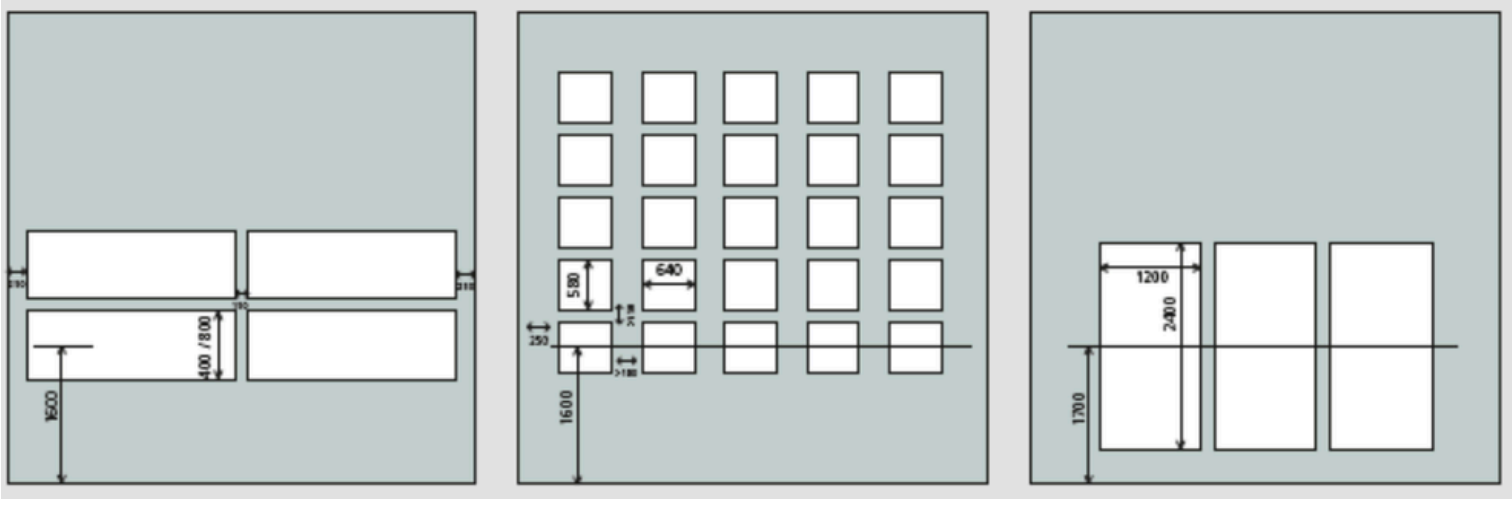
Inside	HS9010P	HS9010PL	HS9020G	HS9030G	HS9110P
Window 640x 580 mm PVC	X	X	X	X	X
Vision panel H 400 mm PVC	X		X	X	X
Vision panel H 800 mm PVC	X		X	X	X
Large Vision panel +/- 1200 x 2400 mm PVC	X				X

Transparency Outside Applications



Outside	HS8010P	HS8010PL	HS8020P	HS8020PL	HS8110P
Window 640x 580 mm PVC	X	X	X	X	X
Vision panel H 400 mm PVC	X		X		X
Vision panel H 800 mm PVC	X		X		X
Large Vision panel +/- 1200 x 2400 mm PVC	X		X		X

Transparency Application doors



Applications	HS9020GHY	HS9030GHY	HS9020GAT	HS9030GAT	HS9010PEM	HS9030GEM	HS9010PFR
Window 640x 580 mm PVC	X	X	X	X	X	X	X
Vision panel H 400 mm PVC	X	X	X	X			
Vision panel H 800 mm PVC	X	X	X	X			
Large Vision panel +/- 1200 x 2400 mm PVC							

Transparency Windows 640 x 580

- DLW

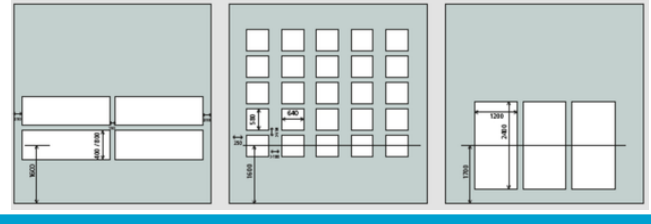
  one window min 1140 mm - One extra window per 790 mm

1140-1929	1930-2719	2720-3509	3510-4299	4300-5089	5090-5500
1	2	3	4	5	6

- Up to 9 windows

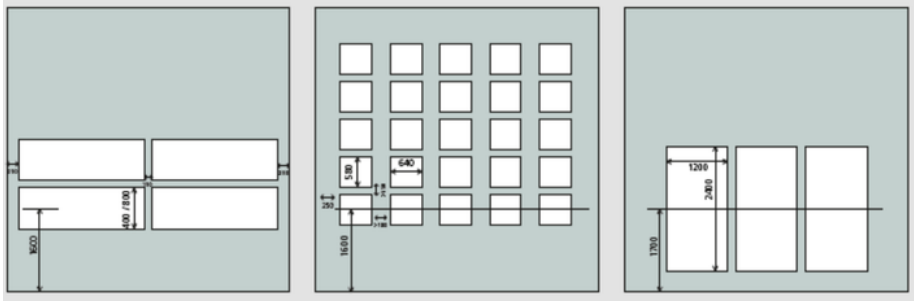
  DLH
  - Row 1 minimum height 2040 mm cc 1600 mm
  - Row 2 between 2041 - 2770 mm cc 2330 mm
  - Row 3 between 2771 - 3500 mm cc 3060 mm
  - Row 4 between 3501 - 4230 mm cc 3790 mm
  - Row 5 between 4231 - 4960 mm cc 4520 mm
  - Row all configurations possible
  - Row 1 and 4
  - Row 2 and 4

  All rows



Transparency Vision Panel – All doors

- **DLW**
  - one up to 2200 mm
  - Two more than 2200 mm



- **DLH**
- Vision panel 400 mm max 2 rows
  - One up to 1950 mm cc 1600 mm
  - Two more than 2500 mm cc 2150 mm

- **DLH**
- Vision panel 800 mm max 2 rows
  - One up to 2150 mm cc 1600 mm
  - Two more than 3100 mm cc 2550 mm

Product overview Inside Applications

Albany HS9010P Designed for medium-sized, heavy-duty operations
Albany HS9010PL Designed for large-sized, heavy-duty operations
Albany HS9020G Suitable for a wide variety of applications
Albany HS9030G The most space-efficient door model
Albany HS9110P Designed for medium-sized, heavy-duty operations

	HS9010P	**HS9010PL**	**HS9020G**	**HS9030G**	**HS9110P**
Speed Opening Closing	2.4 m/s 1.2 m/s	0.8 m/s 0.4 m/s	1.2 m/s 0.5 m/s	2.7 m/s 0.5 m/s	2.4 m/s 1.2 m/s
Size DW DH	1000 x 5500 2000 x 5500	5500 x 11000 2000 x 5500	1000 x 4000 2000 x 4000	1000 x 4000 2000 x 4000	1000 x 5500 2000 x 5500
Windows	4 types	640 x 580	3 types	3 types	4 types
Wind load	Class 1	Class 1	Class 1	Class 1	Class 1

Product overview Outside Applications

Albany HS8010P Designed for medium-sized heavy-duty operations
Albany HS8010PL Designed for large-sized heavy-duty operations
Albany HS8020P Designed for medium-sized heavy-duty operations
Albany HS8020PL Designed for large-sized heavy-duty operations
Albany HS8110P Designed for medium-sized heavy-duty operations

	HS8010P	**HS8010PL**	**HS8020P**	**HS8020PL**	**HS8110P**
Speed Opening Closing	2.4 m/s 1.2 m/s	0.8 m/s 0.4 m/s	2.4 m/s 1.2 m/s	0.8 m/s 0.4 m/s	2.4 m/s 1.2 m/s
Size DW DH	1000 x 5500 2000 x 5500	5500 x 9000 2000 x 5500	1000 x 5500 2000 x 5500	5500 x 8000 2000 x 5500	1000 x 5500 2000 x 5500
Windows	4 types	640 x 580	4 types	640 x 580	4 types
Wind load	Class 3 (4)	Class 2 (3)	Class 4 (5)	Class 3 (4)	Class 2 (3)

Product overview Special Applications

Albany HS9020GHY	Complete door structure made of stainless steel
Albany HS9030GHY	Complete door structure made of stainless steel
Albany HS9020GAT	Airtight door, suitable for inside environments, type cleanroom
Albany HS9030GAT	Airtight door with integrated motor for inside environments, type cleanroom
Albany HS9010PEM	Designed for inside heavy duty operations combined with emergency exit
Albany HS9030GEM	Designed for normal inside application combined with emergency exit
Albany HS9010PFR	Designed for freezer application

	HS9020GHY	HS9030GHY	HS9020GAT	HS9030GAT	HS9010PEM	HS9030GEM	HS9010PFR
Speed Opening Closing	1.2 m/s 0.5 m/s	2.7 m/s 0.5 m/s	1.2 m/s 0.5 m/s	2.7 m/s 0.5 m/s	2.0 m/s 1.4 m/s	2.7 m/s 0.5 m/s	2.4 m/s 1.2 m/s
Size DW DH	1000 x 4000 2000 x 4000	4000 2000 x	1000 x 4000 2000 x 2000	1000 x 4000 2000 x 4000	1000 x 5500 2000 x 5500	1000 x 3500 2000 x 3500	1000 x 4000 2000 x 5500
Windows	3 types	3 types	3 types	3 types	640 x 580	640 x 580	640 x 580
Wind load	Class 1	Class 1	Class 1	Class 1	Class 1	Class 1	Class 1
Air Permeability	Class 1	Class 2	Class 3/1	Class 3	Class 0	Class 0	Class 1

9-series	HS9020G	HS9030G	HS9010P	HS9010PL	HS9110P
Inside Doors					
	Suitable for a wide variety of applications	The most space-efficient door model	Designed for medium-sized, heavy-duty operations	Designed for large-sized, heavy-duty operations	Designed for medium-sized, heavy-duty operations
Usage	Inside	Inside	Inside	Inside	Inside
Operation	Gravity	Gravity	Unique door drive system	Unique door drive system	Unique door drive system
Min Size	1000 x 2000	1000 x 2000	1000 x 2000	5500x2000	1000 x 2000
Max Size	4000 x 4000	4000 x 4000	5500x5500	11000x5500	5500x5500
Opening speed	Up to 1,2 m/s (option 2,0 m/s)	Up to 2,7 m/s	Up to 2,4 m/s if A < 20 m² Up to 2,0 m/s if 20-24 m² Up to 1,6 m/s if A > 24 m²	0,8 m/s	Up to 2,4 m/s if A < 20 m² Up to 2,0 m/s if 20-24 m² Up to 1,6 m/s if A > 24 m²
Closing speed	0,5 m/s	0,5 m/s	1,2 m/s	0,4 m/s	1,2 m/s
Colours RAL	1003 Yellow 6005 Green 2004 Orange 7035 Gray 3000 Red 9005 Black 5002 Blue 9010 White	1003 Yellow 6005 Green 2004 Orange 7035 Gray 3000 Red 9005 Black 5002 Blue 9010 White	1003 Yellow 6005 Green 2004 Orange 7035 Gray 3000 Red 9005 Black 5002 Blue 9010 White	1003 Yellow 6005 Green 2004 Orange 7035 Gray 3000 Red 9005 Black 5002 Blue 9010 White	1003 Yellow 6005 Green 2004 Orange 7035 Gray 3000 Red 9005 Black 5002 Blue 9010 White
Construction	Galvanised steel	Galvanised steel	Galvanised steel Option: Stainless steel 304	Galvanised steel	Galvanised steel Option: Stainless steel 304
Covers	Option	Standard, glasfiber reinforced plastic	Option: Galvanised steel or Stainless steel 304	Option: Side and motor Drum only if DLW up to 5900	Standard: Side covers Option: Motor and drum Galv or stainless steel
Control box	Steel	Steel	Steel	Steel	Steel
Visibility	Windows 640x580 Vision panel H=400 Vision panel H=800	Windows 640x580 Vision panel H=400 Vision panel H=800	Windows 640x580 Windows +/-1200x2400 Vision panel H=400 Vision panel H=800	Windows 640x580	Windows 640x580 Windows +/-1200x2400 Vision panel H=400 Vision panel H=800
Chain hoist	Yes	No	Yes	Yes	Yes
Power supply	230V / 16A / 50Hz-60Hz	230V / 16A / 50Hz-60Hz	< 7m² 230V / 16A / 50Hz-60Hz > 7m² 400V 3 phase 1,5 kW 3x16A / 50Hz-60Hz		< 7m² 230V / 16A / 50Hz-60Hz
Wind load	22 m/s – Class 1 Max Windload in operation / closing: 9 m/s - 50 Pa	22 m/s – Class 1	22 m/s – Class 1	22 m/s – Class 1	22 m/s – Class 1
Thermal Res.	6,02 W/m²K	6,02 W/m²K	6,02 W/m²K	6,02 W/m²K	6,02 W/m²K
Side space	65	81	135 / 145	135 / 145	135 / 145
Side space Oper.	230	85	340	360	340
Headroom	390	402	500 (650 cover, H > 4000)	650	450 (650 cover, H > 4000)
Depth	255	328	540 (650 cover, H > 4000)	680	540 (650 cover, H > 4000)
Placement Oper.	Right / Left	Right (Inside the drum)	Right / Left	Right / Left	Right / Left

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8-series	HS8010P	HS8010PL	HS8020P	HS8020PL	HS8110P
Outside Doors					
	Designed for medium-sized heavy-duty operations	Designed for large-sized heavy-duty operations	Designed for medium-sized heavy-duty operations	Designed for large-sized heavy-duty operations	Designed for medium-sized heavy-duty operations
Usage	Outside	Outside	Outside	Outside	Outside
Operation	Unique door drive system	Unique door drive system	Unique door drive system	Unique door drive system	Unique door drive system
Min Size	1000 x 2000	5500x2000	1000 x 2000	5500x2000	1000 x 2000
Max Size	5500x5500	9000x5500	5500x5500	8000x5500	5500x5500
Opening speed	Up to 2,4 m/s if A < 20 m² Up to 2,0 m/s if 20-24 m² Up to 1,8 m/s if A > 24 m²	0,8 m/s	Up to 2,4 m/s if A < 20 m² Up to 2,0 m/s if 20-24 m² Up to 1,8 m/s if A > 24 m²	0,8 m/s	Up to 2,4 m/s if A < 20 m² Up to 2,0 m/s if 20-24 m² Up to 1,8 m/s if A > 24 m²
Closing speed	1,2 m/s	0,4 m/s	1,2 m/s	0,4 m/s	1,2 m/s
Colours RAL	1003 Yellow 6005 Green 2004 Orange 7035 Gray 3000 Red 9005 Black 5002 Blue 9010 White	1003 Yellow 6005 Green 2004 Orange 7035 Gray 3000 Red 9005 Black 5002 Blue 9010 White	1003 Yellow 6005 Green 2004 Orange 7035 Gray 3000 Red 9005 Black 5002 Blue 9010 White	1003 Yellow 6005 Green 2004 Orange 7035 Gray 3000 Red 9005 Black 5002 Blue 9010 White	1003 Yellow 6005 Green 2004 Orange 7035 Gray 3000 Red 9005 Black 5002 Blue 9010 White
Construction	Galvanised steel Option: Stainless steel 304	Galvanised steel	Galvanised steel Option: Stainless steel 304	Galvanised steel	Galvanised steel Option: Stainless steel 304
Covers	Option: Galvanised steel or Stainless steel 304	Option: Side and motor Drum only if DLW up to 5900	Option: Galvanised steel or Stainless steel 304	Option: Side and motor Drum only if DLW up to 5900	Standard: Side covers Option: Motor and drum Galv or stainless steel
Control box	Steel	Steel	Steel	Steel	Steel
Visibility	Windows 640x580 Windows +/-1200x2400 Vision panel H=400 Vision panel H=800	Windows 640x580	Windows 640x580 Windows +/-1200x2400 Vision panel H=400 Vision panel H=800	Windows 640x580	Windows 640x580 Windows +/-1200x2400 Vision panel H=400 Vision panel H=800
Chain hoist	Yes	Yes	Yes	Yes	Yes
Power supply	400V 3 phase 1,5 kW 3x16A / 50Hz-60Hz				
Wind load	35 m/s – Class 3 Max 4.0x5,5 m: 40 m/s Class 4	28 m/s – Class 2 Max 6,0x5,5 m: 35 m/s Class 3	40 m/s – Class 4 Max 5.0x5,5 m: 55 m/s Class 5	35 m/s – Class 3 Max 6.0x5,5 m: 40 m/s Class 4	28 m/s – Class 2 Max 4.0x5,5 m: 35 m/s Class 3
Thermal Res.	6,02 W/m²K	6,02 W/m²K	6,02 W/m²K	6,02 W/m²K	6,02 W/m²K
Side space	135 / 145	135 / 145	135 / 145	135 / 145	135 / 145
Side space Oper.	340	360	340	360	340
Headroom	500 (650 cover, H > 4000)	650	500 (650 cover, H > 4000)	650	450 (650 cover, H > 4000)
Depth	540 (650 cover, H > 4000)	680	540 (650 cover, H > 4000)	680	540 (650 cover, H > 4000)
Placement Oper.	Right / Left	Right / Left	Right / Left	Right / Left	Right / Left

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9-series	HS9020GHY	HS9020GAT	HS9030GHY	HS9030GAT	HS9030GEM
Application Doors					
	Complete door frame made of stainless steel	Airtight door, suitable for inside environments, type cleanroom.	Complete door frame made of stainless steel.	Airtight door with integrated motor for inside environments, type cleanroom.	Designed for normal inside application combined with emergency exit
Usage	Inside	Inside	Inside	Inside	Inside
Operation	Gravity	Gravity	Gravity	Gravity	Gravity
Min Size	1000 x 2000	1000 x 2000	1000 x 2000	1000 x 2000	1000 x 2200
Max Size	4000 x 4000	4000 x 4000	4000 x 4000	4000 x 4000	3500 x 3500
Opening speed	Up to 1,2 m/s (option 2,0m/s)	Up to 1,2 m/s (option 2,0m/s)	Up to 2,7 m/s	Up to 2,7 m/s	Up to 2,0 m/s
Closing speed	0,5 m/s	0,5 m/s	0,5 m/s	0,5 m/s	0,5 m/s
Colours RAL	1003 Yellow 6005 Green 2004 Orange 7035 Gray 3000 Red 9005 Black 5002 Blue 9010 White	1003 Yellow 6005 Green 2004 Orange 7035 Gray 3000 Red 9005 Black 5002 Blue 9010 White	1003 Yellow 6005 Green 2004 Orange 7035 Gray 3000 Red 9005 Black 5002 Blue 9010 White	1003 Yellow 6005 Green 2004 Orange 7035 Gray 3000 Red 9005 Black 5002 Blue 9010 White	1003 Yellow 6005 Green 2004 Orange 7035 Gray 3000 Red 9005 Black 5002 Blue 9010 White
Construction	Stainless steel 304	Standard: Galvanised steel Option: Stainless steel 304	Stainless steel 304	Standard: Galvanised steel Option: Stainless steel 304	Standard: Galvanised steel Option: Stainless steel 304
Covers	Stainless steel 304	Option: Galvanised steel or Stainless steel	Standard, glasfiber reinforced plastic	Standard, glasfiber reinforced plastic	Standard, glasfiber reinforced plastic
Control box	Standard: Steel Option: Stainless steel 304	Steel	Standard: Steel Option: Stainless steel 304	Steel	Steel
Visibility	Windows 640x580 Vision panel H=400 Vision panel H=800	Windows 640x580 Vision panel H=400 Vision panel H=800	Windows 640x580 Vision panel H=400 Vision panel H=800	Windows 640x580 Vision panel H=400 Vision panel H=800	Vision panel H=400 mm
Chain hoist	Yes	No	No	No	No
Power supply	230V / 16A / 50Hz-60Hz	230V / 16A / 50Hz-60Hz	230V / 16A / 50Hz-60Hz	230V / 16A / 50Hz-60Hz	230V / 16A / 50Hz-60Hz
Wind load	22 m/s – Class 1	22 m/s – Class 1	22 m/s – Class 1	22 m/s – Class 1	22 m/s – Class 1
	Max Windload in operation / closing: 9 m/s - 50 Pa				
Thermal Res.	6,02 W/m²K	6,02 W/m²K	6,02 W/m²K	6,02 W/m²K	6,02 W/m²K
Air Permability	Class 1	Class 3/1	Class 1	Class 4 Overpres, 3 Underpress	Class 0
Side space	65	65	81	81	81
Side space Oper.	230	230	85	85	85
Headroom	390	390	402	402	402
Depth	255	255	328	328	408
Placement Oper.	Right / Left	Right / Left	Right (Inside the drum)	Right (Inside the drum)	Right (Inside the drum)

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9-series	HS9010PEM	HS9010PFR
Application Doors		
	Designed for inside heavy duty operations combined with emergency exit	Designed for freezer application
Usage	Inside	Inside
Operation	Unique door drive system	Unique door drive system
Min Size	1000 x 2200	1000 x 2000
Max Size	5500 x 5500	4000 x 5500
Opening speed	Up to 2,0 m/s	Up to 2,4 m/s
Closing speed	1,0 m/s	1,2 m/s
Colours RAL	1003 Yellow 6005 Green 2004 Orange 7035 Gray 3000 Red 9005 Black 5002 Blue 9010 White	1003 Yellow 6005 Green 2004 Orange 7035 Gray 3000 Red 9005 Black 5002 Blue 9010 White
Construction	Standard: Galvanised steel Option: Stainless steel 304	Standard: Galvanised steel Option: Stainless steel 304
Covers	Option: Galvanised steel or Stainless steel 304	Option: Galvanised steel or Stainless steel **NO DRUM COVER**
Control box	Steel	Steel
Visibility	Vision panel H=400 mm	Windows 640x580
Chain hoist	Yes	No
Power supply	400V 3 phase 1,5 kW 3x16A / 50Hz-60Hz	
Wind load	22 m/s – Class 1	22 m/s – Class 1
Thermal Res.	6,02 W/m²K	6,02 W/m²K
Side space	135 / 145	135 / 145
Side space Oper.	340	340
Headroom	500 (650 cover, H > 4000)	500
Depth	620 (735 H > 4000)	550
Placement Oper.	Right / Left	Right / Left
Remark		Opens every 20-25 minutes Do not switch Power off Do not lock the door Ice to be removed occasionally

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Content

1. Product overview
2. “Gravity” driven doors
3. “Power Drive” driven doors
4. Applications/Options
5. Technical summaries
6. **Sales**

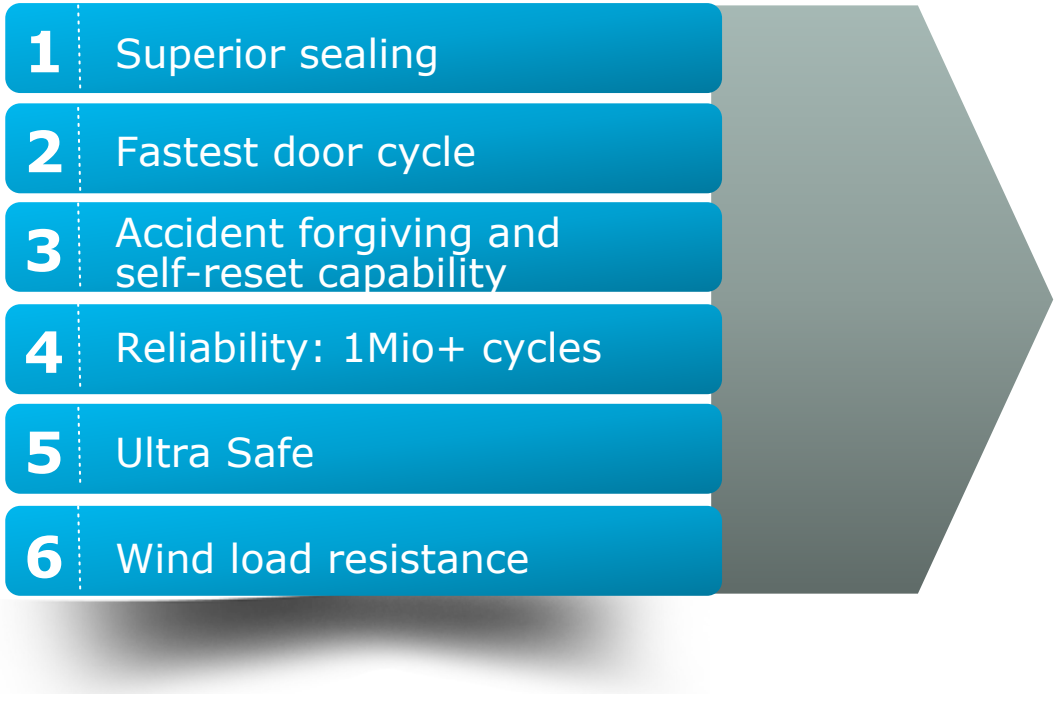
Unique Selling Points

- Tightness/sealing
- Opening and closing speed
- Self resetting
- Reliability
- Safety in use
- Wind load resistance

- Energy and cost savings
- User safety
- Low total cost of ownership
- Better environment control
- Increased operational efficiency
- No operational down time
- Reduced risk profile
- Regulatory compliance
- Use in extreme environments

Best choice

Special characteristics



Customer advantages

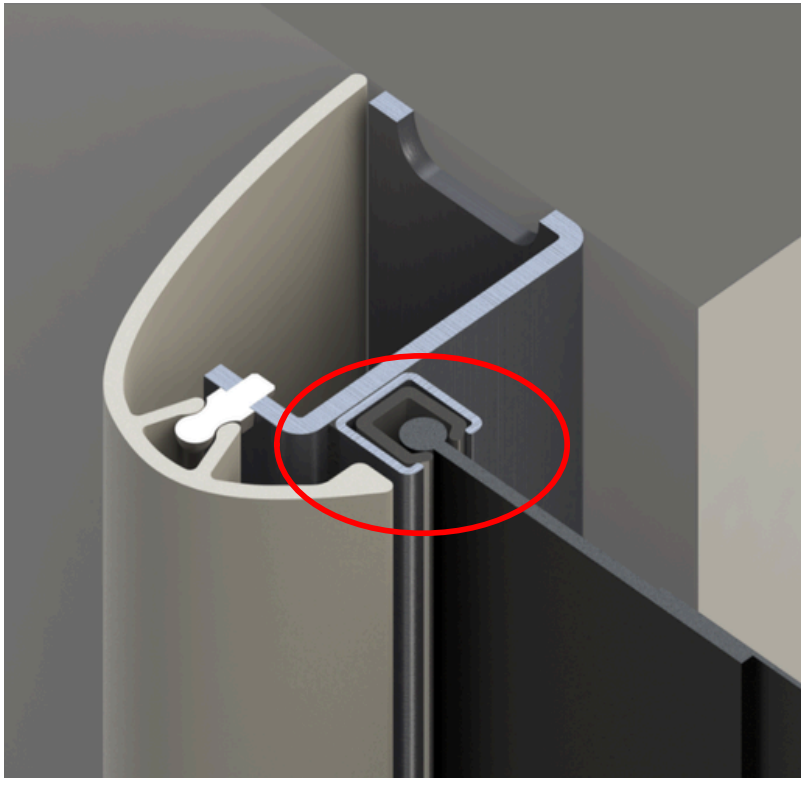
- Energy and cost savings
- User safety
- Low total cost of ownership
- Better environment control
- Increased operational efficiency
- No operational down time
- Reduced risk profile
- Regulatory compliance
- Use in extreme environments

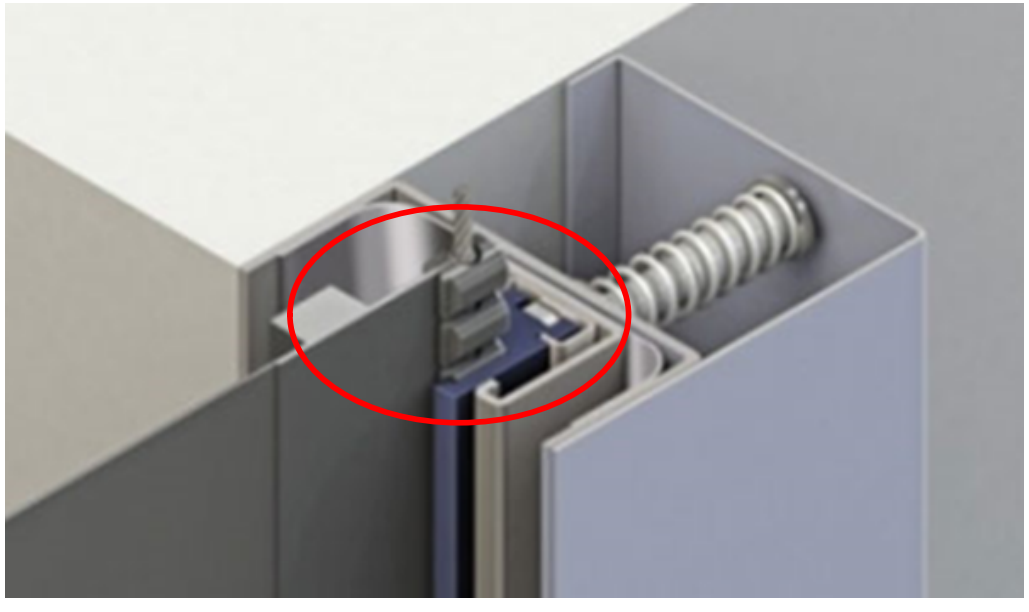
Best Choice

Focus on characteristics

1. Superior sealing
2. Fastest door cycle
3. Accident forgiving and self-reset capability
4. Reliability: 1Mio+ cycles (D3)
5. Ultra safe
6. Wind load resistance

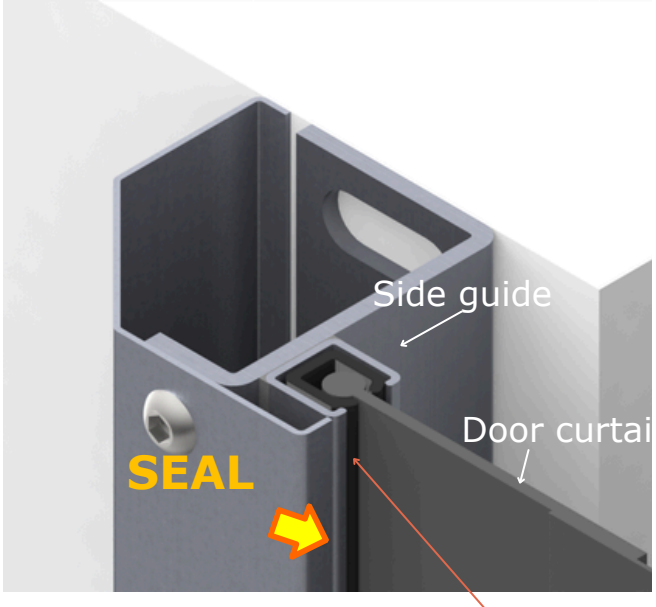
1. Superior sealing



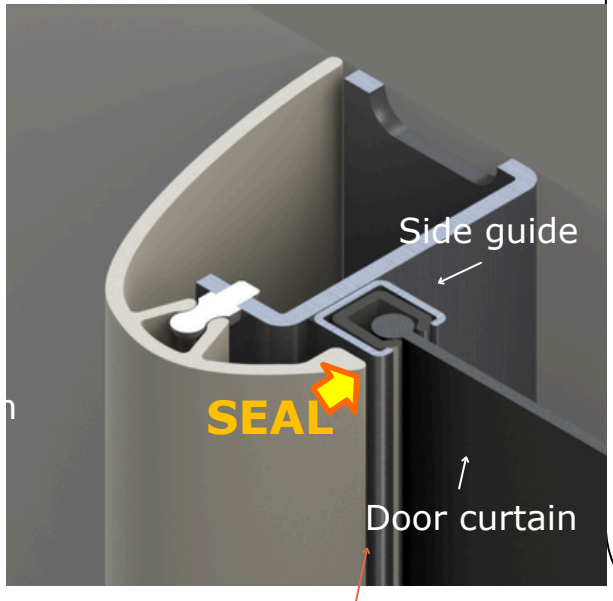


1. Superior sealing

Perfect seal between door curtain and side guide on each HS door



Type HS9020 G



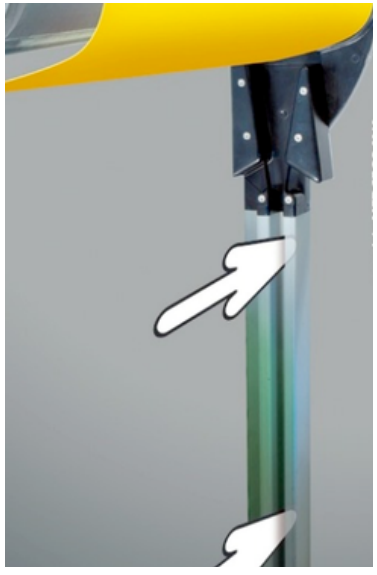
Type HS9030 G

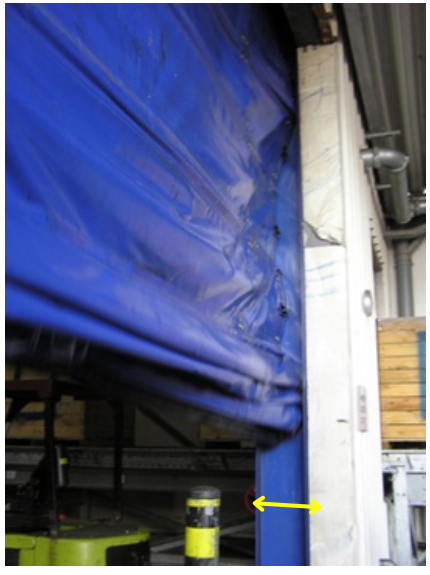
Type HS9010/8010/8020 P and PL

Virtually no air can pass

1. Superior sealing

Competitors: open side guides
= NO seal between door curtain and side guide

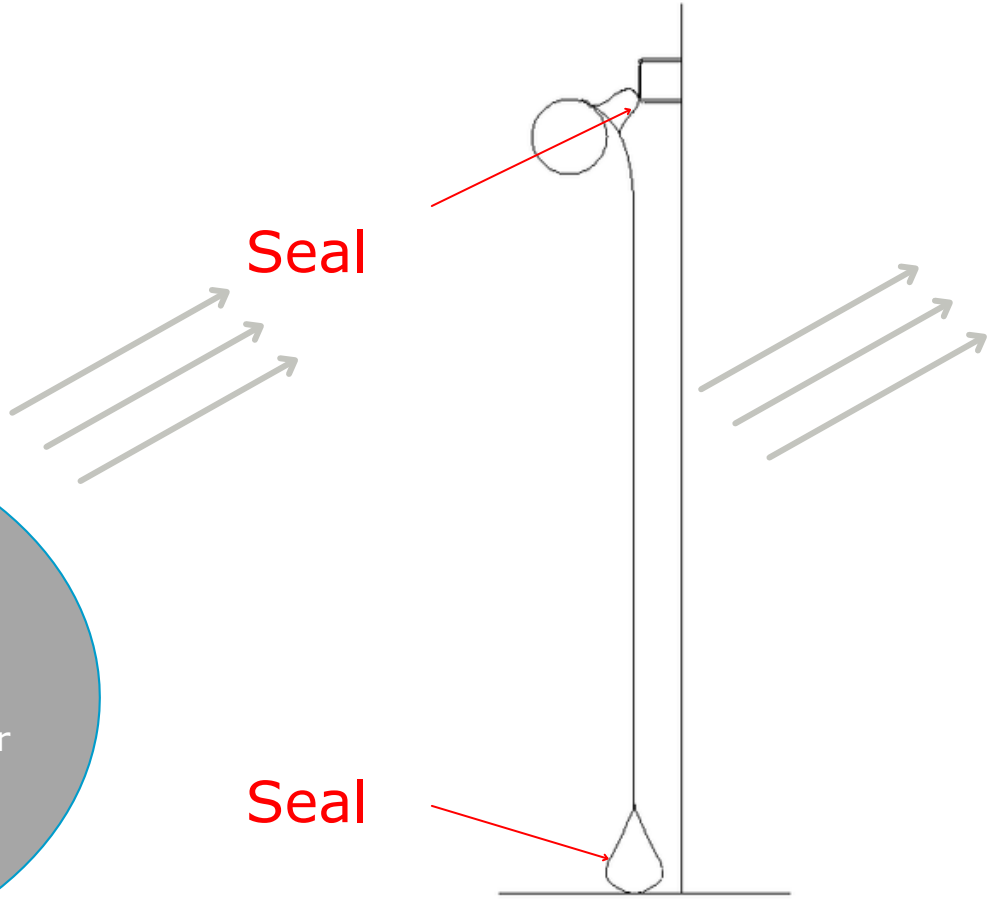




CONSEQUENCE:
SIGNFICANT ENERGY LOSS

1. Superior sealing

- Save on energy
- Keep out dust, dirt and insects
- Minimize transfer of conditioned air



1. Superior sealing: advantages

1. Saves costs (heat/ cool) ge
2. Environment friendly
3. Keeps insects out
4. Keeps odours inside
5. Respect cold chain

Best Choice

Focus on characteristics

1. Superior sealing
2. Fastest door cycle
3. Accident forgiving and self-reset capability
4. Reliability: 1Mio+ cycles
5. Ultra safe
6. Wind load resistance

2. Fastest door cycle

Performance

Operating speed:	Opening:up to 2,4 m/s** Closing: 1,2 m/s

2. Fastest door cycle: advantages

1. Time saving for workers
2. Fluid internal traffic
3. Increased production
4. Minimize air leakage
5. Minimize infiltration of hot, moist air in dry, cold zones
6. Respect cold chain

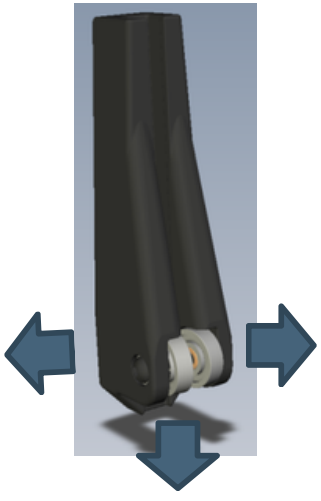
Best Choice

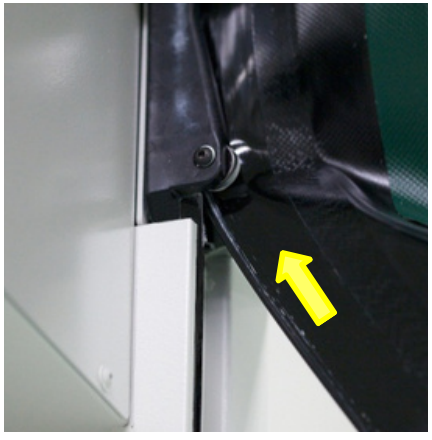
Focus on characteristics

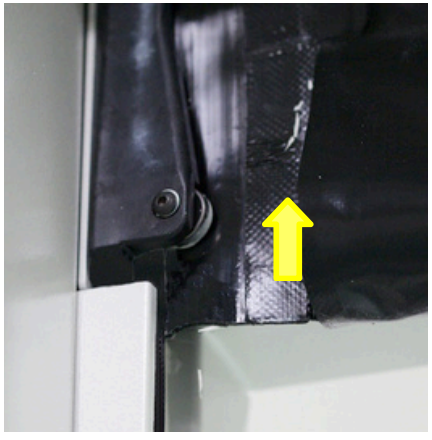
1. Superior sealing
2. Fastest door cycle
3. Accident forgiving and self-reset capability
4. Reliability: 1Mio+ cycles
5. Ultra safe
6. Wind load resistance

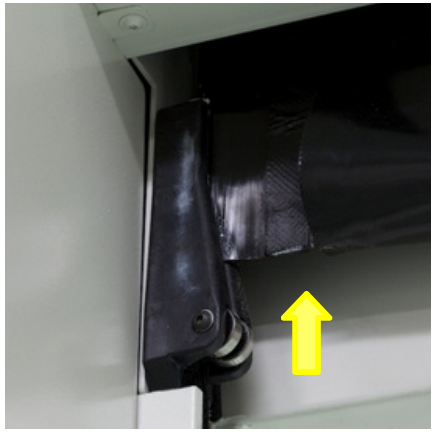
3. Accident forgiving and self-reset capability

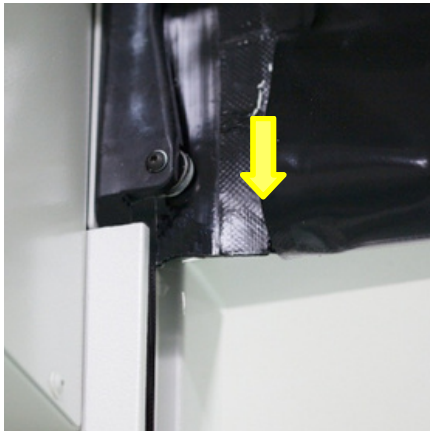
3. Accident forgiving and self-reset











3. Accident forgiving and self reset advantages

1. No costly repairs
2. No interruptions due to blocke
3. No need for technical intervent
4. No time loss for workers
5. No damage to vehicles in case of accident
6. No issues with workers unions

Best choice

Focus on characteristics

1. Superior sealing
2. Fastest door cycle
3. Accident forgiving and self-reset capability
4. Reliability: 1Mio+ cycles
5. Ultra safe
6. Wind load resistance

4. Reliability: tested over 1 million cycles

Few wearing parts in movement

No rigid parts in the curtain

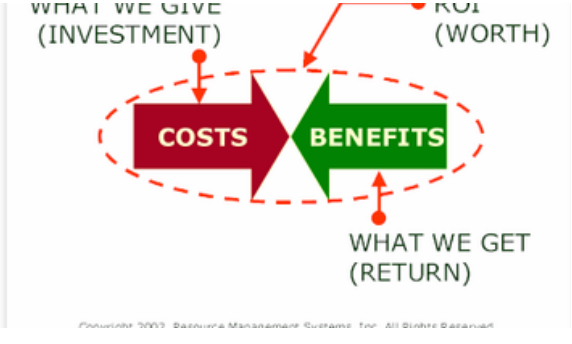
Little maintenance required

Very long lifetime

Attention: not a warranty!

4. Reliability: advantages

1. Return on investment
2. Reduced maintenance costs
3. No production downtime
4. No time loss for workers
5. Minimal need for technical interventions



Best choice

Focus on characteristics

1. Superior sealing
2. Fastest door cycle
3. Accident forgiving and self-rest capability
4. Reliability: 1Mio+ cycles
5. Ultra safe
6. Wind load resistance

5. Ultra Safe

HS Series Doors:

- **Safest Choice on the Market!**
- **Soft and flexible door curtain contains no dangerous elements or mechanics resulting in a safte work place**
- **Power Drive: No straps or cables, no weights**
- **Gravity Drive: no straps or cables, limited weight.**
- **No rigid elements**
- **No bar or stiffeners**
- **WDD (Wireless Detection Device) in the flexible bottom profile and a safety photocell in the side column**

5. Ultra Safe: advantages

1. No risk for injuries
2. Less absenteeism
3. Less insurance costs
4. No issues with workers unions
5. No damage to good
6. No damage to material

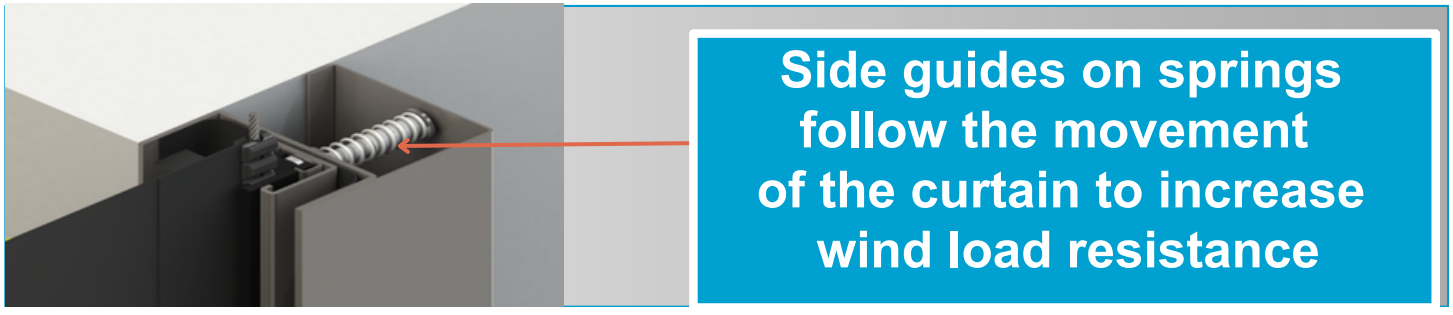
Best choice

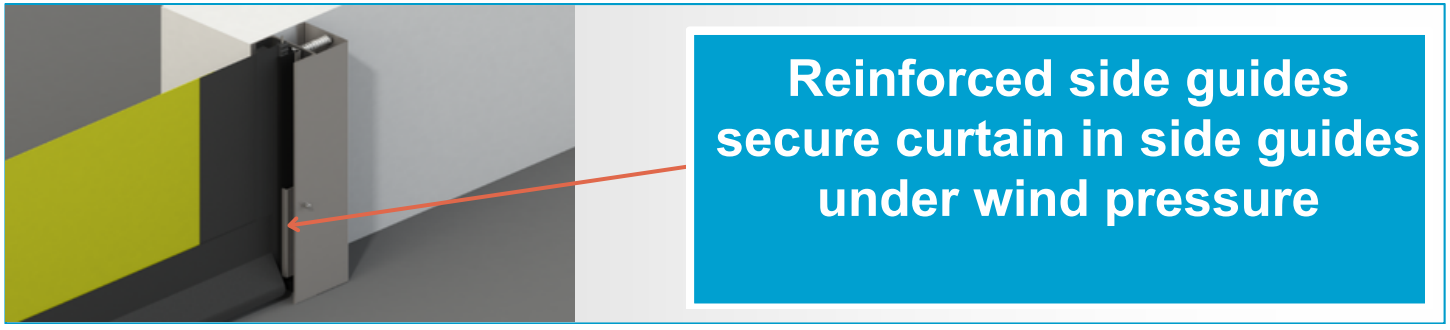
Focus on characteristics

1. Superior sealing
2. Fastest door cycle
3. Accident forgiving and self-reset capability
4. Reliability: 1Mio+ cycles
5. Ultra safe
6. Wind load resistance

6. Wind load resistance

Class 2- 5, depending on door type and size.





- Use under extreme conditions
- Less damage to the door
- Less repair costs
- Better working environment

6. Wind load resistance

European Standard EN 12424:
Wind load resistance of closed door

CLASS 5 = > 1000 Pa

CLASS 4 = 1000 Pa

CLASS 3 = 700 Pa

CLASS 2 = 450 Pa

CLASS 1 = 300 Pa <

HS-Doors approved for:

CLASS 5 : HS8020 P (up to 5x5,5m)

CLASS 4 : HS8020 P (between 5x5,5 and 5,5x5,5m)
HS8020 PL (up to 6x5,5m)
HS8010 P (up to 4x5,5m)

CLASS 3 : HS8010 P (up to 5,5x5,5m)
HS9010 PFR (up to 4x5,5m)
HS8010 PL (up to 6x5,5m)
HS8020 PL (between 6x5,5 and 8x5,5m)
HS8110 P (up to 4 x 5,5m)

CLASS 2 : HS9010 P (up to 5,5x5,5m)
HS8010 PL (up to 9x5,5m)
HS8110 P (up to 5,5x5,5m)

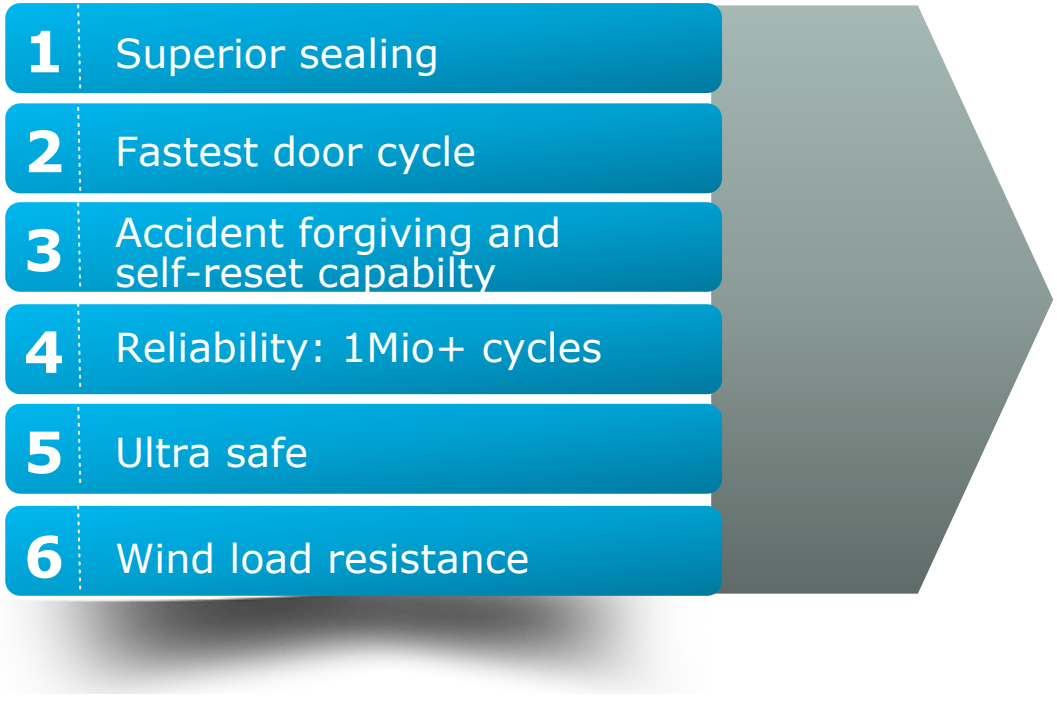
CLASS 1 : all others

6. Wind load resistance advantages

1. Use under extreme conditions
2. Less damage to the door
3. Less repair costs
4. Better working environment

Safe choice

Special characteristics



Customer advantages

- Energy and cost savings
- User safety
- Low total cost of ownership
- Better environment control
- Increased operational efficiency
- No operational down time
- Reduced risk profile
- Regulatory compliance
- Use in extreme environments

The global leader in door opening solutions